Buyer-Initiated Auctions: Nominations in Oil & Gas

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Abstract

Auctions fail when too few bidders participate. In oil and gas leasing, a common remedy is pre-auction nomination: a tract is offered only if some bidder pays a non-refundable fee and commits to buying at the minimum bid. This paper asks how nomination compares with seller-initiated auctions when markets are thin. I develop a two-stage nomination-auction game in which bidders' values are correlated but costly to learn, and each bidder chooses whether to learn early and nominate or to learn only after someone else nominates. Using novel data from North Dakota oil and gas lease sales, I find nomination raised seller revenue by about $3.05 million (44%) across the 728 tracts in the 2017–2018 estimation sample, relative to a seller who chooses arbitrarily which tracts to offer. Nomination is valuable mainly because it guarantees a bidder when attention is scarce: every nominated tract sells, against about 90 percent without nomination, and the average winning bid rises by about 30 percent through the competition it adds. The gains accrue almost entirely to the seller. Bidders gain mainly by avoiding learning costs on tracts they would not buy anyway: nomination steers their learning toward tracts already screened as worth at least the minimum bid.

1 Introduction

Auctions are a primary mechanism for allocating resources, from private goods to government assets. Before any auction can be designed, however, the seller must decide which items to

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offer. When running each auction is costly, that decision matters: an item that draws no bidder still has to be prepared and offered. And when demand is uncertain, the seller cannot tell in advance which items those are. The problem is illustrated by recent U.S. government attempts to auction oil and gas leases in Alaska's Arctic National Wildlife Refuge, where the January 2025 Bureau of Land Management (BLM) lease sale offered about 400,000 acres and received zero bids.[1] When demand for an item may not exist at all, the standard design instruments—reserve prices and entry fees—cannot help, because they act only on bidders who have already shown up.

This paper studies a mechanism built for exactly this problem: pre-auction nomination. Rather than the seller deciding which items to auction, interested buyers nominate the items they wish to see offered, paying a small fee and committing to purchase the item at a minimum price if no other bidder appears. Only items that receive a nomination proceed to auction.[2] Nomination is widely used in mineral leasing: between 2009 and 2019 about 87 million acres were nominated for federal onshore oil and gas leasing [U.S. Government Accountability Office, 2021, p. 18], many U.S. states lease their mineral rights this way, and a similar system determined which Outer Continental Shelf tracts the federal government offered before area-wide leasing was adopted in 1983 [Hendricks and Porter, 2014].

I ask what nomination is worth, and to whom. I compare the single-nomination rule used in practice against a benchmark in which the seller chooses which tracts to offer without input from buyers, and measure the difference along five outcomes: the seller's revenue; the *sale rate*, i.e. how often an offered tract draws any bidder at all; the surplus of the bidder who nominates; the surplus of the bidders who bid without nominating; and the learning costs that bidders sink to evaluate tracts. Nomination can move these outcomes through four channels. First, it supplies a *guaranteed bidder*: every offered tract has at least one bidder present, who has already paid to evaluate the tract and committed to buy it. Second, *more valuable tracts are offered*: a bidder nominates only after finding the tract worth enough. Third, nomination *concentrates attention*: bidders with limited attention who cannot monitor thousands of candidate tracts can attend to the short list of nominated ones,

[1] For such BLM-run oil and gas lease auctions, environmental reviews required under the National Environmental Policy Act must be completed before the auction, regardless of whether or not those lands are ultimately leased. Across twelve reviews conducted between June 2020 and January 2021, the cost ranged from $17,000 to $178,000 per lease sale, and BLM Utah reported an average of about $2,100 per parcel across its two most recent auctions [U.S. Government Accountability Office, 2021, pp. 28–29, n.42].

[2] In U.S. federal onshore oil and gas leasing, this same mechanism is officially termed an *expression of interest* (EOI): a member of the public may nominate a parcel for a lease sale by submitting an EOI, and BLM may also select parcels directly [U.S. Government Accountability Office, 2021]. I use *nomination* throughout to match the terminology of my empirical setting (North Dakota state leasing). Section 1.1 situates this mechanism relative to related but distinct notions of "expressions of interest" and "indications of interest" in the finance literature (IPO bookbuilding, M&A sale processes).

so more of them arrive per tract. Fourth, when bidders' values are correlated but learned only at a cost, a nomination creates an *information spill-over*: it tells other bidders that the tract is likely valuable to them too, so more of them pay to learn. The spill-over has a countervailing force, *under-provision*: a bidder who can wait to learn from someone else's nomination has less reason to learn early and nominate herself, so nominations may become rarer. Only this last channel requires correlated values and costly learning; the other three operate in any nomination system.

To quantify these channels I build a two-stage nomination-auction game. Potential bidders arrive over time according to two Poisson processes, one before nomination and one after, whose rates are allowed to differ. Values are correlated private values: firms draw on the same geology and commodity prices, so their values move together, but each firm's value depends on its own technology and lease portfolio and can be learned only by paying a privately known learning cost. A bidder arriving before nomination chooses among learning early (and nominating if her value is high enough), waiting to learn only if someone else nominates, and exiting; a bidder arriving after nomination decides whether to learn and enter. A tract with at least one nomination is sold by a second-price auction, and the number of nominations is not revealed. In equilibrium, bidders follow cutoff rules: low-cost bidders learn early, intermediate-cost bidders wait, high-cost bidders exit, and an early learner nominates when her value exceeds a threshold.

I estimate the model with lease auctions run by the State of North Dakota in 2017–2018: 878 auctions and 2,740 bids, combined with nomination records. Estimation proceeds in two steps. First, I recover the value distribution and the correlation level from bids. Because North Dakota uses proxy bidding, the winner pays one increment above the second-highest bid and the winning bid is never observed.[3] I develop a copula-based estimator that matches the observed order statistics to their theoretical distribution under exchangeable copula dependence. Second, I estimate the arrival rates and the learning-cost distribution by GMM with the equilibrium conditions imposed as constraints. Because bidders' values scale with tract characteristics while the fee, the reserve, and learning costs are fixed in dollars, the equilibrium differs across tracts, and I solve it separately for 19 groups of tracts ordered by value.

The estimates show that values are strongly correlated ($\hat{\rho} = 0.86$). The mean learning cost is about \$1,161, small against an average bid of about \$11,800, so on valuable tracts essentially every firm that considers a tract finds it worth evaluating. I also find that few firms

[3]Under proxy bidding, each bidder submits a maximum willingness to pay (the proxy bid) to the online platform, which then bids on her behalf in minimum increments up to that maximum. The auction therefore works like an ascending button auction played by the computer, and the bidder with the highest proxy bid wins at one increment above the second-highest proxy bid. See Section 2.

look at any given tract before nomination: the pre-nomination arrival rate is about 0.005 per tract, against about 2.9 after nomination. This is consistent with the low nomination rate (on average about 98 nominations per season across roughly 28,000 available tracts, about 0.35% per tract) against a high auction participation rate (about 3.3 bidders per auctioned tract, with roughly 90 tracts auctioned per season).[4]

I then use the estimated model to construct a sequence of counterfactuals that switch off one channel at a time: the information spill-over (late entrants decide on their prior rather than updating on the nomination), the selection of more valuable tracts (the early bidder's value need only clear the reserve rather than the nomination threshold), and the guaranteed bidder (the early bidder, whose value clears the reserve, is no longer certain to be present). The last world is the benchmark without nomination, in which the seller offers tracts arbitrarily, and the three steps sum to the total value of nomination. To measure under-provision, I also switch the signal off and re-solve the early-learning cutoff.

Because concentrated attention changes how many bidders look at a tract in the first place, an additional experiment isolates it: a one-stage market with a single arrival process, evaluated first at the arrival rate the estimated attention function implies when attention is spread over all available tracts, and then at the rate it implies when the same attention is concentrated on the much smaller set of offered tracts.

I find that, across the 728 tracts in the estimation sample, nomination raises seller revenue by about \$3.05 million, or 44% (from \$6.86 million to \$9.91 million), relative to a seller who chooses arbitrarily which tracts to offer.[5] Almost all of this gain (\$3.01 million) comes from the *guaranteed bidder*: every nominated tract sells, while about 10% of arbitrarily offered tracts draw no bidder, and the winning bid on a sold tract is about 30% higher because the guaranteed bidder adds competition. The *selection of more valuable tracts* adds essentially nothing, because the nomination threshold barely exceeds the reserve. Total surplus rises by about \$3.3 million: nominators gain \$2.1 million, and non-nominators lose \$1.9 million as the guaranteed nominating bidder competes away their rents. The information spill-over is small: it raises participation by 0.1 percentage points on average and by 1.4 points on the least promising 4% of tracts, and thus the under-provision it induces is essentially zero (0.04 points).

The extensive margin is decisive for tracts that look less valuable on observable characteristics, such as being far from existing wells or small: an arbitrarily offered tract of this

[4]There can be multiple nominations received for a given tract, so the total nominations received per season exceeds the count of total auctions.

[5]These totals value each tract at its own scaler s_t (Appendix Table 20). Section 7 values each bin at its median tract, which gives \$6.40 million and \$9.28 million (Table 3); the split across channels is essentially unchanged.

kind draws a bidder only 1.4% of the time, and nomination shifts its sale rate to one.

Nomination also lowers the learning costs that bidders might sink by concentrating learning on the tracts most likely to be worth it. On the tracts it offers, wasted learning—the cost paid by bidders who learn but draw $v < r$ and never bid—roughly halves, from about \$272 to \$140 per auctioned tract.

The apparent opportunity cost of nomination—tracts left unoffered because the nominator drew a low value, though others might have valued them—is in fact a saving. Because values are correlated, a low early draw means that everyone's value is likely low: auctioning such a tract would spend about \$3,965 on learning to realize only about \$1,571 in value, roughly an eighth of the $\approx$ \$12,756 winning value on a nominated tract.

The largest channel is concentrated attention. I find that per-tract attention falls roughly in proportion to the size of the pool. When the attention-implied arrival rate is spread across all $\approx$28,000 available tracts, under 1% of tracts sell. Concentrating the same attention on a smaller set of tracts raises arrivals so 98% sell—worth about \$8,130 per tract to the seller and \$18,318 to bidders. Nomination is valuable mainly because it secures a committed bidder when attention is scarce.

Finally, I ask how much the seller would need to know to do without nomination. A seller that observed which tracts are most promising on observable characteristics, and offers tracts arbitrarily only from the top of that ranking, would match nomination's revenue only by restricting itself to roughly the top 8–10% of the $\approx$ 28,000 available tracts. Even then it would attract fewer bidders (2.7 versus 3.7 per tract) and sell fewer tracts (92% versus 100%), winning only by selling a few high-value tracts at high prices. Nomination delivers, without any information on the seller's side, what the seller could otherwise achieve only by knowing which tracts are the most promising.

1.1 Contributions

This paper makes several contributions. First, I provide the first structural analysis of pre-auction nomination, quantifying how it creates value through a guaranteed bidder, the selection of more valuable tracts, an information spill-over to other bidders, and the concentration of bidders' limited attention, and how much a seller would need to know to replicate it without nomination. The closest formal model is Gorbenko and Malenko [2024] that studies the initiation of takeover auctions. In their model, two potential buyers who already know their signals can send the seller a costless, non-binding indication of interest, after which the seller decides whether to hold an auction. An indication reveals that the bidder is optimistic. When values have a large common component, the rival revalues the asset upward and bids

more aggressively, so bidders prefer not to initiate and sales are seller-initiated. When values are mostly private, the rival barely revalues the asset, so revealing interest costs the initiator little. A bidder with a high enough signal then approaches the seller to trigger the sale now rather than wait, since waiting risks delay or letting the rival initiate first. Nomination differs in three ways. First, it is costly and binding: the nominator pays a fee and commits to buy at the reserve, and the nominated tract is offered automatically. Second, potential bidders arrive over time and must pay to learn their values, so a nomination affects whether rivals learn and enter, not only how they bid. Third, values are correlated private values, so what a rival infers from a nomination is about her own value rather than a shared common value. In an extension, Gorbenko and Malenko [2024] let an uninformed rival pay to learn her signal and show that an initiating bid can preempt that rival. In my setting a nomination works in the opposite direction, encouraging rivals to learn (the information spill-over), although the option to wait for someone else's nomination works against it (under-provision). But like a preemptive bid, the nominator then enters as a committed, guaranteed competitor and competes away the rents of the rivals it draws in.

A nomination thickens the market by putting a tract onto bidders' short lists. Klemperer and Bulow [1996] show that an auction with one more bidder and no reserve raises more revenue than the optimal mechanism with one fewer bidder. Coey et al. [2019] turn this result into an empirical bound, the bidder exclusion effect. My decomposition follows the same ranking: almost all of what nomination is worth to the seller comes from the guaranteed bidder—one more committed bidder on every offered tract—rather than from the selection of more valuable tracts.

The screening role of nomination also connects it to work on non-binding signals of interest sent before a sale. In auction design, non-binding indicative bidding serves as a device for screening bidders under costly entry [Ye, 2007, Quint and Hendricks, 2018], with experimental support in Kagel et al. [2008]: a preliminary interest signal helps sellers select higher-value bidders for a costly second stage, though Ye [2007] shows fully separating equilibria generally fail to exist when the signal is non-binding. M&A sale processes have an analogous non-binding preliminary screening phase [Hansen, 2001, Boone and Mulherin, 2007], in which a cheap early signal feeds the later, binding sale stage.

Nomination differs from these signals along two dimensions. First, the signal is costly: the $10 fee is small but nonrefundable, and a bidder must learn her value before nominating. Second, and more distinctively, nomination carries a commitment: the nominator must complete the purchase at the minimum price if no one else bids. This entitles the nominator to win at the reserve when there are no other competing bidders, a payoff that non-nominating entrants, who enter only after someone else has triggered the auction, can never obtain.

The structural estimates (Section 6) show that the binding constraint on the overall nomination rate is the scarcity of early, pre-nomination arrivals rather than the height of any single cost. The learning cost, by contrast, does screen participation on the lower-value tracts, where the early learning share falls from 100% in the high-value bins to 3.5% in the lowest-value bin. Nomination therefore remains a costly signal in the sense of Spence [1978] for low-value tracts: the fee and learning cost deter nomination, and the purchase commitment makes it unprofitable for a bidder who has learned a low value.

This places nomination closer to the costly-entry tradition of Levin and Smith [1994] and Bulow and Klemperer [2009] than to either non-binding indicative bidding or a competitively bid entry-rights auction: thin early arrival limits how many tracts are nominated, while the learning cost shapes which tracts are nominated. Nomination is also a form of sequential participation. Roberts and Sweeting [2013] show that when entry is costly and selective, letting later buyers condition their entry on earlier entrants' decisions can raise both revenue and efficiency. Under nomination, later bidders condition on the fact that someone has nominated.

My decomposition of the value of nomination by party (Section 7, Table 3) shows who gains and who pays when the early signal is binding: the seller and the nominator gain, while the rivals that the nomination draws in lose, because they must face a guaranteed strong competitor. This resonates with the entry-inducement logic of Bulow and Klemperer [2009]: most of what nomination is worth to the seller comes from securing the presence of a committed bidder.

I also extend affiliated private value (APV) empirical methodologies, developed for settings where all bids are observed [Li et al., 2002, Athey and Haile, 2002], to a setting with both a censored highest bid and positively dependent values. When the full bid profile is not observed, values are not nonparametrically identified under standard affiliation assumptions [e.g., Athey and Haile, 2002]; some parametric or semiparametric structure is needed to estimate the dependence parameter and the marginal value distribution from the observed lower order statistics alone. Under independence, Espín-Sánchez et al. [2025] show that the number of bidders and the marginal value distribution are nonparametrically identified from any two order statistics, so the highest bid need not be observed. Their framework also identifies dependence, but only when the copula is max-stable (e.g. CES or Gumbel-type extreme-value copula), and only from the two highest order statistics. My setting departs on both margins. The positive dependence estimated from the natural resource auction data need not be max-stable, and the highest order statistic is censored. No general nonparametric identification result covers dependence and a censored highest bid at the same time. I address this gap by pairing a parametric copula for the dependence structure with a flexible

approximation to the marginal.

The findings also connect to the literature on information design in auctions. Bergemann and Pesendorfer [2007], Board [2009], and Ganuza and Penalva [2010] study how much information the *seller* should disclose prior to an auction to maximize revenue. My setting inverts this question: it is *buyers* who choose what information to reveal through nomination, and the seller's instruments (the nomination threshold, the fee, and the purchase commitment) determine how costly demand revelation becomes. The cost of generating this information is borne by buyers so a nomination is a public good among bidders: each would rather another firm pay to learn and nominate and then learn later, an incentive that does not arise in standard seller-disclosure frameworks.

A motive to delay learning in order to observe rivals also appears at the exploration stage of this industry. Hodgson [2024] estimates a dynamic model of UK offshore drilling in which firms delay costly exploratory drilling to observe rivals' outcomes. There, the information spilling over is a signal about a commonly-valued unknown (the location of oil), so firms learn from realized well results of rivals. In my setting, what a bidder infers from a rival's nomination is instead something about their own private value, transmitted through the positive dependence of values rather than through any revision of a common object.

The concentrated-attention channel also connects nomination to work on how limited attention is spread across many alternatives.[6] In the sparse-max model of Gabaix [2014], a decision maker attends only to the parts of a problem that matter enough relative to the cost of attention and ignores the rest entirely. Read as a model of which tracts a firm arrives at, it predicts that most tracts receive no attention at all. When the choice set is large, decision makers seriously consider only a small subset of it, and an option's chance of entering that subset depends on whether something brings it to their notice, such as advertising, not only on its value [Goeree, 2008, Abaluck and Adams-Prassl, 2021, Crawford et al., 2021]. A nomination plays that role here: it places the tract in the sale notice that every potential bidder reads.

Financial markets offer a close parallel: the immediate response to a firm's earnings announcement is weaker on days when more other firms announce [Hirshleifer et al., 2009]. In my setting, a potential bidder facing roughly 28,000 available tracts faces the same problem. In rational-inattention models of consideration sets, alternatives with higher prior expected value are more likely to be considered [Caplin et al., 2019], and more generally attention rises with the stakes involved [Gabaix, 2019]. Consistent with this, the share of available tracts that are nominated falls by about two orders of magnitude from the most to the least valuable tracts (Appendix C.1).

[6] For surveys, see Gabaix [2019] and Loewenstein and Wojtowicz [2025].

Coey et al. [2020] estimate that eBay buyers encounter roughly thirteen relevant auctions a month but attend to only about 6% of them, and that auctions become unviable when few buyers pay attention. Bodoh-Creed et al. [2021] show that on a decentralized platform with many simultaneous auctions, consolidating trade into fewer auctions removes much of the allocative inefficiency, though at some cost to seller revenue. Nomination achieves a similar consolidation through buyers rather than the platform: shrinking the offered set from roughly 28,000 available tracts to a few dozen per season raises the per-tract average arrival rate from ≈ 0.005 to ≈ 2.9.

2 Institutional Details

The ND state government holds quarterly auctions for oil and gas lease sales through the Department of Trust Lands. Each cycle, referred to as a *season* throughout this paper, begins almost immediately after the previous auction concludes. At this time, the state announces the date of the next auction and calls for nominations from interested parties along with the deadline for nominations. The list of tracts open for leasing is posted on its website. Tracts need at least one nomination to be considered for auction. The size of the tract can vary, ranging from 0.01 acres to a maximum of 160 acres as specified by state regulations. The nomination process requires a nonrefundable $10 fee per tract. Nominators can determine the minimum opening bid. If not specified, the opening bid is $2 per acre.

Following the nomination period, the state conducts a sequence of ascending price auctions to individually sell nominated tracts. Each tract opens for bidding individually, with auctions closing after a 7-day period, sequentially at two-minute intervals. Bids are received at per-acre level. If no one bids, the nominator becomes the automatic winner and has to pay the minimum opening bid they specified.

Upon winning a lease auction, the successful bidder must pay several upfront costs. For example, a $2 per-acre bid on 40 net acres incurs the following costs: an one time bonus payment determined by the bid $2 and size of 40 acres($80), an one time ND administrative fee ($10), a five-year prepaid rental fee ($1 per net acre annually, totaling $200), and a 2% buyer premium on the bonus payment ($1.60) combine for a total initial cost of $291.60. Once a well is drilled and begins producing, the leaseholder pays a royalty rate, which is the percentage of oil or gas revenue paid to the mineral rights owner, of either 3/16ths or 1/6th, depending on lease terms.[7]

The winning bidders obtain exclusive rights to explore, drill, and produce oil and gas

[7]Royalty rate differs by county, and most counties are one/sixth (1/6) except for counties like Billings, Divide, Dunn, Golden Valley, McKenzie, Mountrail and Williams.

from their tracts for a maximum lease period of five years. Extensions beyond this period are only permitted if the tract is being actively developed and producing.

Bidders have access to what is called *proxy bidding*, an online system that bids on behalf of the user until the highest competing bid exceeds the value the user has set as their *proxy bid*. In other words, as long as the competing bidders are bidding lower than the proxy bid value, the proxy bidder will keep its winning status with a bid incrementally higher ($1) than the highest competing bid. If the competing bid is equal to the proxy bid, say at $10, but was placed after the proxy bid was placed, the proxy bidder will show up as the winner at $10 and the bidder who bid $10 but placed after the proxy bidder will lose with their bid recorded as $9.

Under a proxy bidding system and the private value assumption, the optimal bidding strategy of the bidders is to place a proxy bid at their private value. For this reason, most winning bids are a dollar higher than the second highest bid. In this sense, the winning bid is *censored*. Sometimes bidders will not use this proxy bid system. When this happens, a distinct *jump* is observed in the data. For example, instead of placing a proxy bid of $10 after the previous bidder placed a proxy bid of $5, the next highest recorded bid will be $10 instead of a bid incrementally higher than $5, which in this case is $6. Such instances are rare and informative in that they reveal the winner's true bid directly. However, I do not explicitly use these jump-bid auctions and they enter the estimation on equal footing with all other auctions. In addition, I abstract away from the fact that the losing bidder's bid may be lower than what is recorded due to this system.

While the broad geological formation underlying a given region is public knowledge, translating that shared information into a firm-specific dollar value for a particular tract is not costless. For instance, identifying whether a given tract sits within such a reservoir – a localized pocket within the broader formation where the rock is naturally fractured or under sufficiently high pressure to make extraction economical – and how much it would cost a specific firm to develop it requires firm-specific analysis. Phone interviews with market participants conducted for this project confirm that this evaluation step is a real and heterogeneous cost across firms: larger firms typically maintain in-house land men and geologists (or even their "bosses" with years of experience are capable of such evaluation on the spot from "looking at the map") to evaluate prospective tracts, while smaller firms rely on third-party consultants whose availability, and hence cost, is uncertain from one tract to the next. This heterogeneity in the cost of acquiring firm-specific information motivates modeling this cost, which I refer to as the *learning cost*, as privately observed and drawn independently across bidders, rather than as a fixed or publicly known cost of participation.

3 Data

The main data for estimation are auction-level bid data and nomination information for each auction in 2017–2018. They combine (1) auction-level bid data for all bidders, scraped from the online platform that runs the auctions, and (2) nomination data. Ever since the Mineral Leasing Act was amended by Congress to allow for internet-based lease sales as part of the National Defense Authorization Act (NDAA) for Fiscal Year 2015, many state-government-run mineral rights sales auctions adopted internet-based auctions through the bidding platform EnergyNet. Past bidding history from open auctions (English auctions) conducted on this platform is available to the public. Additional information about each auction, such as royalty rate, size in acres of the tract, identity of nominator, and start and end times of the auction publicized during the auction, is available as well. Detailed information on nominations received per tract was acquired through a FOIA request. Bidder identity during auction is anonymized using bidder id numbers. The winning bidder's identity is revealed. Using the bidder id and winning bidders' names, bidders who have won at least once during the given season can be traced within the season. Bidders who never won stay anonymous. This means firm-level characteristics (size, drilling technology, or existing lease portfolio) are observed for only a small subset of bidder-tract pairs (winners within a season). Consequently, I do not model them explicitly as bidder-level heterogeneity.

Supplementary data are needed to understand which lands were available for nomination for each auction. This information is reconstructed using (1) the comprehensive list of tracts available for leasing and (2) the auction history for 2012–2016. Potential nominators can check current lease status and available tracts for nomination through the public records of the ND Department of Trust Lands. These records include detailed geographical information according to the World Geodetic System 1984 (WGS 84) coordinate system for tracts that are currently leased and tracts available for lease. They also include information on the most recent past lease associated with each tract and whether the lease was extended beyond its maximum five-year lease period.[8] Past auction history for both offline (before the NDAA was passed) and online auctions is acquired from public data of the ND Department of Trust Lands. Combining these two data sources, I reconstruct the list of tracts that would have been available for nomination by the time a potential nominator arrived to nominate. Figure 7 in Appendix A maps the geographic distribution of available tracts alongside the locations of nearby producing wells, illustrating the variation in well proximity across tracts that motivates including distance to the nearest well as a covariate in the bid-homogenization

[8] To the best of my knowledge, based on the most current lease data, most leases terminate after the five-year period unless a new well was drilled.

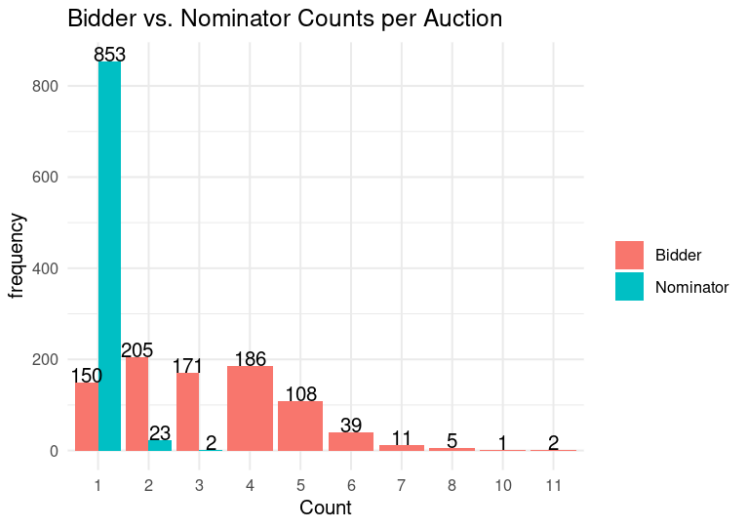


Figure 1: Histogram of per Auction Bidder vs. Nominator

Notes: Histogram comparing the number of bidders versus nominators per auction. The stark discrepancy shows extensive delayed-learning: while auctions receive 1–11 bidders (median of 3), the vast majority are triggered by a single nominator, with only a few receiving 2–3 nominations.

regression.

There are 878 auctions in total, covering 2,740 bids. Below, I report the summary statistics from the entire data unless specified otherwise. The number of auctions used for estimation is 728 covering 1,875 bids. I keep only the auctions where there is more than one competitor and auctions where the size is greater than 0.01 acres.

Figure 1 shows the number of bidders and nominators per tract, showing a stark discrepancy in the number of the two. While number of bidders per auction ranges from one to 11 with the median being three, those who nominated and triggered the auction is at most three with majority of auctions being triggered by a single nominator. This shows how much delayed-learning takes place, despite the monetary cost of triggering an auction being very low ($10 per nomination). Appendix A reports the full distribution of bidders per auction (Figure 8) and the number of unique bidders active in each season (Table 13).

Figure 2 documents this late-learning pattern at the individual bidder level. Because bidder identity is unobserved unless a bidder wins and their winning status is announced by the seller, the figure is restricted to bidders who won at least once during the auction season. Within a season each bidder carries a unique identifier, making their participation across tracts traceable conditional on at least one win. Each point is a nominator-auction pair for this identified sample: the horizontal axis counts all tracts the bidder participated in during that auction, and the vertical axis counts the tracts they actually nominated. Points on the 45-degree dashed line represent bidders who nominated every tract they participated

in; points below the line represent late-learning. The vast majority of observations in green circles lie along the horizontal axis with almost zero nominations, despite participating in dozens or even over a hundred tracts. By contrast, the small cluster of orange triangles near the 45-degree line represents the handful of nominators who actually trigger the auction. Even among nominators, some still defer learning until later on a substantial share of tracts they enter. With Figure 1, these two figures together establish that late-learning is pervasive.

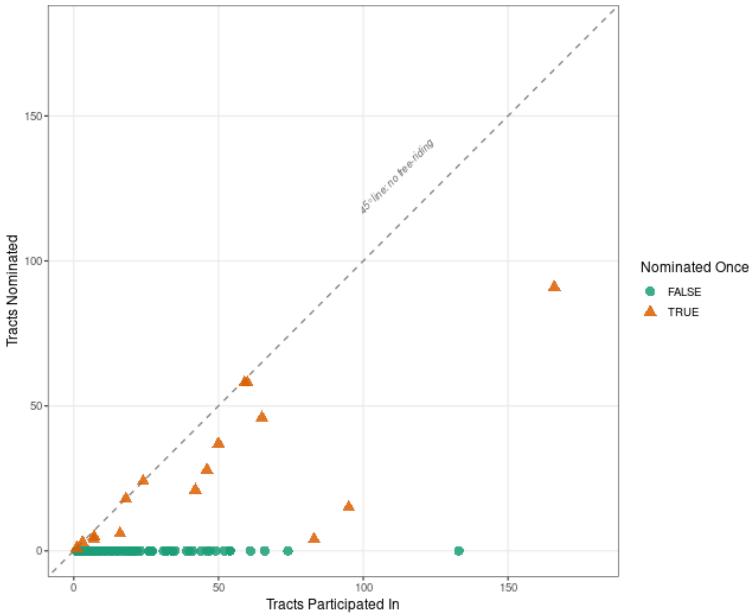


Figure 2: Tracts Participated vs. Tracts Nominated, by Nominator-Auction Pair

Notes: Each point is a bidder-auction pair restricted to bidders whose identities are known. The horizontal axis is the total number of tracts the bidder participated in (bid on); the vertical axis is the number of tracts they nominated.

Beyond the raw magnitude of late-learning documented in Figures 1 and 2, phone interviews with market participants suggest that non-nomination is a deliberate strategic choice. One non-nominating bidder reasoned that it is "unlikely there's anything valuable if no one else nominated it" and described nominating as "a waste of time" when they can instead "rely on other nominators" to trigger the auction on their behalf. This paper directly models this delayed learning mechanism: nomination requires paying the learning cost upfront and the $10 nomination fee, while late-learning lets a bidder defer that cost until the auction is confirmed to occur.

I do not model tract quality explicitly, as not all tracts have realized oil and gas production data available. Instead, I use bids as a revealed-preference measure of tract value: each bid reflects the bidder's perceived value of holding the exclusive right to explore, drill, and extract from that tract over the five-year lease period, incorporating their private assessment of

geological prospects, extraction costs, and future commodity prices. The bid homogenization procedure in Appendix B.1 controls for observable tract characteristics, including geographic location, isolating the private value component.

Table 1 reports the same number but broken down by season. The second row indicates the number of auctions that were triggered, meaning these are the number of tracts that received at least one nomination. The 'Tracts Available' column reports the number of tracts that would have been open for nomination, calculated using the tract list data and auction history data from five years before. The table also reports the number of parcels that received two and three nominations, obtained through a FOIA request. Total number of bids received for that season is reported in the last column. During 2017–2018, a total of 878 tracts were nominated and auctioned, about 110 per season, or roughly 0.4% of the approximately 28,000 tracts available in a given season.

Table 1: Summary of Available Tracts, Nomination Counts and Bids Received

Season	**# Auctions**	**Tracts Available**	**Nom. ×2**	**Nom. ×3**	**# Bids Received**
2017-Q1	173	27,867	21	2	361
2017-Q2	234	27,885	0	0	661
2017-Q3	166	27,885	0	0	504
2017-Q4	67	27,965	0	0	270
2018-Q1	52	28,062	2	0	240
2018-Q2	58	28,064	0	0	191
2018-Q3	29	28,408	0	0	148
2018-Q4	99	28,461	0	0	365
Total	**878**		**23**	**2**	**2,740**

Notes: Table summarizes the number of auctions and nominations by season during 2017–2018. The 'Tracts Available' column shows the total number of tracts that could have been nominated. The 'Nom.×2' and 'Nom.×3' columns indicate the number of tracts that received multiple nominations.

Summary statistics on bid values, parcel sizes, royalty rates, nearest-well characteristics, and starting bids are reported in Appendix A (Tables 5–12).

4 Model

I model nomination and auction as two separate stages. Potential bidders, or "buyers", arrive sequentially and must incur a cost to learn their private values. They face strategic decisions about when to learn their values and nominate. The auction only proceeds if sufficient nominations are received, creating coordination and late-learning incentives.

4.1 Game Setup

I model the sale of a single tract t to a population of potential bidders who arrive over time; the total number that ever arrive, I_t, is random and determined endogenously by the Poisson arrival processes described below. Bidders are symmetric, and conditional on N bidders arriving, their private values $\mathrm{V}_t = (V_{1t}, \ldots, V_{I_t t})$ are exchangeable and jointly distributed according to H_V on a common support $[0, \infty)^N$.[9] The marginal CDF $F_V(v_{it})$ is then defined as $H_V(\infty, \ldots, v_i, \ldots, \infty)$.

I allow for positive dependence across values so that for $V_{it}, V_{jt}\,(i \neq j)$, $F_V(v_{jt} \mid v'_{it}) \leq F_V(v_{jt} \mid v_{it})$ for all $v'_{it} > v_{it}$. This captures the reality that firms observing similar information (production from nearby wells or market conditions) form positively correlated value estimates. [10]

This modeling choice is consistent with a broader shift in the recent empirical auction literature away from common-value frameworks for onshore oil and gas lease sales [Covert and Sweeney, 2023].[11] Improvements in horizontal drilling and hydraulic fracturing have made oil extraction far less dependent on locating a single, uncertain geologic reservoir. Instead, modern techniques target a broad source rock rather than a narrow, hard-to-locate point deposit, reducing the production uncertainty that historically justified a common-value treatment of these auctions [Kong, 2020]. Consequently, differences in bid values are driven primarily by bidder-specific factors (development strategy, drilling technology, portfolio fit, and cost of capital) rather than by private information about a shared underlying quantity, which is precisely the private-values structure this paper models. At the same time, allowing for zero correlation across bidders would be too strong of an assumption in this setting: bidders drawing on the same regional geological and market information are still likely to reach correlated, if not identical, conclusions about a tract's value.[12] The correlated private values structure takes a middle path appropriate to this setting where bidder values remain genuinely private and require costly exploration to learn, but are permitted to be positively correlated through a common factor reflecting shared uncertainty about the region's geology and prevailing oil and gas prices.

No bidder observes any realization of V_t ex ante, including her own, and must incur a

[9] H_V denotes this consistent family of exchangeable distributions, indexed implicitly by the realized number of arrivals.

[10] This is the affiliated private values (APV) property of Milgrom and Weber [1982]; see Appendix D.1 for the formal definition and for verification that the copula adopted in Section 5 satisfies it.

[11] See, e.g., Hendricks and Porter [1988] for the classic common-value treatment of offshore auctions.

[12] Krasnokutskaya and Seim [2011] make a related point in the context of highway procurement auctions, arguing that correlation in bids can arise either from heterogeneity that is observable to bidders but not to the econometrician, or from genuine correlation of private costs, and that distinguishing between the two is generally not possible without strong auxiliary assumptions.

learning cost ($c_{it} \sim F_C$, i.i.d.) to learn her value, i.e. paying c_{it} reveals v_{it} to buyer i alone. This learning cost is privately observed and independent of V_t.

There are three policy instruments exogenously given and common across tracts: the nomination threshold $K \in \{0\} \cup \mathbb{N}$, the per-nomination fee $\phi \geq 0$, and the reserve price $r \geq 0$. These are public.

I describe the model below for a single tract, suppressing tract-level heterogeneity; Section 4.5 extends the model to heterogeneous tracts and introduces the observable tract characteristics X_t.

The sale proceeds in two stages: the nomination (Stage 1) and the auction (Stage 2). In Stage 1, the seller receives nominations for t. If total received nominations reach the nomination threshold ($M_t \geq K$), t is sold in second-price auction in Stage 2. The realized nomination count M_t is not revealed. Bidders arrive according to two independent, homogeneous Poisson processes, one for each stage. I use λ_{pre} and λ_{post} to represent the arrival rates of Stage 1 and Stage 2, respectively.[13]

An important timing distinction clarifies how positive dependence in H_V operates. A bidder who has not yet paid c_{it} to learn can update her beliefs about her own yet-to-be-learned value upon observing nomination signals from others. Let $n(v) \in \{0, 1\}$ denote a bidder's (as yet unspecified) nomination rule, mapping a realized value v to a nomination decision, and let $\mathcal{E}_m \equiv \{n(V_1) = 1, \ldots, n(V_m) = 1\}$ be the event that exactly m other bidders have nominated. By Bayes' rule and the joint distribution H_V, being exchangeable and symmetric across bidders, the posterior density of bidder i's own value conditional on $\mathcal{E}_m$ is

$$f_{V_i|\mathcal{E}_m}(v) \;=\; \frac{\displaystyle\int_{\mathbb{R}_+^m} h_{m+1}(v,\, u_1, \ldots, u_m) \prod_{j=1}^{m} \mathbb{I}\{n(u_j) = 1\}\, du_1 \cdots du_m}{\Pr(\mathcal{E}_m)}, \tag{1}$$

where h_{m+1} is the $(m+1)$-dimensional marginal density of H_V and $\Pr(\mathcal{E}_m) = \mathbb{E}\left[\prod_{j=1}^{m} \mathbb{I}\{n(V_j) = 1\}\right]$ is the normalizing constant. Section 4.4 shows that the equilibrium nomination rule takes the threshold form $n(v) = \mathbb{I}\{v \geq v^*\}$, in which case $\mathcal{E}_m$ reduces to $\{V_1 \geq v^*, \ldots, V_m \geq v^*\}$ and the integral above is taken over $(v^*, \infty)^m$; this is the form used throughout the rest of the paper. Under positive dependence of H_V and any weakly increasing nomination rule $n(\cdot)$, the posterior stochastically dominates the prior:

$$F_{V_i|\mathcal{E}_m}(v) \;\leq\; F_V(v) \qquad \text{for all } v, \tag{2}$$

[13] I do not observe how long each stage lasts. I therefore normalize each stage to one unit of time, so the two arrival rates represent the expected number of arrivals in each stage.

because knowing that others nominated raises the probability that V_i is also high.[14][15]

4.2 Strategy Set

I assume the following:

Assumption A1. *Learning is required before nominating or bidding.*

Under Assumption A1, I restrict the strategy set of buyers accordingly: l_i, or learning timing, for bidders who arrived in Stage 1; n_i, or nomination decision for bidders who learned their values; d_i, or learning decision for bidders who arrived in Stage 2.

Potential bidder i arriving during Stage 1 chooses a learning timing $l_i \in \{1, 2, 3\}$. **Stage 1 (early) learner** ($l_i = 1$) pays learning cost c_{it} immediately and learns private value $v_{it} \sim F_V(\cdot)$. Nomination decision $n_i = n(v_{it}) \in \{0 \text{ (don't nominate)}, 1 \text{ (nominate)}\}$ is made based on v_{it}.

Stage 2 (late) learner ($l_i = 2$) waits without learning v_{it} and pays the exploration cost c_{it} to learn in Stage 2 only if the auction proceeds. **Exit** ($l_i = 3$) means leaving without learning their private value. Let $\pi_{l=1}$ and $\pi_{l=2}$ be the ex-ante probability of a randomly arriving bidder choosing $l = 1$ and $l = 2$ respectively.

If Stage 2 (the auction stage) is triggered ($M_t \geq K$), more potential bidders arrive according to Poisson(λ_{post}). Upon observing c_{it}, they decide whether to enter ($d_i \equiv d(c_{it}) \in \{1 \text{ (Learn)}, 0 \text{ (Exit)}\}$. $d_i = 1$ bidders observe their values. Let $\pi_{d=1}$ be the ex-ante probability of a randomly arriving bidder in Stage 2 choosing $d = 1$. Bidders who chose $l_i = 2$ in Stage 1 also learn their value by paying the learning cost c_{it}. All potential bidders who have learned their value ($l \in \{1, 2\}$ from Stage 1 and $d = 1$ from Stage 2) enter the auction and bid $b_i(v_i)$ if the value exceeds reserve price ($v_{it} \geq r$) and exit otherwise (Figure 3).

4.3 Payoffs

4.3.1 Stage 2: Expected Auction Surplus

A bidder with realized value v who enters the auction bids $b(v) = v$ (weakly dominant strategy under second-price rules) and wins if and only if her value exceeds both the reserve r and the highest competing value. Given the joint value distribution H_V, the expected

[14] This follows directly from positive correlation of H_V and holds for any copula satisfying that property, not only the Gaussian copula adopted in estimation; see Appendix D.1.

[15] A simplified version of this game—stripping out value uncertainty and costly learning—is solved in Appendix E.1 to build intuition for how the nomination threshold K shapes entry and the trigger rate.

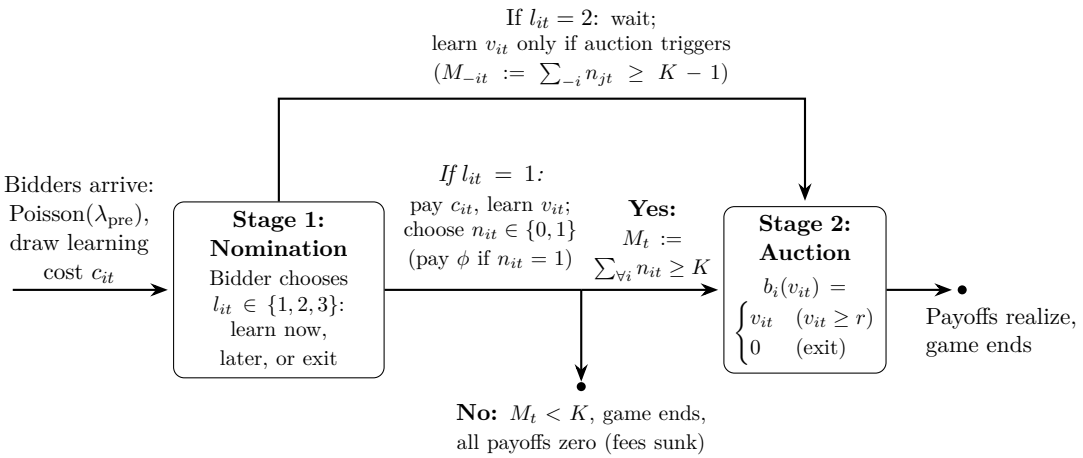


Figure 3: Timing of the model.

surplus given j total competitors is

$$\tilde{\Pi}(v;\, j,\, H_V) \;=\; \int_r^v (v - y)\, f_{Y|v}(y \mid v,\, j,\, H_V)\, dy, \tag{3}$$

where $Y \equiv \max\{V_1, \ldots, V_j\}$ is the highest competing value and $f_{Y|v}$ is its conditional density given own value v, accounting for positive dependence through H_V (derived in Appendix B.4). With $j = 0$, $\tilde{\Pi}(v; 0, H_V) = v - r$ when $v \geq r$ and zero otherwise.

When a bidder faces m *deterministic* competitors (realized nominators) and a Poisson(Λ) number of *stochastic* competitors (non-nominating early arrivals and late arrivals), the expected surplus integrates over the Poisson distribution:

$$\Pi(v;\, m,\, \Lambda,\, H_V) \;=\; \sum_{n=0}^{\infty} \frac{e^{-\Lambda}\Lambda^n}{n!}\, \tilde{\Pi}(v;\, m + n,\, H_V). \tag{4}$$

In this setup,[16] the total expected stochastic competition is the sum of non-nominating bidders with values above the reserve coming from three different groups: (1) later-learners from Stage 1 ($l = 2$); (2) early learners from Stage 1 ($l = 1$) who did not nominate but participate in the auction; and (3) Stage 2 arriving learners ($d = 1$).

4.3.2 Stage 1: Learning and Nomination

Bidders arriving at Stage 1 choose when to learn their values. Payoffs of the three strategies ($l_i \in \{1, 2, 3\}$) depend on three factors: 1. timing when learning cost c_i is paid; 2. difference in number of nominations required to satisfy $M \geq K$; and 3. difference in beliefs about yet-

[16] Appendix D.10 describes how the Poisson sum in equation (4) is truncated in computation.

to-be-learned value under positive dependency. In addition, because I make the restricting assumption that nomination decision can be made only after learning in Stage 1, only those who chose $l_i = 1$ can decide whether to nominate ($n_i = 1$) upon paying nomination fee ϕ.

$l_i = 1$ **(Learning Early) Payoff.** For a bidder with learning cost c_i, the value of learning immediately is

$$\begin{aligned} U_{l_i=1}(c_i) \;=\; -c_i \;+\; \int_{\underline{v}}^{\bar{v}} \mathbb{I}\{n(v)=1\}\cdot \left[-\phi \;+\; \sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\,\Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\right] f_V(v)\,dv \\ +\; \int_{\underline{v}}^{\bar{v}} \mathbb{I}\{n(v)=0\}\cdot \sum_{m=K}^{\infty} \Pr(M_{-i}=m)\,\Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v)\,dv, \end{aligned} \tag{5}$$

where $-c_i$ is the learning cost paid immediately upon choosing $l_i = 1$. Because learning is required for entry and, conditional on having learned, a bidder enters and bids whenever $v \geq r$, an early learner's continuation value has two branches. The first (bracketed) term applies when she nominates ($n(v) = 1$, i.e. $v \geq v^*$):

$$-\phi \;+\; \sum_{m=K-1}^{\infty} \Pr(M_{-i}=m)\,\Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V) \tag{6}$$

is the payoff from nominating given the realized value v: ϕ is the nomination fee paid when $n(v) = 1$, and $\Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)$ is the expected auction surplus for each realized M_{-i}, the number of nominations from other bidders. The sum starts at $m = K-1$ so that the bidder's own nomination brings total nominations to $m + 1 \geq K$, triggering the auction.

The second term applies when the bidder has learned but does *not* nominate ($n(v) = 0$): they pay no nomination fee, and because they did not trigger the auction, auction surplus is collected only when others' nominations reach the threshold ($m \geq K$).

$l_i = 2$ **(Learning Later) Payoff.** A buyer with cost c_i who chooses $l_i = 2$ pays nothing in Stage 1. If the auction triggers ($M \geq K$), she updates her beliefs about her own value using the posterior implied by H_V upon observing the auction was triggered ($\mathbb{I}\{M \geq K\}$, but not M itself), pays c_i to learn v_i, then enters if $v_i \geq r$. The expected payoff is

$$U_{l_i=2}(c_i) \;=\; \sum_{m=K}^{\infty} \Pr(M=m)\left[-c_i \;+\; \int_{r}^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\,dv\right], \tag{7}$$

where M counts only *other* bidders' nominations. The competition argument is Λ_{nn} because conditioning on your own arrival does not change the distribution of the *other* arrivals.[17] $f_V(v \mid \mathcal{E}_m)$ is the posterior over her own value after observing $\mathcal{E}_m$, the event that m other bidders nominated (equation (1)), derived from H_V. By the stochastic dominance result in equation (2), this posterior shifts the non-nominating bidder's value distribution weakly upward relative to the unconditional $f_V(v)$. Section 4.4 shows that the equilibrium nomination rule takes the threshold form $n(v) = \mathbb{I}\{v \geq v^*\}$. Under this rule, $\mathcal{E}_m$ specializes to $\{V_1 \geq v^*, \ldots, V_m \geq v^*\}$, the form used in the rest of the paper.

$l_i = 3$ **(Exit) Payoff.** Bidders who choose to exit never return to the game and earn zero payoff.

$d_i = 1$ **(Late Arrivals' Learning) Payoff.** A Stage-2 post-arrival who observes the auction triggered ($M \geq K$) faces a similar decision as the $l_i = 2$ strategy: pay learning cost c, learn v, and bid if $v \geq r$. The two problems differ only in *when* the learning cost is sunk.

Remark 1 (Post-arrivals share the later-learner cutoff: $c^*_{\text{post}} = c^*_H$). Conditional on exactly m nominations, a post-arrival and an $l = 2$ later learner update their value beliefs to the same posterior $f_V(v \mid \mathcal{E}_m)$ (from H_V) and face the same Poisson pool of other non-nominator rivals. They differ only in the timing of paying the learning cost: a post-arrival exists only in a triggered auction and therefore pays c_i *for sure*, whereas an $l = 2$ bidder pays c_i only if the auction triggers. The post-arrival's expected payoff is thus

$$U_{\text{post}}(c_i) \;=\; -c_i \;+\; \sum_{m=K}^{\infty} \Pr(M{=}m \mid M \geq K) \int_r^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv. \tag{8}$$

Multiplying through by $\Pr(M \geq K)$ recovers the later-learner value, $U_{l=2}(c_i) = \Pr(M \geq K)\, U_{\text{post}}(c_i)$: the two are proportional and therefore share the same zero. Since c^*_H solves $U_{l=2}(c^*_H) = 0$, the post-arrival's cutoff coincides with it, $c^*_{\text{post}} = c^*_H$.

4.4 Equilibrium

I characterize a symmetric Perfect Bayesian Equilibrium (PBE) in which all arriving Stage 1 bidders follow the same cost-cutoff strategy $l_i = l(c_{it}) \in \{1, 2, 3\}$.[18] Bidders form ratio-

[17] This follows from the environmental equivalence property of Poisson games. See Myerson [1998].

[18] With incomplete information over private costs and values, the appropriate solution concept is a symmetric perfect Bayesian equilibrium (PBE). Because Stage 2 has a unique Nash equilibrium (truthful bidding is weakly dominant), backward induction uniquely pins down Stage 2 behavior, and Stage 1 strategies are best responses given correct beliefs. The resulting equilibrium is therefore PBE.

nal expectations about the distribution of competitors, consistent with equilibrium play and derived from the Poisson arrival process and the equilibrium probabilities $(\pi_{l=1}, \pi_{l=2})$. Equilibrium requires these to be a fixed point: given $(\pi_{l=1}, \pi_{l=2})$, optimal behavior for each arriving bidder regenerates the same pair.

I solve the model by backward induction. In **Stage 2**, truthful bidding is the weakly dominant strategy and constitutes the unique Nash equilibrium of the Stage 2 subgame for bidders with bids exceeding reserve price ($b(v_{it}) = v_{it} \cdot \mathbb{I}\{v_{it} \geq r\}$). I characterize expected auction surplus under positively dependent private values (Section 4.3.1).

For **Stage 1**, I employ a threshold strategy. First, the nomination threshold v^* follows from the indifference condition of a marginal bidder who has learned v: v^{**} sets (6) equal to zero, and $v^* = \max\{r, v^{**}\}$. Given v^* and expected competition, the three payoffs $U(c; l = k)$, $\forall k \in \{1, 2, 3\}$ yield two cost cutoffs: bidders with $c < c_L^*$ choose $l = 1$ and those with $c_L^* \leq c \leq c_H^*$ choose $l = 2$. Late arrivals' threshold strategy is defined similarly, where those with $c < c_{\text{post}}^*$ choose to learn $(d = 1)$.

Since payoffs discussed in Section 4.3.2 are increasing linearly in their respective arguments v and c (higher valuation increases 6, or $n = 1$ payoff; lower cost increases $l = 1$ and $l = 2$ payoffs at different rates), the optimal strategies are weakly increasing in those arguments. Therefore, for each realized (and learned) v and c, cutoff strategies are determined using indifference condition.

Nomination Threshold v^*. The nomination threshold is

$$v^* = \max\{r,\, v^{**}\}, \tag{9}$$

where v^{**} satisfies

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \Pi(v^{**};\, m,\, \Lambda_{\text{nn}},\, H_V) = \phi. \tag{10}$$

Intuitively, v^{**} is the value at which a bidder is exactly indifferent between nominating and not nominating given expected competition and the per-nomination fee ϕ. When $v < v^{**}$, the fee exceeds the expected surplus from participation. The reserve r imposes an institutional floor: any bidder with $v < r$ cannot profitably participate even unopposed.

This specializes the general nomination-event definition in equation (1) to the threshold form $\mathcal{E}_m = \{V_1 \geq v^*, \ldots, V_m \geq v^*\}$, with $n(v) = \mathbb{I}\{v \geq v^*\}$. This form is used in the payoffs for $l_i = 2$ and $d_i = 1$ calculations.[19]

The rule that a lone nominator who faces no competing bidders is obligated to complete

[19] The pure-strategy threshold rule is robust to noisy/mixed nomination strategies. See Appendix D.2 for the formal argument.

the purchase at r highlights both the benefit and cost of nominating that later-learners never receive when $K = 1$. Nominators face a higher ex-post surplus since they face a higher probability of triggering the auction, yielding a nonnegative surplus ($v - r \geq 0$) if they are the sole bidder. However, they face a lower ex-ante surplus since $v < v^*$ means they won't nominate but their learning cost c is sunk. Late-learners, by contrast, enter only conditional on $M \geq K \geq 1$, so they can never be the sole bidder at auction.[20] Nomination is discouraged only by the learning cost c_i and the fee ϕ, both of which enter the payoff unconditionally; the fallback purchase rule does not.

Plugging v^* into (5), since $\mathbb{I}\{n(v) = 1\} = 1$ when $v \geq v^*$ and zero otherwise, (5) can be rewritten as $-c_i + \int_{v^*}^{\bar{v}} U_{n_i=1}(v) f_V(v)\, dv$ and nomination fee is paid with probability $1 - F_V(v^*)$:

$$U_{l_i=1}(c_i) \;=\; -c_i - \phi \cdot \big[1 - F_V(v^*)\big] \;+\; \int_{v^*}^{\bar{v}} \sum_{m=K-1}^{\infty} \big[\Pr(M_{-i}{=}m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\big] f_V(v)\, dv \tag{11}$$

$l = 1$, $l = 2$ Thresholds c_L^* and c_H^*. Equilibrium strategy should characterize intervals of cost type c such that one strategy dominates the other. I begin by studying an important property of the two payoff functions $U_{l=1}(c_i)$ and $U_{l=2}(c_i)$ that determines equilibrium structure.

Proposition 1 (Single Crossing)**.** *The payoff functions cross at most once and there exists at most one c_L^* such that $U_{l=1}(c_L^*) = U_{l=2}(c_L^*)$.*

Proof. See Appendix D.7. □

Let $A = \int_{v^*}^{\bar{v}} \sum_{m=K-1}^{\infty} \Pr(M_{-i}{=}m)\, \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V) f_V(v)\, dv$ from (11) be the expected gross auction surplus for a $l = 1$ player and
$B = \sum_{m=K}^{\infty} \Big[\Pr(M{=}m) \int_r^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv\Big]$ from (7) be the expected gross auction surplus for $l = 2$ player.

Since $\Pr(M \geq K) < 1$,

$$|dU_{l=1}/dc_i| = 1 > \Pr(M \geq K) = |dU_{l=2}/dc_i| \tag{12}$$

Two lines with different slopes can intersect at most once. If they intersect at c_L^*, then

[20] The $j = 0$ term inside $\Pi(v; K-1, \Lambda_{\text{nn}}, H_V)$ captures exactly this: a strictly positive contribution to nominators' expected surplus that is structurally unavailable to non-nominators.

$$\begin{cases} U_{l=1}(c_i) > U_{l=2}(c_i) & (l=1) \text{ if } c_i < c_L^* \\ U_{l=1}(c_i) < U_{l=2}(c_i) & (l=2) \text{ if } c_i > c_L^* \end{cases}$$

Depending on the intercepts and slopes, exactly one of the three equilibrium cases holds.

Proposition 2 (Equilibrium Structure)**.** *Define*

$$\Delta_0 := U_{l=1}(0) - U_{l=2}(0) = A - \phi\big[1 - F_V(v^*)\big] - B \quad \textit{(payoff difference at zero cost)} \tag{13}$$

$$\Delta_{l=2} := U_{l=1}(c_H^*) - U_{l=2}(c_H^*) = U_{l=1}(c_H^*) \quad \textit{(payoff from choosing } l=1 \textit{ at } c_H^*\textit{)} \tag{14}$$

Then exactly one of the following holds. The empirically relevant case is Case 1 (interior solution), described below. The two corner solutions in Case 2 (coordination failure: everyone waits and learns later) and Case 3 (everyone learns early) are presented in Appendix D.4.

Case 1: Interior Solution (Single Crossing) *If* $\Delta_0 \geq 0$ *and* $\Delta_{l=2} < 0$*, then there exists a unique* $c_L^* \in (0, c_H^*)$ *where* $U_{l=1}(c_L^*) = U_{l=2}(c_L^*)$ *such that*

$$\begin{cases} c_i < c_L^* : l = 1 \\ c_L^* \leq c_i \leq c_H^* : l = 2 \\ c_i > c_H^* : l = 3 \end{cases}$$

Figure 4 illustrates all three equilibrium cases. Case 1 (left panel, interior solution) is the empirically relevant scenario: low-cost bidders ($c_i < c_L^*$) learn early ($l = 1$), intermediate-cost bidders ($c_L^* \leq c_i \leq c_H^*$) learn later ($l = 2$), and high-cost bidders exit ($l = 3$). This matches the ND data, where both nominators ($l = 1$) and non-nominators ($l = 2$) co-exist in equilibrium. Cases 2 and 3 are corner solutions, arising when the nomination fee is prohibitively high (coordination failure, no nominations) or negligible (all $l = 1$). Which case prevails depends on the auction parameters K and ϕ. See Appendix D.4.

In Case 1 (interior solution), the equilibrium cutoff c_L^* satisfies

$$U_{l=1}(c_L^*) = U_{l=2}(c_L^*) \tag{15}$$

Expanding and solving for c_L^*

$$c_L^* = \frac{A - B - \phi\big[1 - F_V(v^*)\big]}{\Pr(M < K)} \tag{16}$$

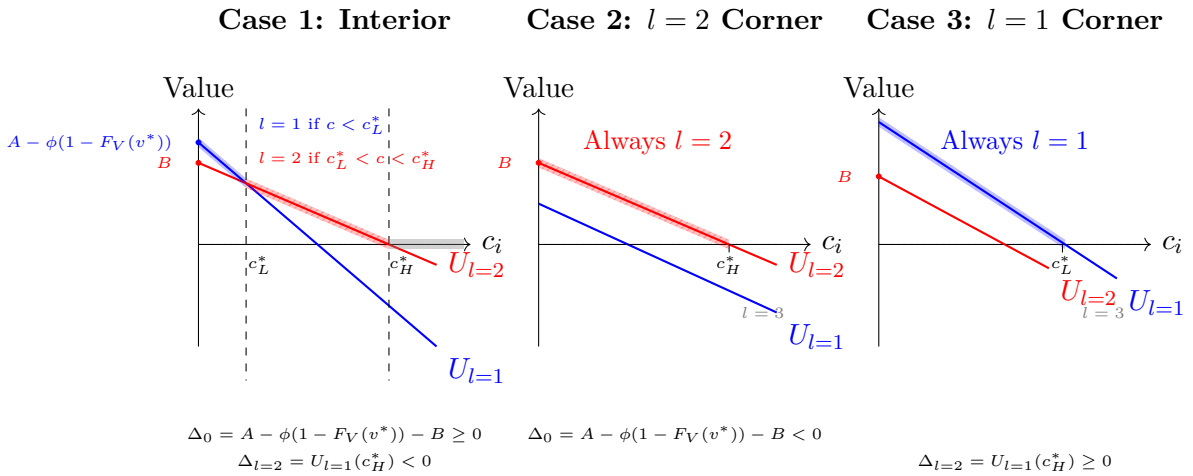


Figure 4: Equilibrium cases based on value function intercepts and slopes.

Notes: Three equilibrium cases based on the value of early learning ($U_{l=1}$) and learning later ($U_{l=2}$) as functions of learning cost c. The intersection points define cutoff costs c_L^* and c_H^*. The single crossing property ensures at most one intersection.

Then, the probability of a randomly arriving bidder to choose $l = 1$ is

$$\pi_0^{l=1} = \Pr[c < c_L^*] = F_C(c_L^*) \tag{17}$$

By independence of c and v, the unconditional nomination probability is the product of the two marginals:

$$\Pr[c < c_L^*, v > v^*] = \pi_0^{l=1} \cdot [1 - F_V(v^*)]$$

Similarly, the cutoff c_H^* satisfies $U(c_H^*; l = 2) = 0$, i.e. the learning cost at which the expected auction surplus of choosing $l = 2$ exactly covers the cost of exploring in Stage 2:

$$c_H^* = \frac{\sum_{m=K}^{\infty} \Pr(M = m) \int_r^{\bar{v}} \Pi(v;\, m,\, \Lambda_{\text{nn}},\, H_V)\, f_V(v \mid \mathcal{E}_m)\, dv}{\Pr(M \geq K)}. \tag{18}$$

Then probability of a random bidder choosing $l = 2$ is

$$\pi_0^{l=2} = \Pr[c_L^* < c < c_H^*] = F_C(c_H^*) - F_C(c_L^*) \tag{19}$$

4.4.1 Competition Structure

The probabilities $\pi_0^{l=1}$ and $\pi_0^{l=2}$ determine the distribution of competitors that any individual bidder expects to face.

Competition from Other Nominators Given nomination probability $\pi_0^{l=1}$, the number of other nominators M_{-i} follows from the Poisson arrival process.

Lemma 1 (Poisson Arrivals of Nominators)**.** *If bidders arrive according to Poisson(λ_{pre}) and each independently choose $l = 1$ and then $n = 1$, meaning the probability of such event happening is $\pi_0^{l=1} \cdot [1 - F_V(v^*)]$, then the number of competing nominators is Poisson with the* nominator rate Λ_{nom}*:*

$$M_{-i} \sim Poisson\left(\Lambda_{\text{nom}}\right), \qquad \Lambda_{\text{nom}} := \lambda_{pre} \cdot \pi_0^{l=1} \cdot [1 - F_V(v^*)] \tag{20}$$

Proof. See Appendix D.6. □

Competition from Non-nominators The expected competition from Stage 1 arrivals choosing $l = 2$ and Stage 2 arrivals choosing $d = 1$ is given by the following two components:

$$\Lambda^{l=2} = \lambda_{pre} \cdot \pi_0^{l=2} \cdot [1 - F_{\text{post}}(r)], \tag{21}$$

$$\Lambda^{d=1} = \lambda_{post} \cdot (\pi_0^{l=1} + \pi_0^{l=2}) \cdot [1 - F_{\text{post}}(r)], \tag{22}$$

$$\Lambda_{\text{nn}} = \Lambda^{l=2} + \Lambda^{d=1}, \tag{23}$$

where Λ_{nn} is the *non-nominator competition rate* (Stage-1 later learners plus Stage-2 post-nomination arrivals), and

$$F_{\text{post}}(r) \;=\; \sum_{m=K}^{\infty} F_{V_i|\mathcal{E}_m}(r)\, \Pr(M = m \mid M \geq K) \tag{24}$$

is the reserve-clearing CDF averaged over the posterior $F_{V_i|\mathcal{E}_m}$ of (2).

All non-nominators enter the auction when $v \geq r$. Because they act only after observing the auction triggered ($M \geq K$), correlation shifts their own-value belief upward, so their reserve-clearing probability is the posterior $[1 - F_{\text{post}}(r)]$ of (24) rather than the prior $[1 - F_V(r)]$. Post-nomination arrivals face the same cost cutoff c_H^* (Remark 1), so the fraction that finds exploration profitable is $F_C(c_H^*) = \pi_0^{l=1} + \pi_0^{l=2}$.

4.4.2 Fixed Point Equilibrium Characterization

Consistency Requirement The payoff functions $U_{l=1}(c)$ and $U_{l=2}(c)$ depend on the expected competition $(M_{-i}, \Lambda_{\text{nn}})$, which depends on the probabilities $(\pi_0^{l=1}, \pi_0^{l=2})$, which depend on the payoff functions through equations (16), (17) and (18), (19). An equilibrium requires these objects to be mutually consistent.

Definition 1 (Equilibrium Consistency)**.** *Probabilities* $(\pi_0^{l=1}, \pi_0^{l=2})$ *are* ***consistent*** *if*

1. *Given* $(\pi_0^{l=1}, \pi_0^{l=2})$*, expected competition is characterized by* $M_{-i} \sim Poisson(\Lambda_{\text{nom}})$ *and* Λ_{nn}*, both of which can be computed.*

2. *Given expected competition,* $U_{l=1}(c)$*,* $U_{l=2}(c)$ *and* $U_{post}(c)$ *can be rewritten via equations* (11)*,* (7) *and* (8)

3. *Given payoff functions, the optimal threshold strategy cutoff points defining* l, n, d *can be derived.*

4. *The optimal strategies generate probabilities* $(\pi_0^{l=1}, \pi_0^{l=2})$ *via equations* (17) *and* (19) *that equal the original probabilities*

Putting everything together, a symmetric equilibrium is defined as the following.

Definition 2 (Symmetric PBE)**.** *A symmetric equilibrium is a tuple* $(v^*,\, c_L^*,\, c_H^*,\, \pi_{l=1},\, \pi_{l=2})$ *such that*

1. ***Nomination threshold.*** $v^* = \max\{v^{**}, r\}$*, where* v^{**} *is the unique solution to*

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \tilde{\Pi}(v^{**};\, m,\, \Lambda_{\text{nn}},\, H_V) \;=\; \phi. \tag{25}$$

2. ***Cost cutoffs.*** c_L^* *solves* $U(c_L^*; l=1) = U(c_L^*; l=2)$*, and* c_H^* *solves* $U(c_H^*; l=2) = 0$*.*

3. ***Strategy probabilities.***

$$\pi_{l=1} = F_C(c_L^*), \qquad \pi_{l=2} = F_C(c_H^*) - F_C(c_L^*;\, \theta_c). \tag{26}$$

4. ***Competition rates (consistency).***

$$M_{-i} \sim Poisson(\lambda_{pre}\, \pi_{l=1}\, [1 - F_V(v^*)])\,, \tag{27}$$

$$\Lambda_{\text{nn}} = \lambda_{pre}\, \pi_{l=2}\, [1 - F_{\text{post}}(r)] \;+\; \lambda_{post}\, (\pi_{l=1} + \pi_{l=2})\, [1 - F_{\text{post}}(r)]. \tag{28}$$

The equilibrium is a fixed point: $\pi_{l=1}$ *and* $\pi_{l=2}$ *determine the competition rates via* (27)–(28)*, which determine the value functions* $U_{l=1}(c), U_{l=2}(c)$*, determining the cutoffs* c_L^*, c_H^* *and recovering* $\pi_{l=1}, \pi_{l=2}$ *via* (26)*.*

4.5 Expansion to Heterogeneous Game

Section 4 characterizes equilibrium behavior for a single, implicitly homogeneous tract. In the data, however, no two tracts are alike: tracts differ in size, royalty rate, proximity to producing wells, and the timing of nearby drilling activity ("spud date"). These differences are not incidental: they are exactly the variables ND's own tract listings advertise to every potential bidder before nomination even begins, and they visibly correlate with how much bidders are willing to pay (Table 14). Before turning to estimation, I extend the model to admit this heterogeneity, and in doing so motivate the multiplicative value decomposition used throughout the empirical analysis.

Let X_t denote the vector of tract characteristics observable to every bidder upon arrival (size, royalty rate, distance to the nearest existing well, and that well's spud date). One might think that, because X_t is public information, bidders already know their value for the tract and the costly-learning structure of Section 4.2 is unnecessary. However, what a given X_t *implies* for a specific firm's payoff depends on firm-specific drilling technology, available machinery, and the firm's existing lease portfolio. [21] Translating X_t into a usable valuation is precisely the costly step that Section 4.2 models via the learning cost c_i.[22] Observability of X_t and costly learning of V_i are therefore not in tension: X_t is the public starting point, and c_i is the cost of figuring out what that starting point is worth to a specific firm.

Because X_t is public and known to shift values in a common direction (a tract close to a recently productive well is known by every bidder to be more promising than a distant one, before anyone pays learning cost) the location of the value distribution differs systematically across tracts. Treating two tracts with very different X_t as though they generate values on the same scale is not innocuous: it would conflate genuine cross-bidder, private heterogeneity (the object of interest) with heterogeneity that is mechanically driven by observable, public differences across tracts.

This is compounded by an asymmetry in units. The policy instruments governing the auction – the nomination fee ϕ and the reserve price r – are fixed in dollar terms by the state, and a bidder's learning cost c_i (land-men's time, third-party consulting fees) is similarly denominated in dollars and does not automatically rise or fall with how lucrative a particular tract happens to be.[23] Values (V_t), by contrast, scale directly with X_t: a tract twice as

[21] Phone interviews with bidders indicate that X_t is necessary, but far from sufficient, to determine a firm's value for a tract: most areas "have been explored" in the sense that their basic geological and locational characteristics are known, but translating X_t into a firm-specific dollar value still requires private, costly evaluation.

[22] For instance, large firms generally maintain in-house land men and geologists for this purpose, while small firms rely on third parties whose availability is itself uncertain.

[23] A flat \$10 nomination fee or reserve prices starting from \$2 per-acre is a rounding error on a highly promising tract but can be a meaningful deterrent on a low-value one.

promising is (by definition) worth roughly twice as much to every bidder who values it. Because V scales with X_t while ϕ, r, and the distribution of c_i do not, the equilibrium derived in Section 4.4, which holds the units of V fixed relative to the units of ϕ, r, c_i, does not simply rescale tract by tract. The game itself takes a different shape as X_t varies, and the model needs to be rewritten to make this dependence explicit before it can be taken to data.

The observable component s_t. I summarize the public, tract-level effect of X_t on values with a single scalar, $s_t \equiv s(X_t)$, which I call the tract's *observable component*. Each bidder's value is then

$$V_{it} = s_t \cdot \tilde{V}_{it}, \tag{29}$$

where $\tilde{V}_{it}$ is bidder i's residual, private value for tract t, distributed according to a tract-invariant $H_{\tilde{V}}$ as in Section 4, again allowing for correlation.[24] [25]

I deliberately call s_t the tract's *observable component* rather than a *common value*: unlike in Milgrom and Weber [1982] where a common value is an unknown payoff state that all bidders learn about through noisy signals, s_t is known with certainty to every bidder the moment they arrive and no exploration is required to learn it. I refer to $\tilde{V}_{it}$ as bidder i's private *value*, not a *signal*: once bidder i pays c_{it} and learns $\tilde{V}_{it}$, that realization is her private payoff-relevant valuation, which is never revised in light of other bidders' information.[26]

Implications for the game. Because ϕ, r, and the (unrealized) distribution of c_i are fixed in dollar terms while V scales with s_t, the equilibrium objects of Section 4.4 (v^*, c_L^*, c_H^*) must in general be solved tract by tract: two tracts with different s_t but otherwise identical primitives face a different relative balance between the fixed-dollar instruments and the value of participating. Section 5.1 operationalizes s_t empirically via a hedonic regression on X_t, and Proposition 3 formalizes the rescaling implied by equation (29): solving the model in *residual* units $\tilde{V} = V/s_t$ is equivalent to solving the dollar-denominated model tract by tract, provided the fixed-dollar instruments are rescaled in tandem $(\tilde{\phi}_t = \phi/s_t,\ \tilde{r}_t = r/s_t,\ \tilde{c} = c/s_t)$.

[24] Formally, H_V in Section 4 is the distribution of $s_t \cdot (\tilde{V}_{1t}, \ldots, \tilde{V}_{Nt})$ induced by $H_{\tilde{V}}$. Nothing in the equilibrium characterization of Section 4.4 relied on H_V being unscaled, so that characterization carries over unchanged once restated in terms of $\tilde{V}$.

[25] Phone interview evidence directly supports treating residual values as private and correlated, rather than common: "value depends on firm-specific tech, machinery, [and] portfolio constraints," consistent with firms drawing correlated information about a tract's prospects from unobservable characteristics while placing a firm-specific, private value on that information.

[26] The only updating that occurs anywhere in the model is the belief revision of an unexplored bidder about her own not-yet-realized $\tilde{V}_{it}$, upon observing others' nomination behavior (equations (1)– (2)) – updating a prior about one's own draw, which is the defining feature of a correlated *private* values model as opposed to a common or interdependent values model

This is the sense in which s_t is the object that matters for equilibrium: any two tracts with the same s_t (even with different underlying X_t) generate the same residual-space equilibrium.

5 Estimation

The model is estimated in two steps. I parameterize the joint value distribution as

$$H_V(v_1, \ldots, v_N) \;=\; C(F_V(v_1), \ldots, F_V(v_N);\, \rho)\,,$$

where $C(\cdot; \rho)$ is a Gaussian equicorrelation copula with parameter $\rho \in [0, 1)$. **Step 1** recovers the private-value distribution $\hat{F}_V$ and the copula parameter $\hat{\rho}$ from observed auction bids. **Step 2** estimates the nomination-stage structural parameters (Poisson arrival rates $(\lambda_{\text{pre}}, \lambda_{\text{post}})$ and the cost distribution $F_c(\cdot; \xi_c, \omega_c, \alpha_c)$) using GMM with equilibrium conditions imposed as explicit constraints (MPEC). Full estimation details are in Appendix B.

5.1 Step 1: Bid Homogenization and Value Distribution Recovery

Raw bids b_{istc} vary multiplicatively with observable tract characteristics (size, royalty rate, well proximity). I estimate a log-linear hedonic regression to isolate the private-value component, recovering auction-specific scaling factors $s_t = \exp(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t)$ and homogenized bids $\tilde{b}_{istc} = b_{istc}/s_t$. An equilibrium translational invariance result (Proposition 3) justifies working in residual space: a uniform shift in log-heterogeneity Z_t shifts all equilibrium actions proportionally, leaving residual-space strategies unchanged. Regression estimates are in Table 14 (reproduced in Appendix B.1); residual diagnostics supporting the log-linear specification are in Appendix B.11.

Note that by Proposition 3, all equilibrium objects discussed in Section 4 can be redefined in residual space. Notably, the model primitives (ϕ, r, F_C) are replaced with $(\tilde{\phi}_t, \tilde{r}_t, F_{\tilde{C}_t}) = (10/s_t, r/s_t, F_{C/s_t})$. This is because the \$10 nomination fee, the reserve, and the learning cost c do not scale with s_t (e.g., v decreases with distance to the nearest well, while the fee, the reserve, and c do not). For computational simplicity, I group the 728 estimation auctions into 19 bins of s_t (Appendix B.6) and assign each bin b a representative fee $\tilde{x}_b = \text{median}_{t \in b}(x/s_t)$ for $x \in \{\phi, r, c\}$.

Given the homogenized residual-unit bids and cross-auction variation in bidder counts I_r, I jointly estimate a monotone B-spline $\hat{F}_V$ and the equicorrelation Gaussian copula parameter $\hat{\rho}$, matching the theoretical order-statistic CDF (Theorem 1) to empirical Weibull ranks on a 200-point grid over 100-fold cross-validated train/test splits. The Gaussian copula achieves the lowest hold-out RMSE (Table 16; $\hat{\rho} = 0.8605$) and is also the theoretically appropriate

choice for a correlated private-values setting (see Appendix B.3 for the economic rationale). A Lognormal tail is fitted to the B-spline and then refined by a truncation-corrected maximum-likelihood step at the lower bid-support boundary v^* (Appendix B.3) to extrapolate F_V below the observed bid support. The non-nominating bidder's conditional value Details are in Appendix B.3.

5.2 Step 2: Nomination-Stage Parameters

GMM moments. I match four moment conditions per bin to model-implied values: (1) the auction trigger rate $\Pr(M_b \geq K)$; (2) mean nominations per triggered tract (zero-truncated Poisson mean); (3) the variance of nominations per triggered tract (zero-truncated Poisson variance); and (4) mean total bidders per triggered auction $\mathbb{E}[I \mid M \geq K]$. Moment (1) is matched in log scale. Full definitions are in Appendix B.5, and Figure 12 shows the fit bin by bin.[27]

Binning. After homogenization, the flat \$10 nomination fee becomes a bin-specific residual-space fee $\phi_b = 10/s_b$ where $s_b = \text{median}_{t \in b}(s_t)$. Because this varies with s_b, the equilibrium indifference conditions, and therefore the cost cutoffs (v^*, c_L^*, c_H^*), differ across bins. Treating all auctions as having the same equilibrium would be incorrect. I address this by estimating bin-specific equilibrium objects $(v_b^*, c_{L;b}^*, c_{H;b}^*)$ while the cost distribution F_C remains shared across all bins. Bin construction details are in Appendix B.6.

MPEC implementation. Following Su and Judd [2012], I replace the nested fixed point [Rust, 1987] with MPEC: the 62-element decision vector

$$\boldsymbol{z} = \left(\lambda_{\text{pre}},\ \lambda_{\text{post}},\ \xi_c,\ \omega_c,\ \alpha_c,\ v_1^*, \ldots, v_{19}^*,\ c_{L,1}^*, \ldots, c_{L,19}^*,\ c_{H,1}^*, \ldots, c_{H,19}^*\right)$$

is optimized subject to 57 equality constraints (three per bin for 19 bins, namely nomination threshold $h_{1,b}$, early learning cutoff $h_{2,b}$, and non-nominator cutoff $h_{3,b}$). Full MPEC implementation details are in Appendix B.8.

[27] The trigger-rate moment m_1 requires linking each auctioned tract to the per-season available-tract universe, which is done by a fuzzy match on parsed legal descriptions (Township/Range, Section, season). This match links roughly 746 of the 878 auctions; the remainder fail the legal-description match, so the trigger rate used in estimation is an undercount of $\approx 15\%$. The match gap is confined to m_1: the auction-level moments (nomination counts and bidder counts, m_2–m_4) are built from the structural estimation sample and do not depend on this match. Recovering the unmatched auctions would raise the trigger rate—and hence $\hat{\lambda}_{\text{pre}}$—by about 15% (to ≈ 0.006), leaving the qualitative thin-arrival conclusion and all other estimates unchanged.

6 Estimation Results

Table 2 reports the five shared structural parameters estimated by the binned MPEC procedure.[28] The skew-normal cost parameters are estimated directly in dollars and shared across bins, so the mean exploration cost implied by these estimates is $\approx$ \$1,161, with implied standard deviation $\approx$ \$416. The negative shape parameter $\hat{\alpha}_c$ makes the cost distribution left-skewed. [29] The equilibrium cost cutoffs are estimated in residual units and convert to dollars through the auction-specific scaler $s_t = \exp(\hat{y}_t)$ as $c^* \cdot s_t$. In the high-value bins the dollar cutoff $\hat{c}_b^{*H} \cdot s_b$ lies well above the cost mean $\hat{\mu}_c$, so the cutoffs are non-binding there: essentially all arriving bidders find learning worthwhile. In the lowest-value bins, by contrast, the dollar cutoffs fall into the body of the cost distribution and cost does screen participation (see below). The mean learning cost of \$1,161 is modest relative to the average total bid of \$11,803 across all bidders (and approximately \$13,900 for winning bidders). The barrier to nomination therefore operates through the low pre-period arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$ rather than through cost screening.

The gap between $\hat{\lambda}_{\text{pre}} \approx 0.005$ and $\hat{\lambda}_{\text{post}} \approx 2.894$ is consistent with the attention asymmetry described in the introduction: before nomination, attention is spread over the full $\approx$ 28,000-tract pool, whereas after nomination it is concentrated on the smaller set of tracts actually offered. Two reduced-form patterns in the raw data support this reading. Figure 5(a) plots, by season, the total number of nominations received against the number of available tracts: despite $\approx$ 28,000 tracts available each season, only $\approx$ 100 nominations arrive, and that count decreases with the size of the available pool. Figure 5(b) plots the number of bids per auctioned tract against the number of tracts auctioned, one point per season: bidders per tract decline as more tracts are offered, so attention is far less diluted once the choice set has been pre-screened by nomination.

To quantify the dilution I fit an *attention function* to each stage—a simple relationship describing how the response received per tract falls as the number of tracts competing for attention grows.[30] I estimate it with a Poisson GLM at the season (s) $\times$ bin (b) level—i.e. $\text{count}_{sb} \sim \text{Poisson}(\mu_{sb})$ with $\log \mu_{sb} = \log A + (1 - \kappa) \log(\text{pool}_{sb})$, so that bin–seasons with

[28] These estimates use the Joint GMM Strategy copula parameter ($\hat{\rho} = 0.861$, Table 16).

[29] Mean exploration cost is calculated as $\hat{\mu}_c = \hat{\xi}_c + \hat{\omega}_c \, \delta \sqrt{2/\pi}$ where $\delta = \hat{\alpha}_c / \sqrt{1 + \hat{\alpha}_c^2}$

[30] The idea that each potential buyer attends to only a fraction of the listings that reach her is also central to Coey et al. [2020], who model eBay participation as a Poisson process thinned by an attention probability and estimate that buyers attend to about 6% of the relevant auctions they encounter. The attention function here plays the same role, letting the thinning depend on the size of the candidate pool. The same dilution appears in financial markets, where the immediate response to an earnings announcement is weaker on days with more competing announcements [Hirshleifer et al., 2009]. The attention function here is a reduced-form, market-level relationship and differs from the individual-level "attention function" of Gabaix [2014], which maps the importance of a decision variable into the weight placed on it.

zero nominations (a genuine response of zero per tract) are retained:

$$\mathbb{E}\big[\text{response per candidate tract}_{sb}\big] \;=\; A \times \big(\#\text{ candidate tracts}_{sb}\big)^{-\kappa}. \tag{30}$$

The response per candidate tract is total nominations per available tract for the nomination stage and bids per auctioned tract for the auction stage; the candidate pool is the set of tracts a participant can view at that stage—available tracts for the nomination stage and auctioned (offered) tracts for the auction stage. The elasticity κ measures attention dilution where a 1% larger candidate pool lowers the expected response per tract by about κ%: $\kappa = 0$ means the per-tract response is independent of pool size (no dilution); $\kappa = 1$ means the *total* response is fixed and merely spread thinner (complete dilution); $\kappa > 1$ means the *total* response itself *falls* as the pool grows (super-dilution). Pooling both stages, each rate against its own candidate pool, yields $\hat{\kappa} \approx 1.15$ (Figure 6; robustness checks in Appendix C.1). Per-tract attention thus falls roughly as 1/pool—the force that keeps $\hat{\lambda}_{\text{pre}}$ orders of magnitude below $\hat{\lambda}_{\text{post}}$.

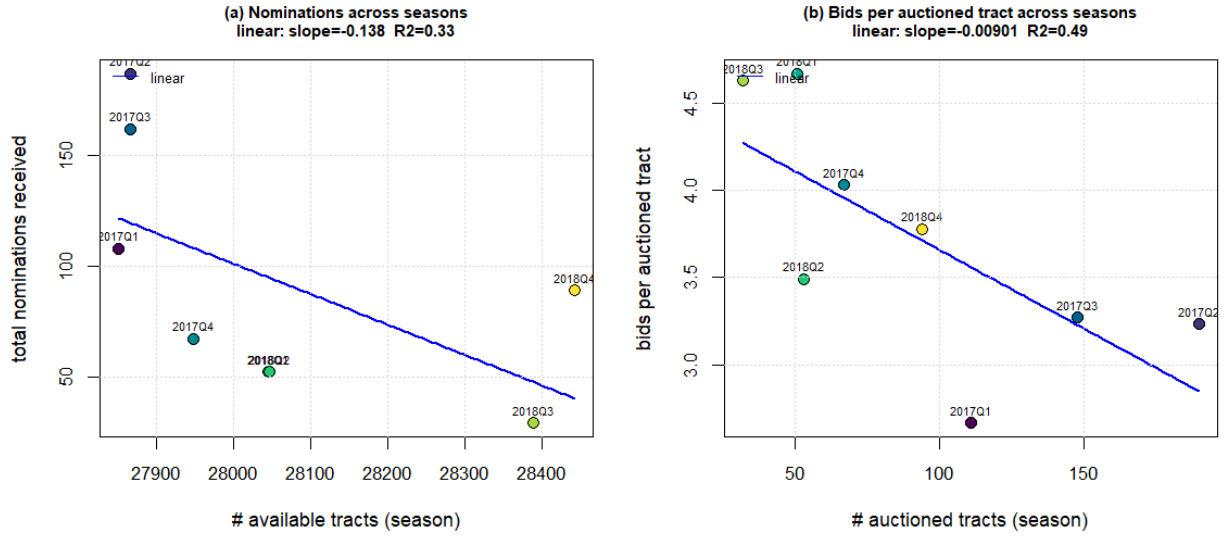


Figure 5: Attention across the two stages (raw data). **(a)** Total nominations received per season against the number of available tracts (one point per season, 2017Q1–2018Q4). **(b)** Bids per auctioned tract (total bids ÷ tracts auctioned) against the number of tracts auctioned, one point per season.

The per-bin equilibrium objects $(\hat{v}_b^*, \hat{c}_b^{*L}, \hat{c}_b^{*H})$ are reported in Appendix Table 18. The nomination threshold $\hat{v}_b^*$ rises with the residual-space fee ϕ_b across bins, while the early learner and non-nominating bidder cutoffs ($\hat{c}_b^{*L} \in [1.51, 2.21]$; $\hat{c}_b^{*H} \in [1.87, 3.05]$, in residual units) imply dollar cutoffs $\hat{c}_b^* \cdot s_b$ in the far right tail of the shared cost distribution in the high-value bins, so almost all arriving bidders find learning worthwhile; in the lowest-value

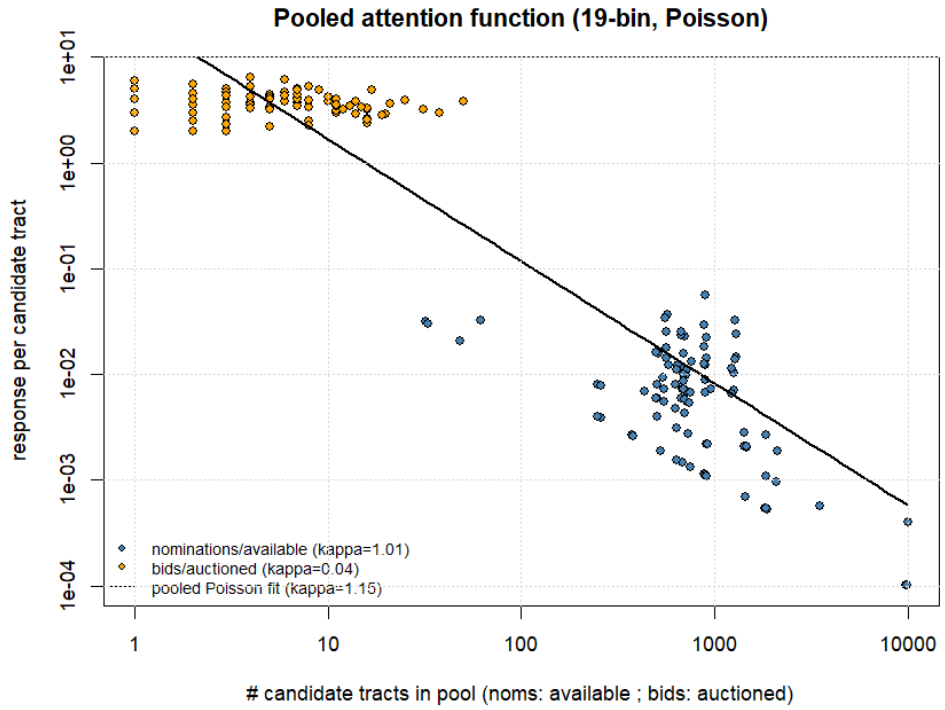


Figure 6: Pooled attention function. Per-candidate response rate against the size of the relevant candidate pool (log–log), stacking the nomination stage (total nominations per available tract vs. # available) and the bidding stage (bids per auctioned tract vs. # auctioned). The solid line is the pooled Poisson fit of equation (30), $\hat{\kappa} \approx 1.15$.

Table 2: MPEC Structural Parameter Estimates

Parameter	Description	Estimate
$\hat{\lambda}_{\text{pre}}$	Pre-auction Poisson arrival rate	0.005
$\hat{\lambda}_{\text{post}}$	Post-auction Poisson arrival rate	2.894
$\hat{\xi}_c$	Skew-Normal location (cost dist., \$)	1,712
$\hat{\omega}_c$	Skew-Normal scale (cost dist., \$)	690
$\hat{\alpha}_c$	Skew-Normal shape (cost dist.)	−126.9
	Implied SN mean (\$)	1,161
	Implied SN std. dev. (\$)	416

Notes: Binned MPEC estimation, $B = 19$ bins of s_t, break-points derived from the full available-tract universe (2017Q1–2018Q4). The skew-normal cost parameters $(\hat{\xi}_c, \hat{\omega}_c, \hat{\alpha}_c)$ are estimated directly in dollars and shared across bins.

bins the dollar cutoffs fall into the body of the distribution. The model's fit to the four GMM moments is reported in Appendix Table 19. The estimated value functions $U_{l=1}(c)$ and $U_{l=2}(c)$ and the implied cost-distribution mass shares are presented in Appendix B.10 (Figure 13).

The average cost of learning reflects cost-based selection into $l = 1$ versus $l = 2$. Early learners ($l = 1$) pay the least, an average of \$979; late learners ($l = 2$) pay the most, \$1,374; late (post-nomination) arrivals ($d = 1$) pay \$1,021; and the unconditional population mean is \$1,161. Nominators being the lowest-cost type and later learners the highest is precisely the sorting the model predicts: cost-efficient firms find early learning worthwhile, whereas higher-cost firms defer and wait on the information a nomination reveals.

Evaluating the cost distribution at the dollar cutoffs $\hat{c}_b^* \cdot s_b$ shows how much learning cost screens participation. In the high-value bins (bins 1–11) the dollar cutoff sits in the far right tail of the cost distribution, so the early learning ($l = 1$) share $\Pr(c < \hat{c}_b^{*L}\, s_b)$ is 100% and cost imposes no screen. In the least promising, low-s bins the dollar cutoff falls into the body of the distribution: the early learning ($l = 1$) share declines to 3.5% in the lowest-s bin, and the share of arrivals who would *not* nominate on cost grounds, $\Pr(c > \hat{c}_b^{*L}\, s_b)$, rises to 96% in that same bin; the population-weighted average across all bins is $\approx 50\%$.

Bidders do exit: the exit share $\Pr(c > \hat{c}_b^{*H}\, s_b)$ is 0% in the high-value bins but reaches 95% in the lowest-s. Population-weighted overall average is $\approx 43\%$, so the model does not imply that no one exits. Cost screening is thus concentrated in low-value tracts due to a cost-relative-to-value effect.

However, the learning cost cannot explain the low nomination rate conditional on learning: even in bins 1–11, where cost imposes no screen and everyone learns upon arriving, the

trigger rate is only $\approx 0.5\%$, because the pre-period arrival rate is so low ($\hat{\lambda}_{\text{pre}} \approx 0.005$). The low arrival rate is therefore the binding constraint on the overall nomination rate, while cost screening governs which tracts are nominated.

As a robustness check on the nomination-count specification, I re-estimate the model replacing the mean and variance of nominations (Moments 2–3) with the conditional multiple-nomination share $\Pr(M \geq 2 \mid M \geq 1)$. The results, and their implications for the single-rate Poisson arrival assumption, are reported in Appendix B.7.

7 Counterfactual Analysis

Nomination does four things at once. It supplies a **guaranteed bidder**: a nomination guarantees that a committed bidder is present and has paid to learn the tract, whereas an arbitrarily offered tract frequently draws no one. It ensures that **more valuable tracts are offered**: a firm nominates only when it has found $v \geq v^*$, so nominated tracts are more valuable than tracts the seller would pick without that signal. Because values are correlated, a nomination creates an **information spill-over**—it reveals that someone found the tract $v \geq v^*$, encouraging bidders who have not yet learned to pay to learn—but it can also cause **under-provision**, since a bidder who can wait for someone else's nomination has less reason to learn early and nominate herself. Finally, and most importantly, offering far fewer tracts **concentrates attention**: more bidders notice and arrive at each tract on offer.

To quantify these channels I compare seven counterfactual worlds, holding the estimated parameters $(\hat{\lambda}_{\text{pre}}, \hat{\lambda}_{\text{post}}, \hat{\xi}_c, \hat{\omega}_c, \hat{\alpha}_c)$ fixed and re-solving each world's equilibrium. In every world a bidder's value for a given auction is drawn with the estimated correlation; Appendix B.13 gives the exact simulation. Beyond each party's surplus, I track two costs that the surplus accounting does not show separately. **Wasteful learning** is the learning cost sunk by a bidder who pays to learn her value, finds it below the reserve ($v < r$), and so never bids. The **opportunity cost** of nomination is the surplus forgone on tracts that nomination leaves unoffered because the early arrival drew a value below v^*, though later arrivals might have valued them.

Worlds $\mathcal{W}_1$–$\mathcal{W}_4$: the value of nomination holding arrival fixed. Each world removes one thing from the one before. World $\mathcal{W}_1$ is the estimated K=1 economy: nominated auctions, in which the early arrival has found $v \geq v^*$ and always bids, and non-nominating entrants draw their value from the *posterior* correlated distribution. World $\mathcal{W}_2$ keeps the same nominated auctions but switches the signal off, so late entrants use the raw prior F_V—the difference $\mathcal{W}_1 - \mathcal{W}_2$ is *information spill-over*. World $\mathcal{W}_3$ removes the nomination

threshold: the early arrival is still present on every tract and still bids, but its value need only clear the reserve ($v \geq r$) rather than v^*—the difference $\mathcal{W}_2 - \mathcal{W}_3$ is the *selection of more valuable tracts*. World $\mathcal{W}_4$ removes the guaranteed bidder: bidders arrive at the natural rate ($\lambda_{\text{pre}} + \lambda_{\text{post}}$), and an early arrival bids only if it happens to be present, its learning cost is low enough for it to learn, and the value it then draws clears the reserve. World $\mathcal{W}_4$ is the no-nomination benchmark: the seller offers an arbitrary tract and hopes someone shows up. The successive differences sum to the value of nomination:

$$\underbrace{(\mathcal{W}_1 - \mathcal{W}_4)}_{\text{value of nomination}} = \underbrace{(\mathcal{W}_1 - \mathcal{W}_2)}_{\text{information spill-over}} + \underbrace{(\mathcal{W}_2 - \mathcal{W}_3)}_{\text{more valuable tracts}} + \underbrace{(\mathcal{W}_3 - \mathcal{W}_4)}_{\text{guaranteed bidder}}.$$

I report every quantity per offered tract.[31] Table 3 and Figure 14 report the decomposition by party. Pooled across tracts, nomination raises **total surplus by about $4,239 per offered tract**. The **seller** gains $3,950 per offered tract, almost all of it from the guaranteed bidder ($3,897); the selection of more valuable tracts adds only $52, with a standard error of $724.[32] Conditional on a sale, revenue rises by about $2,956, roughly 30% above the no-nomination benchmark, and the guaranteed bidder accounts for $2,903 of it. Per offered tract the gain is larger, because every nominated tract sells while 10.2% of arbitrarily offered tracts in $\mathcal{W}_4$ draw no bidder. The guaranteed bidder accounts for the entire rise in the sale rate, from 89.8% to 100%, and adds about 0.8 bidders per sold auction. The selection of more valuable tracts adds essentially nothing because the nomination threshold barely exceeds the reserve (v_b^*/r_b ranges from 1.03 to 1.19 across bins): a bidder who finds a tract worth bidding on is almost always willing to nominate it. What matters is that some bidder whose value clears the reserve is on the tract, not how far above the reserve that value is. On the least promising tracts (Bin 19), an arbitrarily offered tract sells only 1.4% of the time (Figure 16).

The **nominator** gains $2,721, almost entirely from the guaranteed bidder ($2,718). The reason is visible in the table: in $\mathcal{W}_4$ the nominator's surplus is only $11.6 per tract, because without the nomination rule she reaches an arbitrary tract only about 0.5% of the time. Conditional on participating, the nominator's surplus barely changes across worlds ($\approx$ $2,500). The **non-nominators** lose $2,432 per tract: a guaranteed strong nominator

[31] This is because in $\mathcal{W}_4$ a tract the nominator-type never reaches contributes *zero* surplus, and since nomination's central role is to get a committed bidder onto the tract at all, those empty tracts belong in the average.

[32] Table 3 and Figure 14 value each bin at its median tract. Summing over the 728 tracts, this gives seller revenue of $6.40 million without nomination and $9.28 million with it (bottom panel of Table 3). The abstract and the introduction instead value each tract at its own scaler s_t (Appendix Table 20): seller revenue is $9,418 per offered tract without nomination and $13,607 with it, or $6.86 million and $9.91 million over the 728 tracts, a gain of $3.05 million (44%); the split across channels is essentially unchanged.

competes away the surplus they would otherwise capture on a tract with no committed buyer ($\mathcal{W}_4$ non-nominator surplus is \$7,154, the highest in the table). The externality thus runs both ways—the nominator's presence that the seller monetizes is the same presence that competes away non-nominator rents. The selection of more valuable tracts is essentially zero for every party, and information spill-over is essentially zero on average but matters for the least-promising tracts (Appendix C.6). Information spill-over is small because, at the estimated nomination thresholds, almost every draw clears v^*, so observing a nomination barely changes the shape of a late entrant's posterior (Appendices D.3 and F).

Information spill-over versus under-provision. The comparison $\mathcal{W}_1 - \mathcal{W}_2$ holds nomination behavior fixed, so it measures only the spill-over: how many more late arrivals learn and enter once they condition on a nomination. Pooled across tracts, the spill-over raises participation by 0.1 percentage points (98.0% to 98.1%), and by 1.4 points on the least promising 4% of tracts (Bins 15–19; 3.5 points in Bin 19). The same signal could also reduce nomination itself, if bidders who expect to learn from someone else's nomination stop learning early. To measure this *under-provision*, world $\mathcal{W}_{2b}$ switches the signal off and re-solves the early-learning cutoff $c^*_{L,b}$ in each bin, holding the nomination threshold v^*_b fixed. Under-provision is the resulting change in the probability that a tract is nominated, $(1 - F_V(v^*_b))\,[F_{\tilde{C}_b}(c^*_{L,b} \mid \mathcal{W}_{2b}) - F_{\tilde{C}_b}(c^*_{L,b} \mid \mathcal{W}_1)]$. It is essentially zero: 0.04 percentage points pooled. The bin-level effects are small and of mixed sign: the signal lowers nomination by 1–3 points in Bins 14–17 and raises it by 1–2 points in Bins 12–13, and has no effect elsewhere. The option to learn later is valuable because it avoids paying to learn on a tract that is never auctioned: bidders who learn later are waiting for the auction to be triggered, not for the information a nomination carries. Because the spill-over exceeds the under-provision, the net effect of the information channel is weakly positive.

Reading learning costs off the same $\mathcal{W}_1$ and $\mathcal{W}_4$ simulations, nomination saves the resources bidders would otherwise sink into evaluating tracts along two margins.[33] On one hand, nomination roughly halves *wasteful learning*, from about \$272 to \$140 per auctioned tract ('Wasted learning' row of Table 3). In $\mathcal{W}_1$ all of it comes from post arrivals: the nominator has already learned $v \geq v^*$ and always bids, so the only learning wasted is by post arrivals who pay to learn and draw $v < r$. They do so far less often on a nominated tract than on an arbitrary one (the probability that a late learner draws $v < r$ falls from 10.6% to 5.2%; Appendix C.7).

Almost all of the saving comes from the guaranteed bidder (\$128). In $\mathcal{W}_3$ the early

[33] Note that these figures are already carried in the seller's and non-nominators' surplus from the $\mathcal{W}_1$–$\mathcal{W}_4$ difference, re-expressed in learning-cost terms.

Table 3: Value of nomination by party and channel ($ per offered tract; W4 counts non-entered tracts as zero; bootstrap s.e. in parentheses)

	Information spill-over ($\mathcal{W}_1-\mathcal{W}_2$)	More valuable tracts ($\mathcal{W}_2-\mathcal{W}_3$)	Guaranteed bidder ($\mathcal{W}_3-\mathcal{W}_4$)	Total (value of nom.) ($\mathcal{W}_1-\mathcal{W}_4$)
Seller	1 (1)	52 (724)	3,897 (650)	3,950 (693)
Nominator	-1 (0)	4 (430)	2,718 (312)	2,721 (299)
Non-nominator	-1 (1)	25 (608)	-2,456 (638)	-2,432 (641)
Total welfare				**4,239**
Wasted learning	0 (0)	-4 (9)	-128 (12)	-132 (12)
Total over the 728 tracts ($ million)				
Seller				2.88
Nominator				1.98
Non-nominator				-1.77
Total welfare				**3.09**

Parties sum to total welfare. The 'Wasted learning' row reports the change in wasted learning (learning cost sunk by bidders who draw $v < r$ and never bid). It is already netted out of the party rows and is not added to total welfare. The bottom panel multiplies the Total column by the 728 tracts in the estimation sample.

Table 4: Concentrated attention: low- vs high-attention arrival (nominators-only, $ per tract; bootstrap s.e. in parentheses)

	Low attention ($\hat{\lambda}_{\text{low}}=0.006$)	High attention ($\hat{\lambda}_{\text{high}}=4.23$)	Concentrated attention (high − low)
Seller	1	8,132	8,130 (186)
Bidders	53	18,371	18,318 (823)
Sale rate	0.006	0.977	
Bidders/tract	0.01	3.91	

Nominators-only general equilibrium at reserve $r = \$2$. Low/high attention are the arrival rates implied by the estimated attention function spread over the ≈28,000 available tracts vs. concentrated on the ≈73 offered tracts (Appendix C.3). W5/W6 are nominators-only, so all bidders are nominators and there are no non-nominators.

arrival's value clears the reserve, so its own learning is never wasted, and because values are correlated, post arrivals on such a tract draw $v < r$ far less often than on an arbitrary tract in $\mathcal{W}_4$, where the early arrival is usually absent and post arrivals' values are drawn from the prior. Raising the early arrival's value from the reserve to the nomination threshold ($\mathcal{W}_2 - \mathcal{W}_3$) saves only another \$4, and information spill-over leaves wasted learning unchanged.

On the other hand, there is opportunity cost to nomination: tracts unoffered because the early arrival drew $v < v^*$ might still be valuable to later arrivals. It turns out this is not a forgone surplus. Because values are highly correlated ($\hat{\rho} = 0.861$), a low early draw likely means a low value to everyone else. The counterfactual exercise reveals auctioning such a tract would instead waste about \$3,965 of learning per tract (from about three bidders paying to learn) to realize only about \$1,571 of winning value (versus \$12,756 on a nominated tract). This nets to a negative total surplus of about −\$2,400 per tract, in every bin but the least promising. On these skipped tracts 54% of late learners draw $v < r$—against about 5% on nominated tracts—so most of the learning is wasteful. The opportunity cost of the nomination screen is therefore non-positive: nomination saves learning cost both by auctioning higher-value tracts and by never convening costly search on low-value ones.

Worlds $\mathcal{W}_5$–$\mathcal{W}_6$: attention. The decomposition above holds the arrival process fixed, so it isolates what a nomination does given who shows up. To quantify the *concentrated attention* channel, I set up a one-stage auction game where bidders arrive, pay to learn their value, and compete in the auction at the reserve $r = \$2$/acre. I vary the *arrival rate*, that is, how many bidders attend to each offered tract. In world $\mathcal{W}_5$ arrivals follow the *thin* attention-implied rate $\hat{\lambda}_{\text{low},b}$, which aggregates to 0.0058: the response per available tract when bidders' limited attention is spread over the ≈28,000 available tracts (Appendix C.3). In world $\mathcal{W}_6$ that same attention is concentrated on the ≈73 tracts actually offered, raising arrivals to about 4.23 bidders per auction. The difference $\mathcal{W}_5 \to \mathcal{W}_6$ measures concentrated attention.

Table 4 reports the two worlds. Under low attention the market barely functions—0.6% of tracts sell. Concentrating attention revives it completely: 98% sell and about four bidders compete per tract. The seller gains \$8,130 per tract and the bidders \$18,318. The result highlights that bidders' attention, not the cost of learning or the nomination mechanism itself, is the binding constraint on the nomination rate. Nomination is valuable because it secures a committed bidder when attention is otherwise too scarce to convene an auction.

How much would the seller need to know? The benchmark $\mathcal{W}_4$ assumes the seller has no information about which individual tracts are worth offering. A seller could instead use

observable characteristics and offer tracts only from the most promising bins—those with scaler s_b above a cutoff—by choosing arbitrarily among them. Figure 17 plots the seller's revenue in this seller-initiated alternative against the cutoff, holding the number of offered tracts at 728 (Table 21 gives the full breakdown). The offered tracts are spread across the included bins in the same proportions as the nominated tracts, so offering all bins reproduces the $\mathcal{W}_4$ benchmark: revenue is \$6.40 million, against \$9.28 million under nomination.[34] To match nomination, the seller must restrict itself to roughly the top 8–10% (Bins 1–3) of available tracts.[35][36]

The revenue comparison masks three margins that move differently. On the **number of bidders**, nomination draws about 3.7 bidders per tract, against 2.6–2.8 for the seller-initiated alternative at every cutoff: the guaranteed bidder is worth roughly one extra bidder that seller-initiation cannot reproduce. On the **sale rate**, every nominated tract sells, while the seller-initiated alternative sells 90–93%. The **price** in the seller-initiated alternative is higher: since it is very selective it offers the most valuable tracts, so its average winning bid exceeds nomination's (\$14,390 for Bins 1–3 against \$12,743), but the advantage disappears as the offered set widens (\$9,788 with all bins). A selective seller therefore beats nomination only through high prices on a few top tracts, while still fielding fewer bidders and selling a smaller share. Nomination's value is not that it sells valuable tracts at high prices—a well-informed seller could do that—but that it secures a committed bidder and a sale on every offered tract without the seller knowing which tracts are worth offering.

Removing the fee and the purchase commitment Besides the learning cost, two features of the nomination rule could limit how often tracts are nominated: the \$10 nomination fee, and the nominator's commitment to buy the tract at the reserve if no one else bids. To measure how much each matters, I re-solve the equilibrium in every bin with each feature removed in turn and compare the trigger rate, the probability that a tract receives at least one nomination. Removing the fee raises the trigger rate, averaged over bins in proportion to their available tracts, by only 1.6% (from 0.197% to 0.200% of available tracts). Removing the commitment means a nominator need not buy: a bidder who learns $v < r$ could nominate

[34] Both totals value each bin at its median tract, as in Table 3.

[35] Offering only these bins yields \$9.66 million, but adding Bin 4 (the top 10%) lowers revenue to \$8.98 million.

[36] A seller choosing arbitrarily among all available tracts in the included bins would instead draw them in proportion to each bin's available tracts. The break-even point does not move: offering only Bins 1–3 still beats nomination (\$9.65 million), and adding Bin 4 still falls short (\$9.07 million). What changes is the other end of the range. Because most available tracts lie in the least promising bins, offering tracts arbitrarily from all bins yields only \$1.97 million, and the sale rate falls from about 90% to 62%. A sale rate this low is not implausible: the January 2025 lease sale in Alaska's Arctic National Wildlife Refuge offered about 400,000 acres and received no bids at all (Section 1).

and simply walk away, so the nomination threshold becomes v^{**} instead of $v^* = \max\{v^{**}, r\}$. This leaves the trigger rate unchanged in every bin. The value at which nominating is worth the fee, v^{**}, already exceeds the reserve in every bin, so a bidder whose value falls below the reserve never nominates, and every nominated tract still sells.

8 Conclusion

This paper provides the first structural analysis of pre-auction nomination, quantifying four channels through which it can create value: a guaranteed bidder, the selection of more valuable tracts, an information spill-over, and concentrated attention. Using detailed data from ND state mineral rights auctions during 2017–2018, I develop and estimate a two-stage game-theoretic model where bidders decide whether to learn early and nominate tracts or learn later after others' nominations.

The empirical analysis reveals three key findings. First, I document that most bidders participate without nominating: while auctions attract multiple bidders, the vast majority are triggered by a single nominator. Second, using a Gaussian equicorrelation copula I find strong positive dependence in bidders' valuations ($\hat{\rho} = 0.861$), indicating that common geological and market uncertainties create correlated values across competitors. Third, I estimate a left-skewed learning cost distribution (Skew-Normal with $\hat{\alpha}_c = -126.9$) with a modest mean of approximately \$1,161 and standard deviation \$416 (Table 2). At these estimates exploration cost screens participation only in the low-value bins: the early learning share is 100% in the high-value bins but falls to $\approx 3.5\%$ in the lowest value bins, so cost governs which tracts are nominated rather than the overall rate. The low observed nomination rate is instead a consequence of scarcity of early pre-nomination arrivals: the estimated pre-period arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$ is orders of magnitude below the post-arrival rate, so few firms arrive early enough to nominate even though learning early and nominating is individually worthwhile. The mean exploration cost of \$1,161 is small relative to the average bid (\$11,803 over all bids, approximately \$13,900 for winning bids alone), underscoring that the binding friction is thin early arrival rather than the cost of learning.

The structural model characterizes equilibrium nomination and learning strategies under correlated values and costly learning, focusing on the single-nomination rule ($K = 1$) that ND actually uses.

Comparing the current single-nomination policy to a no-nomination benchmark, the value of nomination rests mainly on the guaranteed bidder. Across the 728 tracts in the estimation sample, nomination raises seller revenue by about \$3.05 million, or 44% (from \$6.86 million to \$9.91 million): every nominated tract sells; conditional on a sale revenue rises by about

30%. Almost all of the seller's gain comes from the guaranteed bidder; the selection of more valuable tracts adds essentially nothing. The information spill-over is negligible except on the least promising tracts. A seller without nomination could match its revenue only by knowing which tracts are roughly the top 8–10% of available tracts by observable value, and even then would field fewer bidders and leave more tracts unsold. The largest lever, however, is attention: concentrating bidders' limited attention on the few nominated tracts raises the sale rate from under 1% to 98% and is worth about twice the seller's entire value of nomination ($8,130 versus $3,950 per tract), although the two exercises are not directly comparable because the attention worlds abstract from the nomination stage.

Nomination also economizes on the learning costs that bidders sink. On the tracts it offers, the learning wasted by bidders who learn only to find $v < r$ roughly halves. The tracts it leaves unoffered are not a forgone opportunity: because values are strongly correlated, auctioning them would spend about $3,965 on learning to realize about $1,571 of value, so the opportunity cost of the nomination screen is non-positive.

Three design questions that this framework can address, but that I leave for future work, concern how the nomination rule itself is set. The first is whether the seller should raise the nomination fee to screen tract quality: a higher fee raises the nomination threshold v^{**} and selects better tracts, but taxes the decision to learn early and nominate, affecting the auction trigger rate. The second is the choice between a nomination fee and a reserve price. The two instruments screen at different stages, leading to different outcomes. The nomination fee screens at the earlier, pre-auction stage, but once the fee exceeds the maximum attainable auction surplus, no firm nominates and the market collapses. The reserve preserves this extensive margin because it screens bidder quality at the auction stage, but becomes less reliable when the market is thin and the seller's cost of running the auction is high. A systematic fee-versus-reserve comparison is left for future work. Finally, whether requiring more nominations before initiating an auction improves outcome is an open question. Under the current thin early-arrival rate (≈ 0.005 per tract), single-nomination rule is the only threshold consistent with positive auction activity: two or more coincident nominations essentially never occur, so almost no tract would ever be auctioned. Whether this conclusion survives once firms' arrival and early-learning decisions are allowed to respond to the threshold is a question I leave for future work.

References

Jason Abaluck and Abi Adams-Prassl. What do consumers consider before they choose? identification from asymmetric demand responses. *The Quarterly Journal of Economics*, 136(3):1611–1663, 2021.

Barry C Arnold, Narayanaswamy Balakrishnan, and Haikady Navada Nagaraja. *A first course in order statistics*. SIAM, 2008.

Susan Athey and Philip A Haile. Identification of standard auction models. *Econometrica*, 70(6):2107–2140, 2002.

Adelchi Azzalini. *The skew-normal and related families*, volume 3. Cambridge University Press, 2013.

Dirk Bergemann and Martin Pesendorfer. Information structures in optimal auctions. *Journal of economic theory*, 137(1):580–609, 2007.

Simon Board. Revealing information in auctions: the allocation effect. *Economic Theory*, 38(1):125–135, 2009.

Aaron L Bodoh-Creed, Joern Boehnke, and Brent Hickman. How efficient are decentralized auction platforms? *The Review of Economic Studies*, 88(1):91–125, 2021.

Audra L Boone and J Harold Mulherin. How are firms sold? *The Journal of Finance*, 62 (2):847–875, 2007.

Jeremy Bulow and Paul Klemperer. Why do sellers (usually) prefer auctions? *American Economic Review*, 99(4):1544–1575, 2009.

Andrew Caplin, Mark Dean, and John Leahy. Rational inattention, optimal consideration sets, and stochastic choice. *The Review of Economic Studies*, 86(3):1061–1094, 2019.

Luciano I de Castro. Affiliation, equilibrium existence and the revenue ranking of auctions. 2007.

Dominic Coey, Bradley Larsen, and Kane Sweeney. The bidder exclusion effect. *The RAND Journal of Economics*, 50(1):93–120, 2019.

Dominic Coey, Bradley J Larsen, and Brennan C Platt. Discounts and deadlines in consumer search. *American Economic Review*, 110(12):3748–3785, 2020.

Thomas R Covert and Richard L Sweeney. Relinquishing riches: Auctions versus informal negotiations in texas oil and gas leasing. *American Economic Review*, 113(3):628–663, 2023.

Gregory S Crawford, Rachel Griffith, and Alessandro Iaria. A survey of preference estimation with unobserved choice set heterogeneity. *Journal of Econometrics*, 222(1):4–43, 2021.

Herbert A David and Haikady N Nagaraja. *Order statistics*. John Wiley & Sons, 2004.

Jean-Pierre Dubé, Jeremy T Fox, and Che-Lin Su. Improving the numerical performance of static and dynamic aggregate discrete choice random coefficients demand estimation. *Econometrica*, 80(5):2231–2267, 2012.

José-Antonio Espín-Sánchez, Charles Hodgson, and Kevin O'Neill. Order statistics as finite mixtures. 2025.

Xavier Gabaix. A sparsity-based model of bounded rationality. *The Quarterly Journal of Economics*, 129(4):1661–1710, 2014.

Xavier Gabaix. Behavioral inattention. In *Handbook of behavioral economics: Applications and foundations 1*, volume 2, pages 261–343. Elsevier, 2019.

Juan-José Ganuza and Jose S Penalva. Signal orderings based on dispersion and the supply of private information in auctions. *Econometrica*, 78(3):1007–1030, 2010.

Michelle Sovinsky Goeree. Limited information and advertising in the us personal computer industry. *Econometrica*, 76(5):1017–1074, 2008.

Alexander S Gorbenko and Andrey Malenko. Auctions with endogenous initiation. *The Journal of Finance*, 79(2):1353–1403, 2024.

Robert G Hansen. Auctions of companies. *Economic Inquiry*, 39(1):30–43, 2001.

Kenneth Hendricks and Robert H Porter. An empirical study of an auction with asymmetric information. *The American Economic Review*, pages 865–883, 1988.

Kenneth Hendricks and Robert H Porter. Auctioning resource rights. *Annu. Rev. Resour. Econ.*, 6(1):175–190, 2014.

Kenneth Hendricks, Joris Pinkse, and Robert H Porter. Empirical implications of equilibrium bidding in first-price, symmetric, common value auctions. *The Review of Economic Studies*, 70(1):115–145, 2003.

David Hirshleifer, Sonya Seongyeon Lim, and Siew Hong Teoh. Driven to distraction: Extraneous events and underreaction to earnings news. *The journal of finance*, 64(5):2289–2325, 2009.

Charles Hodgson. Information externalities, free riding, and optimal exploration in the uk oil industry. Technical report, National Bureau of Economic Research, 2024.

John Kagel, Svetlana Pevnitskaya, and Lixin Ye. Indicative bidding: An experimental analysis. *Games and Economic Behavior*, 62(2):697–721, 2008.

Paul Klemperer and J Bulow. Auctions versus negotiations. *The American Economic Review*, 86(1):180–94, 1996.

Yunmi Kong. Not knowing the competition: evidence and implications for auction design. *The RAND Journal of Economics*, 51(3):840–867, 2020.

Elena Krasnokutskaya and Katja Seim. Bid preference programs and participation in highway procurement auctions. *American Economic Review*, 101(6):2653–2686, 2011.

Dan Levin and James L Smith. Equilibrium in auctions with entry. *The American Economic Review*, pages 585–599, 1994.

Tong Li, Isabelle Perrigne, and Quang Vuong. Structural estimation of the affiliated private value auction model. *RAND Journal of Economics*, pages 171–193, 2002.

George Loewenstein and Zachary Wojtowicz. The economics of attention. *Journal of Economic Literature*, 63(3):1038–1089, 2025.

Willi Maurer and Barry H Margolin. The multivariate inclusion-exclusion formula and order statistics from dependent variates. *The Annals of Statistics*, pages 1190–1199, 1976.

Paul R Milgrom and Robert J Weber. A theory of auctions and competitive bidding. *Econometrica: Journal of the Econometric Society*, pages 1089–1122, 1982.

Alfred Müller and Marco Scarsini. Archimedean copulae and positive dependence. *Journal of Multivariate Analysis*, 93(2):434–445, 2005.

Roger B Myerson. Population uncertainty and poisson games. *International Journal of Game Theory*, 27(3):375–392, 1998.

Roger B Nelsen. *An introduction to copulas*. Springer, 2006.

Anh Nguyen. Information design in common value auction with moral hazard: Application to ocs leasing auctions. Technical report, Working Paper, Carnegie-Mellon University, 2022.

Daniel Quint and Kenneth Hendricks. A theory of indicative bidding. *American Economic Journal: Microeconomics*, 10(2):118–151, 2018.

James W Roberts and Andrew Sweeting. When should sellers use auctions? *American Economic Review*, 103(5):1830–1861, 2013.

John Rust. Optimal replacement of gmc bus engines: An empirical model of harold zurcher. *Econometrica: Journal of the Econometric Society*, pages 999–1033, 1987.

Michael Spence. Job market signaling. In *Uncertainty in economics*, pages 281–306. Elsevier, 1978.

Che-Lin Su and Kenneth L Judd. Constrained optimization approaches to estimation of structural models. *Econometrica*, 80(5):2213–2230, 2012.

U.S. Government Accountability Office. Oil and gas leasing: BLM should update its guidance and review its fees. Technical Report GAO-22-103968, GAO, 11 2021. URL https://www.gao.gov/products/gao-22-103968. Publicly released December 9, 2021.

Lixin Ye. Indicative bidding and a theory of two-stage auctions. *Games and Economic Behavior*, 58(1):181–207, 2007.

A Data Summary Statistics

Tables 5 and 6 report summary statistics for per-acre and total bid values. The bid values exhibit large variation and are harmonized via the homogenization regression described in Section B.1.

Table 5: Bid Value (Per Acre) Summary

Variable	Average	Std Dev	Max	Min
Overall ($)	157	752	18,001	2
Highest Bid ($)	170	829	18,001	2
Excl. Winner ($)	151	712	18,000	2

Notes: Summary statistics for per-acre bid values in dollars. The highest bid represents the highest bid observed in each auction, which is what the winner has to pay. "Excl. Winner" excludes the winning bid, whose recorded value reflects the ascending price proxy bidding mechanism rather than the winner's true valuation.

Table 6: Bid Value (Total) Summary

Variable	Average	Std Dev	Max	Min
Overall ($)	12,346	44,200	1,120,320	0.02
Highest Bid ($)	13,874	48,372	1,120,320	0.02
Excl. Winner ($)	11,626	42,084	1,120,160	0.12

Notes: Summary statistics for total bid values (per-acre bid × tract size) in dollars. "Excl. Winner" excludes the winning bid, whose recorded value is mechanically determined by the ascending price proxy bidding system rather than reflecting the winner's true valuation.

Tables 7 and 8 report parcel sizes (acres) and royalty rates by season. Royalty rates are determined by the county in which the tract is located.

Tables 9–11 report information on the nearest previously drilled well: distance to the tract centroid, start of drilling date, or 'spud date', in calendar terms, and spud date relative to the auction date.

Table 12 summarizes starting bids (minimum bids set by nominators) by season. Most starting bids are \$2; all auctions are modeled with $r = \$2$.

Table 7: Auction Size (acres) by Season

Season	# Auctions	Avg Tract Size (acres)	Min	Max
2017-Q1	173	151	35.5	160
2017-Q2	234	137	3.3	163
2017-Q3	166	137	0.16	311
2017-Q4	67	127	25.8	161
2018-Q1	52	141	28.6	160
2018-Q2	58	131	0.06	203
2018-Q3	29	152	80	160
2018-Q4	99	80.6	0.01	161

Notes: Average, minimum, and maximum tract sizes in acres by auction season. Tract sizes are determined by state regulations and geographic characteristics but cannot exceed 160 acres.

Table 8: Count of Royalty Rates by Season

Season	# Auctions	Royalty Rate 1/16	Royalty Rate 3/16
2017-Q1	173	51	122
2017-Q2	234	58	176
2017-Q3	166	52	114
2017-Q4	67	1	66
2018-Q1	52	25	27
2018-Q2	58	17	41
2018-Q3	29	8	21
2018-Q4	99	2	97

Notes: Distribution of royalty rates (1/16 or 3/16) by season. Royalty rates vary by county location and determine the percentage of production revenue paid to the state.

Table 9: Nearest Well Distance (miles) by Season

Season	Avgerage (miles)	Max	Min
2017-Q1	1.55	9.12	0.0881
2017-Q2	1.15	4.08	0.0105
2017-Q3	1.18	2.63	0.129
2017-Q4	0.912	2.13	0.0475
2018-Q1	0.821	2.41	0.0355
2018-Q2	0.672	1.94	0.0665
2018-Q3	0.722	2.22	0.107
2018-Q4	2.46	4.32	0.0452

Notes: Average, maximum, and minimum distance in miles from each tract to the nearest existing well, by season. Proximity to existing wells provides information about geological characteristics and production potential.

Table 10: Nearest Well Spud Date

Season	Earliest	Oldest	Missing
2017-Q1	2012-10-14	1953-12-15	30
2017-Q2	2014-08-30	1954-05-13	23
2017-Q3	2014-01-28	1955-10-06	23
2017-Q4	2013-07-03	1963-06-17	1
2018-Q1	2014-08-07	1956-05-22	10
2018-Q2	2014-06-03	1954-02-01	5
2018-Q3	2012-03-17	1960-12-02	3
2018-Q4	2014-06-12	1955-04-01	13

Notes: Date range of spud dates (when drilling began) for nearest wells by season. Some tracts have missing spud date information for their nearest well.

Table 11: Nearest Well Spud Date (How Close)

Season	Avg (days)	Max (days)	Min (days)
2017-Q1	11,949.392	23,065	1,577
2017-Q2	11,274.384	23,000	976
2017-Q3	11,988.860	22,580	1,281
2017-Q4	14,514.848	19,867	1,588
2018-Q1	13,457.952	22,540	1,279
2018-Q2	14,075.811	23,465	1,428
2018-Q3	12,373.654	21,067	2,334
2018-Q4	7,561.453	23,230	1,608

Notes: Average, maximum, and minimum time (in days) between the nearest well's spud date and the auction date. Longer time periods indicate less recent drilling activity in the area.

Table 12: Starting Bids by Season

Season	Starting at $2	$3	$4	$21	Total Auctions
2017-Q1	171	2	0	0	173
2017-Q2	210	16	6	2	234
2017-Q3	161	5	0	0	166
2017-Q4	67	0	0	0	67
2018-Q1	52	0	0	0	52
2018-Q2	58	0	0	0	58
2018-Q3	29	0	0	0	29
2018-Q4	98	1	0	0	99
Total	**846**	**24**	**6**	**2**	**878**

Notes: Distribution of starting bids (minimum bids) specified by nominators. The vast majority (846 out of 878) specified a minimum bid of $2, indicating nominators set the minimum at the lowest allowed level.

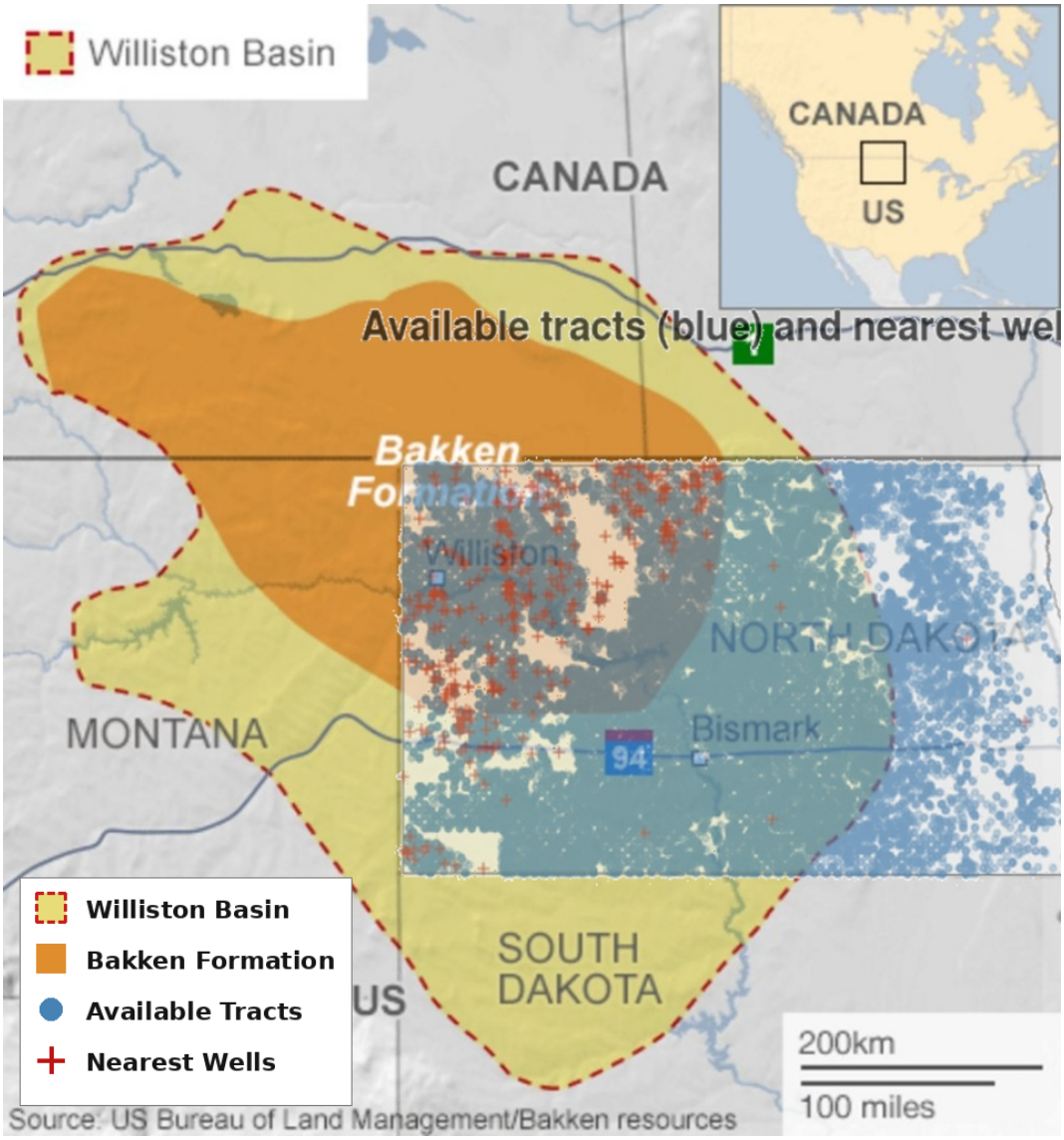


Figure 7: Available Tracts and Nearest Wells in the Williston Basin

Notes: The map overlays the boundary of the Williston Basin and the Bakken Formation with the tracts available for nomination in the estimation sample (blue) and the locations of the nearest producing wells (red crosses). Available tracts vary substantially in their proximity to an existing producing well, motivating the inclusion of distance to the nearest well and its spud date as a regressor in the bid-homogenization procedure (Section B.1): tracts closer to a demonstrated producer or tracts close to a recently developed well carry less geological uncertainty and are expected to command systematically higher bids, all else equal.

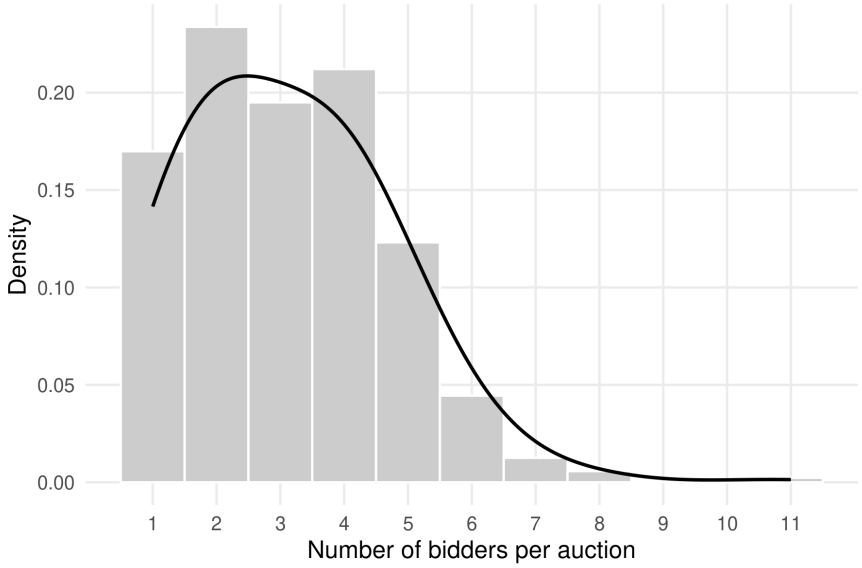


Figure 8: Distribution of the Number of Bidders per Auction

Notes: Histogram of the number of bidders per auction (one observation per auction), with a Gaussian kernel density estimate overlaid. The kernel bandwidth is $h = 2\,\hat{\sigma}\,n^{-1/5}$, an oversmoothed Gaussian-reference choice; Silverman's rule of thumb was visibly undersmoothed for these integer counts.

Table 13: Unique Bidders per Season

Season	Unique bidders
2017-Q1	23
2017-Q2	37
2017-Q3	31
2017-Q4	19
2018-Q1	29
2018-Q2	21
2018-Q3	20
2018-Q4	23

Notes: Number of distinct bidders that submitted at least one bid in each season. Counts are per season; because many bidders are active in more than one season, the column does not sum to the overall number of distinct bidders in the sample.

B Estimation Details and Results

This appendix provides full details for the two-step estimation procedure summarised in Section 5.

B.1 Bid Homogenization

Raw bid amounts b_{istc} (bid by bidder i in auction season s for tract t in cluster c) vary multiplicatively with observable tract heterogeneity. To isolate the private-value component, I implement a two-step log-linear homogenization procedure. The key identifying assumption is an *equilibrium translational invariance* property: a uniform shift in the game-level heterogeneity Z shifts all equilibrium actions by the same amount, leaving the mapping between markup values and markup actions unchanged . Formal justification is given in Proposition 3 below, and a proof is in Appendix B.2.

I first verify that a multiplicative (log-linear) specification is appropriate. Appendix B.11 compares residual diagnostic plots for the linear and log-linear specifications. In sum, the linear model exhibits severe heteroskedasticity (residual variance fans out with fitted values) while the log-linear model stabilizes variance across the distribution. The formal model-selection rationale is discussed in the Appendix.

Appendix B.11 presents the residual diagnostic plots confirming the suitability of the log-linear specification.

Homogenization regression. I estimate the following log-linear hedonic regression:

$$\begin{aligned}\log(b_{istc}) =& \alpha_s + \alpha_c \\ &+\beta_1 \,\text{size}_t + \beta_2 \,\mathbb{I}[\text{royalty}_t = 3/16] + \beta_3 \,\text{distance}_t \\ &+\beta_4 \,\text{spudAge}_t + \beta_5 \,\mathbb{I}[\text{spudAge missing}] + \varepsilon_{istc}\end{aligned} \tag{31}$$

where α_s and α_c are season and cluster fixed effects (clusters are defined in Appendix B.12); size_t is tract acreage; royalty_t is the royalty rate (1/6 or 3/16); distance_t is distance to the nearest well in miles; and spudAge_t is the number of years elapsed between the spud date of the nearest well and the auction date (with an indicator for missing spud dates). Table 14 reports the estimates; the average fixed effect (approximate constant) is 7.757, and all slope coefficients carry their expected signs.

The auction-specific scaling factor is

$$s_t \;=\; \exp\big(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t\big) \;=\; \exp(\widehat{\log b_{istc}} - \hat{\varepsilon}_{istc}), \tag{32}$$

i.e. the exponentiated fitted value from regression (31). Homogenized bids are then the exponentiated residuals:

$$\tilde{b}_{istc} \;=\; \exp(\hat{\varepsilon}_{istc}) \;=\; \frac{b_{istc}}{s_t}. \tag{33}$$

Because every bidder in the same auction faces the same X_t, all bids in auction t are divided by the same s_t and the *relative* ranking of bids within an auction is preserved, as is the within-auction positive dependence structure arising from unobservable common shocks (geological uncertainty, commodity prices).

Descaling the nomination fee and cost. The state's flat \$10 nomination fee, $\phi = \$10$, does *not* scale with s_t. It is charged per tract regardless of its acreage or location. In residual space, this fee becomes

$$\phi \;=\; \frac{10}{s_t}. \tag{34}$$

I assume learning costs for bidder i at tract t are largely *fixed in dollar terms* and do not scale the same way as value does to observables. For example, while value decreases in distance to nearby wells, the cost of geological survey or analysis of a nearby well stays constant or, if anything, increases with distance. The estimation maintains that $C_{it} \sim \text{SN}(\xi_c, \omega_c, \alpha_c)$ is drawn from a *common* distribution across all bins so that the bin-specific residual-space distribution is $\tilde{C}_{it} = C_{it}/s_b \sim \text{SN}(\tilde{\xi}_c, \tilde{\omega}_c, \tilde{\alpha}_c)$ where $s_b > 0$.

The per-bin equilibrium structure (Section B.6) partially captures the resulting cross-bin heterogeneity in effective cost barriers through the bin-specific fees $\tilde{\phi}_b = \phi/s_b$.

Dollar-space *values* scale linearly with s_t, meaning auctions over a more valuable tract has proportionally larger bids, competing maxima, and surplus, so $\Pi = s_t \cdot \tilde{\Pi}$. On the other hand, the flat fee ϕ, the reserve r, and the cost distribution F_C are fixed in dollar terms and therefore do not scale. Therefore, when expressed in residual units they become auction-specific objects $\tilde{\phi}_t = \phi/s_t$, $\tilde{r}_t = r/s_t$, and $\tilde{F}_{C,t}(\tilde{c}) = F_C(s_t\tilde{c})$. Under the Skew-Normal specification, $\tilde{F}_{C,t} = \text{SN}(\xi_c/s_t,\, \omega_c/s_t,\, \alpha_c)$. The cross-auction heterogeneity in residual-space equilibria is entirely driven by this asymmetry between scaling values s_t and non-scaling parameters ϕ, r and F_C parameters. The Poisson rates Λ_{nom} and Λ_{nn} are dimensionless counts and are unaffected.

This implies two things for estimation. First, auctions sharing the same residual-space triple $(\tilde{\phi}_t,\, \tilde{F}_{C,t},\, \tilde{r}_t)$ share the same residual-space equilibrium and may be pooled. Section B.6 groups auctions into 19 bins of s_t and solves one equilibrium per bin. Second, the dollar primitives $(\phi,\, r,\, \xi_c,\, \omega_c,\, \alpha_c)$ and the residual value distribution $F_{\tilde{V}}$, along with other structural primitives (the Poisson rates $(\lambda_{\text{pre}},\, \lambda_{\text{post}})$ and the copula parameter ρ) are estimated as common to all 728 auctions. Their bin-specific residual-space counterparts are recovered

by dividing by s_b. The formal proof is in Appendix B.2.

Table 14 reports the regression results used for bid homogenization. The dependent variable is log total bid $\log(b_{istc})$. The reported constant is the average of the absorbed cluster fixed effects. The auction-specific scaling factor is $s_t = \exp(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t)$. Size enters as a quadratic (size + size^2) to allow a concave bid-size relationship.

All slope coefficients carry their expected signs: the higher royalty rate (3/16 > 1/16) commands a premium; bids decline with distance to the nearest well and with the age of that well's spud date (older wells are less informative about current reservoir conditions); larger tracts attract higher total bids up to the size at which the quadratic term dominates. The spud-date missing indicator is not significant.

Table 14: Log-Bid Homogenization Regression Result

	Dependent variable:
	log(Total Bid)
Constant	7.757***
	(0.205)
Royalty Rate (3/16)	0.324***
	(0.099)
Distance to Nearest Well (miles)	−0.310***
	(0.044)
Size (acres)	0.013***
	(0.003)
Size^2 (acres^2)	−0.0000313**
	(0.0000141)
Spud Date Age (years)	−0.026***
	(0.002)
Observations	1,875

Notes: Bid homogenization regression (equation 31). Dependent variable is log(per-acre bid × acreage), i.e. log total bid in dollars. Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table 15 summarizes the homogenization result. The first row reports overall log bids before homogenization ($\log b_{istc}$); subsequent rows report exponentiated residuals $\tilde{b}_{istc}$ =

$\exp(\hat{\varepsilon}_{istc}) = b_{istc}/s_t$ after homogenization, separately for the estimation sample (auctions where the winner's bid is censored) and the verification sample (auctions where the winner's bid violates the proxy-bid rule and is therefore observed). Homogenization reduces the standard deviation from 1.77 to 1.58, reflecting the removal of observable heterogeneity. The remaining dispersion captures private-value variation and unobservable common shocks.

Table 15: Homogenized Bid (Residuals)

Variable	Average	Std Dev	Max	Min
Per Acre Bid (Log Before Homogenization)	3.25	1.77	9.80	0.693
After Homogenization (Overall)	0.02095452	1.583303	5.493611	-3.973427
For Estimation	0.0305	1.59	5.49	-3.97
For Verification	-0.434	1.39	3.35	-3.74

Notes: Summary statistics of homogenized bids (residuals from the regression in Table 14). Homogenization reduces the standard deviation from 1.77 to 1.58, isolating the private value component from observable heterogeneity.

B.2 Proposition: Residual-Space Reduction

Setup. Each auction t has scaling $s_t = \exp(\hat{\alpha}_s + \hat{\alpha}_c + \hat{\beta}' X_t)$ recovered from the regression in Appendix B.1. Under private values, dollar quantities scale as $v_{jt} = s_t \tilde{v}_{jt}$ and $V_{(n:n),t} = s_t \tilde{V}_{(n:n),t}$. The fee $\phi = \$10$, reserve $\$r$, and the dollar-cost distribution $F_C \sim \text{SN}(\xi_c, \omega_c, \alpha_c)$ are common to all auctions (none scales with s_t). Define

$$\tilde{\phi}_t := \phi/s_t, \qquad \tilde{r}_t := r/s_t, \qquad \tilde{F}_{C,t}(\tilde{c}) := F_C(s_t \tilde{c}).$$

Under the Skew-Normal, $\tilde{F}_{C,t} = \text{SN}(\xi_c/s_t,\, \omega_c/s_t,\, \alpha_c)$: the location and scale parameters depend on s_t, while the shape α_c is invariant. This follows from the following Lemma.

Lemma 2 (Skew-normal closure under scaling)**.** *Let $C \sim \text{SN}(\xi, \omega, \alpha)$ following the Azzalini parameterization [Azzalini, 2013, eq. 2.3], with density*

$$F_C(c) = \frac{2}{\omega}\, \varphi\left(\frac{c-\xi}{\omega}\right) \Phi\left(\alpha\, \frac{c-\xi}{\omega}\right),$$

where φ and Φ are the standard normal density and CDF. For any $s > 0$, the rescaled variable $\tilde{C} := C/s$ satisfies

$$\tilde{C} \sim \text{SN}(\xi/s,\, \omega/s,\, \alpha)\,.$$

The location and scale parameters rescale by $1/s$ while the shape parameter α is invariant.

Proof. By the standard change-of-variables formula, $\tilde{C} = C/s$ has density

$$f_{\tilde{C}}(\tilde{c}) \;=\; s \cdot F_C(s\tilde{c}) \;=\; \frac{2s}{\omega}\,\varphi\!\left(\frac{s\tilde{c}-\xi}{\omega}\right)\Phi\!\left(\alpha\,\frac{s\tilde{c}-\xi}{\omega}\right).$$

This can be rewritten as

$$f_{\tilde{C}}(\tilde{c}) \;=\; \frac{2}{\omega/s}\,\varphi\!\left(\frac{\tilde{c}-\xi/s}{\omega/s}\right)\Phi\!\left(\alpha\,\frac{\tilde{c}-\xi/s}{\omega/s}\right),$$

which is the density of $\mathrm{SN}(\xi/s,\, \omega/s,\, \alpha)$ by [Azzalini, 2013, eq. 2.3]. □

Surplus scaling. The Gaussian equicorrelation copula depends only on ranks, so the conditional density of the maximum competing value satisfies $f_{V_{(n:n)}|v}(s_t\tilde{y}) = s_t^{-1} f_{\tilde{V}_{(n:n)}|\tilde{v}}(\tilde{y})$. Substituting $y = s_t\tilde{y}$ into the dollar-space surplus integral yields

$$\Pi(s_t\tilde{v};\, n,\, \rho) = \int_{\tilde{r}_t}^{\tilde{v}} \underbrace{s_t(\tilde{v}-\tilde{y})}_{v-y}\,\underbrace{s_t^{-1} f_{\tilde{V}_{(n:n)}|\tilde{v}}(\tilde{y})}_{f_{V_{(n:n)}|v}(s_t\tilde{y})}\,\underbrace{s_t\,d\tilde{y}}_{dy} = s_t \cdot \tilde{\Pi}(\tilde{v};\, n,\, \rho), \tag{35}$$

where the residual-space surplus is $\tilde{\Pi}(\tilde{v};\, n,\, \rho) := \int_{\tilde{r}_t}^{\tilde{v}} (\tilde{v}-\tilde{y})\, f_{\tilde{V}_{(n:n)}|\tilde{v}}(\tilde{y})\, d\tilde{y}$. The Jacobian from dy cancels the Jacobian from the density, leaving a single factor of s_t from the integrand.

Nomination threshold scaling. Before turning to the value functions, I establish how the nomination threshold v^* transforms under the change of variables. Recall from (9) that $v^* = \min\{r,\, v^{**}\}$, where v^{**} satisfies the pointwise indifference condition

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \Pi(v^{**};\, m,\, \Lambda_{\mathrm{nn}},\, \rho) = \phi. \tag{36}$$

By the surplus scaling (35), $\Pi(s_t\tilde{v};\, m,\, \cdot) = s_t \cdot \tilde{\Pi}(\tilde{v};\, m,\, \cdot)$. Substituting $v^{**} = s_t\tilde{v}_0$ into (36) and dividing both sides by s_t:

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m)\, \tilde{\Pi}(\tilde{v}_0;\, m,\, \Lambda_{\mathrm{nn}},\, \rho) = \tilde{\phi}_t. \tag{37}$$

By definition, $\tilde{v}^{**} = \tilde{v}_0$, that is, $\tilde{v}_0 = v^{**}/s_t$ satisfies the residual-space pointwise indifference condition with $\tilde{\phi}_t = \phi/s_t$. Therefore, $\tilde{v}^{**} = v^{**}/s_t$.

Next, since $\tilde{r}_t = r/s_t$,

$$\tilde{v}_t^* := \min\{\tilde{r}_t,\, \tilde{v}^{**}\} = \min\{r/s_t,\, v^{**}/s_t\} = (1/s_t)\min\{r, v^{**}\} = v_t^*/s_t.$$

The nomination threshold scales as $\tilde{v}_t^* = v_t^*/s_t$.

Value-function scaling. With the threshold scaling established, the value functions transform as follows. The dollar-space early learner value is $U_{l=1}(c) = -c - \phi[1 - F_V(v^*)] + \sum_m \Pr(M_{-i} = m) S_{l=1}(m)$, where $S_{l=1}(m) = \int_{v^*}^{\bar{v}} \Pi(v;\, m + \Lambda,\, \rho)\, f_V(v)\, dv$. Substituting $v = s_t\tilde{v}$, $v^* = s_t\tilde{v}^*$, $f_V(v) = s_t^{-1} f_{\tilde{V}}(\tilde{v})$, and using the surplus scaling (35):

$$\begin{aligned} S_{l=1}(m) &= \int_{\tilde{v}^*}^{\bar{v}/s_t} \underbrace{s_t\, \tilde{\Pi}(\tilde{v};\, \cdot)}_{\Pi}\, \underbrace{s_t^{-1} f_{\tilde{V}}(\tilde{v})}_{f_V}\, \underbrace{s_t\, d\tilde{v}}_{dv} \\ &= s_t \int_{\tilde{v}^*}^{\bar{v}/s_t} \tilde{\Pi}(\tilde{v};\, \cdot)\, f_{\tilde{V}}(\tilde{v})\, d\tilde{v} = s_t \cdot \tilde{S}_{l=1}(m). \end{aligned} \tag{38}$$

Therefore

$$U_{l=1}(c) = s_t \left[-\underbrace{\tfrac{c}{s_t}}_{\tilde{c}_t} - \underbrace{\tfrac{\phi}{s_t}}_{\tilde{\phi}_t} [1 - F_{\tilde{V}}(\tilde{v}^*)] + \sum_m \Pr(M_{-i} = m)\, \tilde{S}_{l=1}(m) \right],$$

and dividing through by s_t defines the residual-space value function

$$\tilde{U}_{l=1}(\tilde{c}_t) \;:=\; \frac{U_{l=1}(c)}{s_t} \;=\; -\tilde{c}_t - \tilde{\phi}_t[1 - F_{\tilde{V}}(\tilde{v}^*)] + \sum_m \Pr(M_{-i} = m)\, \tilde{S}_{l=1}(m). \tag{39}$$

The identical calculation for the late-learning value yields

$$\tilde{U}_{l=2}(\tilde{c}_t) \;:=\; \frac{U_{l=2}(c)}{s_t} \;=\; \Pr(M \geq K) \Big[\mathbb{E}[\tilde{S}_{l=2}(m) \mid M \geq K] - \tilde{c}_t \Big]. \tag{40}$$

In sum,

$$U_l(s_t\tilde{c};\, \cdot) = s_t \cdot \tilde{U}_l(\tilde{c};\, \cdot), \qquad l \in \{1, 2\}. \tag{41}$$

Proposition 3 (Residual-Space Reduction)**.** *Fix auction t with scaling $s_t > 0$. The dollar-space equilibrium tuple $(v^*, c_L^*, c_H^*, \Lambda_{\text{nom}}, \Lambda_{\text{nn}})$ satisfies the equilibrium system $\mathcal{S}_t$ if and only if the candidate*

$$\tilde{v}^* := v^*/s_t, \quad \tilde{c}_L^* := c_L^*/s_t, \quad \tilde{c}_H^* := c_H^*/s_t,$$

$$\tilde{\Lambda}_{\text{nom}} := \Lambda_{\text{nom}}, \quad \tilde{\Lambda}_{\text{nn}} := \Lambda_{\text{nn}}$$

satisfies the residual-space system $\tilde{\mathcal{S}}_t$. In residual space, auctions differ only through the triple $(\tilde{\phi}_t, \tilde{F}_{C,t}, \tilde{r}_t)$; all other primitives $(\lambda_{\text{pre}}, \lambda_{\text{post}}, F_{\tilde{V}}, \rho, \alpha_c, T, K)$ are common across auctions.

Proof. $\mathcal{S}_t$ comprises the nomination threshold condition (1), the cost-cutoff indifference conditions (2)–(3), and the Poisson-rate consistency conditions (4)–(5):

$$\sum_{m=K-1}^{\infty} \Pr(M_{-i} = m;\, \Lambda_{\text{nom}})\, \Pi(v^{**};\, m,\, \Lambda_{\text{nn}},\, \rho) = \phi, \quad v^* = \max\{v^{**}, r\} \tag{1}$$

$$U_{l=1}(c_L^*) = U_{l=2}(c_L^*); \quad U_{l=2}(c_H^*) = 0 \tag{2,3}$$

$$\Lambda_{\text{nom}} = \lambda_{\text{pre}}\, F_C(c_L^*)[1 - F_V(v^*)];$$

$$\Lambda = \lambda_{\text{pre}} \left[F_C(c_H^*) - F_C(c_L^*)\right] [1 - F_{\text{post}}(r)] \;+\; \lambda_{\text{post}}\, F_C(c_H^*)\, [1 - F_{\text{post}}(r)] \tag{4,5}$$

The residual-space system $\tilde{\mathcal{S}}_t$ has the same form with all quantities tilded.

Verification of (1'). By the threshold-scaling derivation above, $v^{**} = s_t \tilde{v}^{**}$ satisfies (36) if and only if $\tilde{v}^{**}$ satisfies (37), the residual-space pointwise indifference condition. The corresponding nomination threshold is $\tilde{v}^* = \min\{\tilde{r}_t, \tilde{v}^{**}\} = v^*/s_t$.

Verification of (2'), (3'). By (41), $U_l(s_t \tilde{c}) = s_t \tilde{U}_l(\tilde{c})$ for $l \in \{1, 2\}$. Evaluating (2) at $c = s_t \tilde{c}_L^*$ and dividing by s_t yields $\tilde{U}_{l=1}(\tilde{c}_L^*) = \tilde{U}_{l=2}(\tilde{c}_L^*)$. Similarly, (3) at $c = s_t \tilde{c}_H^*$ divided by s_t gives $\tilde{U}_{l=2}(\tilde{c}_H^*) = 0$.

Verification of (4'), (5'). By definition of $F_{\tilde{C}}$ as the CDF of $\tilde{C} = C/s_t$,

$$F_{\tilde{C}}(\tilde{c}_L^*) = F_C(s_t \cdot c_L^*/s_t) = F_C(c_L^*); \qquad F_{\tilde{V}}(\tilde{v}^*) = F_V(v^*).$$

So the RHS of (4') equals the RHS of (4), which equals $\Lambda_{\text{nom}} = \tilde{\Lambda}_{\text{nom}}$ by construction. (5') follows analogously.

The converse direction follows by the same argument with $s_t \to 1/s_t$. □

B.3 Stage 2 Estimation: Value Distribution Recovery

For simplicity, this section will work with the following data structure. The data consists of R independent replicates of auctions, each containing I_r observations. Bids observed within each replicate r is denoted as $X_1^{(r)}, X_2^{(r)}, \ldots, X_{I_r}^{(r)} \sim F(x)$ and are positively correlated within auction through a symmetric Archimedean copula $C_{I_r}(\cdot; \rho)$. I assume bids are exchangeable within each auction, consistent with symmetric bidder equilibria.

Order statistics of the bids will be denoted as $X_{(1:I_r)}^{(r)} \le X_{(2:I_r)}^{(r)} \le \cdots \le X_{(I_r:I_r)}^{(r)}$. Often the highest order statistic $X_{(I_r:I_r)}^{(r)}$ is missing (censored). The goal is to first estimate the de-

pendence parameter ρ using censored data and pseudo-maximum likelihood (Section B.3.2). Second, I recover the marginal CDF $\hat{F}_V(x)$ in two steps: I invert the Inclusion-Exclusion Formula using $\hat{\rho}$ to obtain copula-adjusted quantile estimates for each observed bid, then fit a parametric Log-normal distribution to those estimates by least squares (Section B.3.2). This yields a smooth $\hat{F}_V$ defined on the full support, including the region below the minimum observed bid.

B.3.1 Copula Representation

By Sklar's theorem, the probability integral transforms $U_i^{(r)} = F(X_i^{(r)})$ are marginally Uniform$(0,1)$ but dependent through the copula $C_{I_r}(u_1, \ldots, u_{I_r}; \rho)$ where ρ is the copula parameter. The CDF of the k-th order statistic then follows from the Inclusion-Exclusion Formula, stated here in the form used for estimation. First define the key symmetry condition that makes the formula tractable.

Definition 1 (Exchangeable Copula)**.** An N-dimensional copula C_N is **exchangeable** if it is invariant under permutation of its arguments:

$$C_N(u_1, \ldots, u_N) = C_N(u_{\pi(1)}, \ldots, u_{\pi(N)})$$

for all $(u_1, \ldots, u_N) \in [0,1]^N$ and all permutations π.

In the bivariate case, exchangeability reduces to symmetry of the copula: $C(u,v) = C(v,u)$ for all $(u,v) \in [0,1]^2$, which holds if and only if the associated random variables are identically distributed and exchangeable [Nelsen, 2006, Theorem 2.7.4].

Bivariate Archimedean copulas, generated via $C(u,v) = \varphi^{-1}(\varphi(u) + \varphi(v))$, are always symmetric and hence exchangeable, since addition is commutative [Nelsen, 2006, Theorem 4.1.5].[37]

An N-dimensional Archimedean copula is constructed by the analogous extension $c_L^*(\mathbf{u}) = \varphi^{-1}(\varphi(u_1) + \cdots + \varphi(u_N))$, which is symmetric under permutations by the same argument. However, this extension yields a valid N-copula for all $N \geq 2$ if and only if φ^{-1} is completely monotonic on $[0, \infty)$ [Nelsen, 2006, Theorem 4.6.2]. The Gumbel and Clayton families used in this paper both satisfy this condition and are therefore valid and exchangeable in any dimension.

The Gaussian copula $C(u_1, \ldots, u_N; \Sigma) = \Phi_\Sigma(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_N))$, by contrast, is exchangeable if and only if all off-diagonal entries of the correlation matrix Σ are equal, i.e., $\Sigma_{ij} = \rho$ for all $i \neq j$ (the equicorrelated case). For a general Σ with unequal pairwise

[37] Bivariate Archimedean copulas are generated via $C(u,v) = \varphi^{-1}(\varphi(u) + \varphi(v))$. Swapping u and v leaves the sum $\varphi(u) + \varphi(v)$ unchanged.

correlations, the Gaussian copula is not exchangeable. For this reason, this paper focus on equicorrelated case of the Gaussian copula.

Theorem 1 (David and Nagaraja [2004], Section 5.3, formula (5.3.1)). *Let $Y_1, \ldots, Y_n$ have arbitrary joint CDF $F(y_1, \ldots, y_n)$ with common marginal F. Then:*

$$F_{r:n}(y) = \sum_{j=r}^{n} (-1)^{j-r} \binom{j-1}{r-1} H_{j:j}(y) \tag{42}$$

where $H_{j:j}(y)$ is the sum of the CDFs of the maximum over all $\binom{n}{j}$ subsets of size j. In the exchangeable case with copula C_n, all $\binom{n}{j}$ subsets share the same joint CDF, so $H_{j:j}(y) = \binom{n}{j} C_j(u, \ldots, u)$ where $u = F(y)$, giving:

$$F_{r:n}(y) = \sum_{j=r}^{n} (-1)^{j-r} \binom{j-1}{r-1} \binom{n}{j} C_j(u, \ldots, u) \tag{43}$$

which is eq. (1.2) of Maurer and Margolin [1976], who cite it back to David (1970), formula 5.5.3.

.[38]

Each term $C_{j+m}(u, \ldots, u)$ corresponds to a single evaluation of the copula diagonal $\varphi^{-1}((j+m)\varphi(u))$, which can be computed directly in closed form.

Validity. Exchangeability of C_N is *necessary*: without it, different subsets of size r yield different joint probabilities and the copula diagonal alone is insufficient to characterise the formula.

While the copula of order statistics differs from the copula of original unordered bids, for Archimedean families the Inclusion-Exclusion formula depends only on diagonal copula values, which are preserved under order statistics transformations. This allows us to estimate the dependence structure from observed losers and extrapolate to the unobserved winner.

B.3.2 Joint GMM Estimation

The two unknowns in the copula estimation step are the dependence parameter ρ and the marginal value CDF F_V. I estimate both *simultaneously* using what I call the Joint GMM

[38] For comparison, in the i.i.d. case the copula is the independence copula, so $C_j(u, \ldots, u) = u^j = [F(x)]^j$, and the formula reduces to

$$F_{r:n}(x) = \sum_{j=r}^{n} (-1)^{j-r} \binom{j-1}{r-1} \binom{n}{j} [F(x)]^j,$$

which is formula (3.4.3) of Arnold et al. [2008] , p. 46. The exchangeable copula formula in Theorem 1 thus nests the i.i.d. case as a special case.

Strategy: a minimum-distance estimator that matches the theoretical copula order-statistic CDF to empirical Weibull ranks at every observed bid. I perform 100-fold random cross-validation, partitioning auctions (not individual bids) into 90% training and 10% test sets, fitting each copula specification on the training auctions and scoring predictive fit on the held-out auctions. Out-of-sample root mean squared error (RMSE) on the held-out auctions to select among the Gumbel, Clayton, Gaussian, and Independence specifications.

B-spline parameterisation of F_V. I parameterise F_V as a monotone B-spline

$$\hat{F}_V(x;\,\beta) = B(x;\,\beta), \tag{44}$$

with $K = 10$ internal knots placed at quantiles of the observed bids and polynomial degree 3. Monotonicity is enforced by reparameterising the coefficient vector as cumulative softplus steps: $\beta_j = \sum_{\ell \le j} \log(1+e^{\phi_\ell})/(1+\epsilon)$, so the free parameters ϕ are unconstrained while $B(\cdot;\beta)$ is guaranteed non-decreasing and maps into $[0,1]$. The spline basis is built on all training bids; it is identified over the observed bid support $[v^*,\,\bar{v}]$ but not outside it.

Joint sum of squared residuals (SSR) objective. Let $\hat{U}^{(r)}_{(k)} := \frac{n_{k,I_r} - r_{k,I_r}(X^{(r)}_{(k:I_r)}) + 1}{n_{k,I_r}+1}$ denote the Weibull empirical rank of bid $X^{(r)}_{(k:I_r)}$ among all position-k bids in I_r-bidder auctions, and let $G_k(u;\,k,I,\rho)$ be the copula order-statistic CDF from Equation 43. The joint estimator solves

$$(\hat{\phi},\,\hat{\rho}) = \arg\min_{\phi,\,\rho} \sum_{i \in \mathcal{G}_s} \Big[G\big(B(x_i;\beta(\phi));\; k_i,\; I_i,\; \rho\big) - \hat{U}^{(r_i)}_{(k_i)} \Big]^2, \tag{45}$$

where $G(u;\,k,I,\rho)$ is the CDF of the k-th order statistic from an I-sample exchangeable copula (Equation 43), evaluated at $u = B(x_i;\beta(\phi))$, the B-spline CDF at bid x_i. $\mathcal{G}_s \subseteq \mathcal{I}^{\text{train}}_s$ is a 200-point moment grid constructed from training bids in a given train/test split s by mapping an evenly-spaced target grid over $[x^s_{\min},\,x^s_{\max}]$ to the nearest observed training bids, with $\mathcal{I}^{\text{train}}_s$ indexing all bids in the training set of split s. Using 200 grid points rather than all $\sim$1,500 training bids reduces computation by approximately 7$\times$ per optimizer call while preserving dense coverage of the bid support. The B-spline basis itself is built on all training bids, and only the SSR evaluation uses the grid. Equation (45) is minimised via L-BFGS-B with multiple starting values for ρ.

Lognormal parametric tail. The B-spline $B(x;\hat{\beta})$ is identified only for $x \in [v^*,\,\bar{v}]$. For Stage 1 I need $\hat{F}_V$ on the full support $(0,\infty)$—in particular the scalar $1 - \hat{F}_V(v^*)$ and the conditional density below v^*. I therefore fit a Lognormal$(\hat{\mu},\,\hat{\sigma})$ by nonlinear least squares to

the spline values:

$$(\hat{\mu},\, \hat{\sigma}) = \arg\min_{\mu,\sigma} \sum_{i=1}^{T} \Big[\Phi\Big(\tfrac{\log x_i - \mu}{\sigma}\Big) - B(x_i; \hat{\beta})\Big]^2, \tag{46}$$

where T is the total number of training bids. Because bids are strictly positive and the B-spline closely tracts a lognormal shape over the observed support, the nonlinear least squares (NLS) fit is tight; the lognormal then extrapolates smoothly below v^*. The scalar $1-\hat{F}_V(v^*) = 1-\Phi\big((\log v^* - \hat{\mu})/\hat{\sigma}\big)$ and the full $(\hat{F}_V, \hat{f}_V)$ grid are saved for use in the Stage 1 equilibrium computation.

Copula family selection. I estimate Equation (45) for four copula families—Gaussian, Gumbel, Clayton, and Independence—and select by hold-out RMSE. For each of 100 random 90/10 train/test splits, the test-set RMSE is computed by evaluating the copula diagonal $D(\hat{F}_V(x_{\max});\, I_r{-}1,\, \hat{\rho})$ for each test-auction maximum bid $x_{\max}$ in an I_r-bidder auction and comparing to its empirical Weibull rank.

Copula	$\hat{\rho}$ (mean $\pm$ SD)	**RMSE** (mean $\pm$ SD)
Gaussian	**0.866 $\pm$ 0.096**	**0.183 $\pm$ 0.037**
Gumbel	2.404 ± 0.661	0.210 ± 0.036
Clayton	17.31 ± 4.71	0.212 ± 0.066
Independence	—	0.364 ± 0.043

Table 16: Joint GMM copula selection: hold-out RMSE across 100 random 90/10 splits (728 unviolated auctions, $I_r \in \{2, \ldots, 11\}$, 200-point moment grid). Mean $\pm$ SD across splits. Bold row is the selected specification. Full-data fit (Gaussian): $\hat{\rho} = 0.8605$; the lognormal tail, refined by truncation-corrected MLE, has $\hat{\mu} = 0.032$, $\hat{\sigma} = 1.614$ (NLS stage: -0.715, 3.863; see Appendix B.3).

The Gaussian copula achieves the lowest RMSE (0.128 ± 0.072). Independence is clearly dominated (RMSE = 0.299, 2.3× larger), confirming that positive dependence is a significant feature of the bid distribution. Clayton is near its upper boundary ($\hat{\rho} = 12.51 \pm 6.74$), indicating instability rather than well-identified upper-tail dependence; Gumbel is close but inferior (0.160 vs. 0.128). I therefore select the Gaussian equicorrelation copula $C_n(u_1, \ldots, u_n; \rho) = \Phi_n(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_n); \rho)$ as the preferred specification. Instead of using the mean (0.866), I use the median across the splits (0.861)).[39][40]

[39] The wide spread reflects that ρ is identified by cross-auction-size variation. Appendix D.9 reports subsample checks restricting to $I_r = 3$ and $I_r \in \{3, 4\}$: in both cases all copula families produce identical hold-out RMSE (within 0.001) and parameters collapse to the independence boundary, confirming that the dependence parameter is not identified without sufficient variation in I_r. Auctions with $I_r = 2$ are excluded from these checks because a single non-winner bid contributes no information about ρ, since the copula diagonal evaluates to $D(u, 1, \rho) = u$ for any copula and any ρ for $I_r = 2$ case.

[40] Monte Carlo evidence further justifies using the median: Appendix D.9 documents that the distribution

Economic rationale for the Gaussian copula. The empirical selection aligns with the theoretical prior imposed by the correlated private-values (CPV) assumption. In a common-value setting, bidders effectively compete for the same underlying quantity (e.g., total recoverable reserves), so a very high value for one bidder signals that all bidders have very high values—a common signal drives valuations toward the same extreme. The positive correlation setting studied here is different: each bidder forms a genuinely private assessment shaped by their own capital costs, extraction technology, and interpretation of local geological data. Bidder values are positively correlated because all bidders face the same regional geological uncertainty and energy market conditions, but this shared uncertainty acts as a common factor that shifts the *level* of the entire value distribution rather than compressing it toward a single realization.

The Gaussian equicorrelation copula is precisely the copula implied by this one-factor structure. Under the representation $V_i = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \varepsilon_i$, the common factor W captures shared market and geological conditions, while the idiosyncratic term ε_i preserves the private character of each value. A key implication is that the Gaussian copula has *zero upper tail dependence*. Observing one bidder has an extreme value provides no asymptotic information that another bidder also has an extreme value.

By contrast, the Gumbel or Clayton copula has positive tail dependence, implying that extreme valuations cluster together across bidders. That the data favor the Gaussian over the Gumbel or Clayton is therefore consistent with the APV structure: common geological and market uncertainty creates equicorrelation but not extreme-value clustering.

Lognormal tail fit and structural scalars. Fitting Equation (46) to the full training data under the selected Gaussian copula yields a first-stage NLS fit $(\hat{\mu}, \hat{\sigma}) = (-0.715,\ 3.863)$. This NLS projection matches the spline in shape but is not a likelihood estimate of the latent value distribution, and it is sensitive to the behavior of the spline near the lower support boundary. I therefore refine it with a *truncation-corrected lognormal MLE*. Inverting the estimated Gaussian copula at the observed order statistics recovers a sample of latent value draws $\{x_i\}_{i=1}^n$. Because values below the bid-support boundary $v^* = p_{\min}$ are never observed, these draws are left-truncated at v^*. I maximize the corresponding truncation-corrected log-likelihood

$$\ell^*(\mu, \sigma) = \sum_{i=1}^{n} \log\Big[\tfrac{1}{x_i \sigma}\, \phi\big(\tfrac{\log x_i - \mu}{\sigma}\big)\Big] \;-\; n\, \log\Big[1 - \Phi\big(\tfrac{\log v^* - \mu}{\sigma}\big)\Big], \tag{47}$$

of $\hat{\rho}$ across replications is right-skewed for Gumbel and Clayton families at moderate sample sizes, so the median provides a less biased summary of the typical single-run estimate than the mean.

where the second term rescales the density by the mass above the truncation point v^*. Maximizing (47) gives the final tail parameters $(\hat{\mu}, \hat{\sigma}) = (0.032, 1.614)$, which are the values used in all downstream structural computations.

The lower boundary of the observed bid support is $v^* = 0.000291$, and the fitted lognormal assigns probability essentially one to bids above this threshold ($1 - \hat{F}_V(v^*) = \Phi\big((\log v^* - \hat{\mu})/\hat{\sigma}\big) \approx 0.9999998$) indicating that v^* lies deep in the left tail of the estimated value distribution. Economically, this means virtually all bidder types draw values strictly above the cost threshold, so selection into bidding is not a quantitatively important margin in this sample. The final MLE pair $(\hat{\mu}, \hat{\sigma})$ and the scalar $1 - \hat{F}_V(v^*)$ are passed directly to the Stage 1 equilibrium computation.

Corroboration: pairwise pseudo-MLE. As a cross-check, Table 17 reports results from a two-step estimator: pairwise pseudo-MLE on position-based pseudo-observations (Weibull formula) to estimate $\hat{\rho}$, followed by copula inversion and lognormal WLS to recover F_V. The Gaussian copula again achieves the lowest hold-out RMSE (0.131), with $\hat{\rho} = 0.543$. The agreement between the joint GMM estimate ($\hat{\rho} = 0.861$) and the two-step estimate ($\hat{\rho} = 0.543$) on the same copula family strengthens confidence in the specification choice.[41]

Method	$\hat{\rho}$ / $\hat{\rho}$	**MAE**	**RMSE**
Gumbel (pairwise)	1.5732	0.1214	0.1506
Clayton (pairwise)	0.7581	0.1288	0.1613
Clayton (n-dimensional)	0.6462	0.1342	0.1680
Independence		0.1470	0.1789
Gaussian (pairwise)	**0.5425**	**0.1068**	**0.1312**

Table 17: Pairwise pseudo-MLE comparison by hold-out MAE and RMSE on unviolated auctions (100 random 90/10 splits; position-based ranking, Weibull formula). MAE and RMSE are computed on lognormal CDF predictions. Bold row indicates the specification also selected by the Joint GMM Strategy (Table 16).

Copula Parameter Consistency. A key identification assumption is that the copula parameter ρ governing the dependence among ranked losers also governs the distribution of values of all bidders.

Formally, if the joint distribution of all I_r order statistics is

$$(X^{(r)}_{(1:I_r)}, \ldots, X^{(r)}_{(I_r:I_r)}) \sim C_{I_r}(\cdot;\, \rho)$$

[41] The two estimates differ because the joint GMM identifies ρ through the order-statistic CDF evaluated at the winner bid, while pairwise pseudo-MLE uses only loser bids. Both are consistent estimators of the same population parameter under the maintained copula model.

then the marginal distribution of losers

$$(X^{(r)}_{(1:I_r)}, \ldots, X^{(r)}_{(I_r-1:I_r)}) \sim C^{\text{marginal}}_{I_r-1}(\cdot;\, \rho)$$

has the **same parameter** ρ for the Gaussian equicorrelation copula, since marginalising an equicorrelation Gaussian simply reduces the dimension while leaving ρ unchanged. This allows the joint GMM estimator to use all loser bids to identify ρ and then predict the censored winner bid using the same $\hat{\rho}$.

B.4 Computing the Conditional Density of the Maximum Order Statistic

The main challenge is computing the conditional density $f_{V_{(n:n)}|v}(y|v,\rho)$. This reflects how knowing the bidder's own value v affects her beliefs about the distribution of competitors' maximum value. Under positive dependence, these variables are not independent: a higher value v suggests competitors are also more likely to have higher values. I first derive the unconditional distribution of the maximum, then compute its density, and finally obtaining the conditional density given the bidder's value. I focus my main results on the **Gaussian copula** specification ($\hat{\rho} = 0.543$ from pairwise pseudo-MLE; $\hat{\rho} = 0.861$ from the robustness check), which achieves the lowest hold-out RMSE in my copula estimation procedure (Table 17), though I implement the Clayton and Gumbel copulas for comparison in Appendix D.9 as well.

Distribution of the Maximum Under Positive Dependence From Equation 43, setting $k = n$, the cumulative distribution function of the maximum order statistic $V_{(n:n)} = \max\{V_1, \ldots, V_n\}$ among n competitors is given by

$$F_{V_{(n:n)}}(y) = C_n(F(y), \ldots, F(y); \rho) \tag{48}$$

I use the n-dimensional Gaussian equicorrelation copula for C_n. $F(\cdot)$ is the marginal CDF of a single bidder's value and $\hat{F}(\cdot)$ estimated from the copula-inversion procedure (Section B.3.2) will be used. For the Gaussian copula, the diagonal evaluates to

$$C_n(u, \ldots, u;\, \rho) = \Phi_n\big(\Phi^{-1}(u), \ldots, \Phi^{-1}(u);\, \rho\big), \tag{49}$$

which is computed via one-factor Gauss-Hermite quadrature (28 nodes) for $n \geq 3$ and the bivariate normal CDF for $n = 2$. The conditional density simulation uses the one-factor

representation $Z_j = \sqrt{\hat{\rho}}\, W + \sqrt{1-\hat{\rho}}\, \varepsilon_j$ as described in Section D.10.

Density of the Maximum Differentiating the CDF yields the unconditional probability density function:

$$f_{V_{(n:n)}}(y) = \frac{d}{dy} F_{V_{(n:n)}}(y) = n^{1/\rho} \cdot [F(y)]^{n^{1/\rho}-1} \cdot f(y) \tag{50}$$

$f(\cdot)$ is estimated by applying the central difference method on $\hat{F}(\cdot)$ estimated with the procedure in Appendix B.3.[42]

Conditional Density Given Bidder's Value Finally, the conditional density $f_{V_{(n:n)}|v}(y|v,\rho)$ is derived using the Bayes rule ($f_{V_{(n:n)}|v}(y|v,\rho) = \frac{f_{V,V_{(n:n)}}(v,y;\rho)}{f_V(v)} = \frac{f_{V,V_{(n:n)}}(v,y;\rho)}{f(v)}$). This calls for specifying $f_{V,V_{(n:n)}}(v,y;\rho)$, the joint density of one's bid and the maximum of competitors' bid. The $(n+1)$-dimension joint distribution of $(V, V_1, ..., V_n)$, where V is the given bidder's value and the remaining n arguments are that of the competitors, can be characterized as

$$f_{V,V_1,\ldots,V_n}(v,v_1,\ldots,v_n;\rho) = \frac{\partial^{n+1} F_{V,V_1,\ldots,V_n}}{\partial v\, \partial v_1\, \partial v_2 \cdots \partial v_n} = c_{n+1}(F(v),F(v_1),\ldots,F(v_n);\rho)\cdot f(v)\cdot \prod_{i=1}^{n} f(v_i) \tag{51}$$

where c_{n+1} is the $(n+1)$-dimensional copula density.[43]

I implement Monte Carlo simulation to estimate the conditional density directly from the full $(n+1)$-dimensional Gaussian equicorrelation copula with parameter $\hat{\rho} = 0.738$. The simulation uses the one-factor representation $Z_j = \sqrt{\hat{\rho}}\, W + \sqrt{1-\hat{\rho}}\, \varepsilon_j$, where W and ε_j are i.i.d. $N(0,1)$, and $U_j = \Phi(Z_j)$. I draw $N_{\text{sim}} = 100{,}000$ such $(n+1)$-dimensional vectors. I transform competitor uniforms $U_1, \ldots, U_n$ using the truncated inverse CDF $F^{-1}_{V|\text{trunc}}(u) = F_V^{-1}(F_V(v^*) + u \cdot (1 - F_V(v^*)))$, which maps $[0,1]$ to $[v^*, \infty)$, reflecting that all auction participants had $v \geq v^*$. The own value U_0 is transformed using the full inverse CDF F_V^{-1}. I compute the maximum competitor value $V_{(n:n)} = \max\{V_1, \ldots, V_n\}$ for each draw, retain draws where $|U_0 - F_V(v)| < \varepsilon$ (tolerance $\varepsilon = 0.02 \cdot \max(F_V(v), 1 - F_V(v))$, doubled if fewer than 100 draws survive), and estimate the conditional density from the retained draws using Gaussian KDE with Silverman bandwidth, normalized to integrate to one over $[r, v]$. More details can be found in Section D.10.

[42] $f(x) \approx \frac{F(x+h)-F(x-h)}{2h}$

[43] $F_{V,V_1,\ldots,V_n}(v,v_1,\ldots,v_n;\rho) = C_{n+1}(F(v),F(v_1),\ldots,F(v_n);\rho)$, where C_{n+1} is a $(n+1)$-dimensional copula and $F(\cdot)$ is the common marginal CDF.

B.5 Stage 1 Estimation: GMM Moments and Identification

Having estimated $(\hat{F}, \hat{\rho})$ from Stage 2, I now estimate the nomination stage parameters, arrival rates λ_{pre} (pre-auction period) and λ_{post} (post-auction period) and the cost distribution parameter θ_c from $F_C(c; \boldsymbol{\theta}_c)$. These parameters determine the equilibrium nomination probabilities $(\pi_0^{l=1}, \pi_0^{l=2})$ through the fixed point system characterized in Section 4.4. I estimate them using generalized method of moments (GMM), matching model predictions to five data moments.

I specify the learning cost c_i to be drawn i.i.d. from a Skew-Normal distribution $c_i \sim \text{SN}(\xi_c, \omega_c, \alpha_c)$, with CDF $F_C(c; \xi_c, \omega_c, \alpha_c) = \Phi\left(\frac{c-\xi_c}{\omega_c}\right) - 2\,T\left(\frac{c-\xi_c}{\omega_c}, \alpha_c\right)$, where $T(\cdot,\cdot)$ is Owen's T-function, ξ_c is the location, $\omega_c > 0$ is the scale, and α_c controls skewness. The Skew-Normal nests the symmetric Normal ($\alpha_c = 0$) and allows the right-skewed cost distributions suggested by the low observed nomination rates. The cost c_i is specified and estimated in residual units, exactly as the \$10 nomination fee is descaled to $\tilde{\phi}_t = 10/s_t$ before entering the model, the learning cost is the descaled cost; the actual dollar cost for a bidder i at auction t is $c_{it} \cdot s_t$. This means F_C is a single distribution invariant to observable auction characteristics: it captures the residual-space cost after hedonic scaling has been absorbed. In the binned estimator there is no separate cost distribution per bin. The same $\text{SN}(\xi_c, \omega_c, \alpha_c)$ applies uniformly across all 19 bins, and all cross-bin heterogeneity in the equilibrium cutoffs (c^*_{Lb}, c^*_{Hb}) is driven entirely by the bin-specific nomination fees ϕ_b, not by different cost distributions.

B.5.1 Equilibrium Objects

For any candidate parameter vector $\boldsymbol{\psi}$, the inner fixed point loop solves for the equilibrium tuple $(c^*_L, c^*_H, \pi_0^{l=1}, \pi_0^{l=2})$ satisfying:

$$\pi_0^{l=1} = F_C(c^*_L; \alpha, \beta) \tag{52}$$

$$\pi_0^{l=2} = F_C(c^*_H; \alpha, \beta) - F_C(c^*_L; \alpha, \beta) \tag{53}$$

where c^*_L solves $U_{l=1}(c^*_L) = U_{l=2}(c^*_L)$ and c^*_H solves $U_{l=2}(c^*_H) = 0$.

Given these equilibrium objects, define the Poisson rate governing the number of nominations per tract:

$$\Lambda_{\text{nom}} := \lambda_{pre} \cdot \pi_0^{l=1} \cdot [1 - \hat{F}(v^*)] \tag{54}$$

Note that Λ_{nom} depends on $(\lambda_{pre}, \alpha, \beta)$ through $\pi_0^{l=1} = F_C(c^*_L; \alpha, \beta)$, where the equilibrium cutoff c^*_L itself depends on the full competition structure including $\Lambda^{l=2}$ and $\Lambda^{d=1}$.

The non-nominator and post-arrival competition rates are:

$$\Lambda^{l=2} := \lambda_{pre} \cdot \pi_0^{l=2} \cdot [1 - \hat{F}(r)] \tag{55}$$

$$\Lambda^{d=1} := \lambda_{post} \cdot (\pi_0^{l=1} + \pi_0^{l=2}) \cdot [1 - \hat{F}(r)] \tag{56}$$

A key structural feature is that the pre-arrival pool splits into nominators and non-nominators via *independent thinnings* of the same Poisson process. By the independence of Poisson thinnings, M and the non-nominator pool are independent random variables even though both originate from the same arrival process, because they represent disjoint outcomes for each arriving bidder. Post-nomination arrivals are exogenous and independent of M by assumption. Crucially, post-nomination arrivals face the same cost-screening decision as non-nominators: each must pay the learning cost c_i before deciding whether to bid. Since the auction surplus available to a post-period arrival is identical to that of a non-nominator (both observe the same nominators and compete in the same auction), their cost cutoff is c_H^*, identical to that of a non-nominator. Only the fraction $F_C(c_H^*; \xi_c, \omega_c, \alpha_c)$ of post-nomination arrivals find it worthwhile to pay the learning cost and enter the auction.[44]

This independence means that conditional on the auction triggering ($M \geq K$), the conditional mean of total auction participants decomposes as:

$$\mathbb{E}[I \mid M \geq K] = \mathbb{E}[M \mid M \geq K] + \Lambda^{l=2} + \Lambda^{d=1} \tag{57}$$

where $\mathbb{E}[M \mid M \geq K] = \sum_{m=K}^{\infty} m \cdot \Pr(M = m) / \Pr(M \geq K)$, which for $K = 1$ simplifies to the zero-truncated Poisson mean $\Lambda_{\text{nom}}/(1 - e^{-\Lambda_{\text{nom}}})$ [45].

B.5.2 Identification Strategy

I exploit four moment conditions per bin, organized into two groups: three moments from the nomination process—Moment 1 (the auction trigger rate), Moment 2 (the mean number of nominations), and Moment 3 (the variance of nominations)—and one moment from bidder counts, Moment 4 (the mean number of bidders).

Moment 1: Auction Trigger Rate. The trigger rate in bin b is defined using the full universe of available tracts—not just the estimation sample—so that the denominator

[44] See Remark 1 for the formal argument.

[45] The naive approximation $\mathbb{E}[I \mid M \geq K] \approx \Lambda_{\text{nom}} + \Lambda^{l=2} + \Lambda^{d=1}$ is incorrect because $\mathbb{E}[M \mid M \geq K] > \Lambda_{\text{nom}}$ for any finite Λ_{nom}, so ignoring the truncation systematically understates competition.

correctly reflects the population of tracts that *could* have been nominated:

$$\bar{p}_{\text{trigger},b} = \frac{1}{|S|} \sum_{s \in S} \frac{\text{tracts auctioned in bin } b \text{ during season } s}{\text{tracts available for auction in bin } b \text{ during season } s},$$

where S is the set of auction seasons in the sample. A tract is "available" in season s if it has not been leased within the prior five years and its current lease (if any) has expired by the auction date. Averaging over seasons reduces noise from season-specific variation in tract supply. I use the full tract universe provided through the ND Department of Trust Lands, with the same observable characteristics of tracts (size, royalty rate and nearest-well information) used to predict s_t (and hence $\phi_b = 10/s_t$) for each available tract, so that tracts outiside the estimation sample can be assigned to the closest bin.

Under the model, $M_t \sim \text{Poisson}(\Lambda_{\text{nom},b})$, so the bin-$b$ trigger probability is:

$$p_{\text{trigger},b}(\boldsymbol{\psi}) = \Pr(M \geq K) = 1 - e^{-\Lambda_{\text{nom},b}} \quad (K = 1).$$

The moment condition matched in log scale is:[46]

$$g_{1,b}(\boldsymbol{\psi}) = \log \bar{p}_{\text{trigger},b} - \log p_{\text{trigger},b}(\boldsymbol{\psi}) = 0. \tag{58}$$

Moment 2: Mean Nominations per Nominated Tract. Among tracts that received at least one nomination, the average number of nominations provides additional information about the nomination rate:

$$\bar{m}_{\text{nom}} = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} M_t$$

where $\mathcal{A}$ is the set of tracts with $M_t \geq K = 1$. Under the model, conditional on $M \geq 1$, the number of nominations follows a zero-truncated Poisson distribution:

$$\mathbb{E}[M \mid M \geq 1] = \frac{\Lambda_{\text{nom}}}{1 - e^{-\Lambda_{\text{nom}}}}$$

The moment condition is:

$$g_2(\boldsymbol{\psi}) = \bar{m}_{\text{nom}} - \frac{\Lambda_{\text{nom}}(\boldsymbol{\psi})}{1 - e^{-\Lambda_{\text{nom}}(\boldsymbol{\psi})}} = 0$$

Identification from Moments 1–2. Although Moments 1–2 are all functions of the single Poisson rate Λ_{nom}, the key insight is that $\Lambda_{\text{nom}} = \lambda_{pre} \cdot F_C(c_L^*) \cdot [1 - \hat{F}(v^*)]$ is not a free parameter but a composite function of (λ_{pre}, F_C). The equilibrium cutoff c_L^* depends on F_C

[46] Moment 1 is matched in log scale since the trigger rate is a small positive quantity.

through the fixed point system. In other words, different cost distributions generate different equilibrium nomination probabilities $\pi_0^{l=1} = F_C(c_L^*)$ and different late-learning rates $\pi_0^{l=2}$, which feed back into the competition structure and therefore into c_L^* itself. Consequently, Moments 1–2 collectively pin down Λ_{nom}.The equilibrium consistency conditions imposed in the inner loop create the nonlinear mapping that separates λ_{pre} from F_C.

Moment 3: Variance of Nominations per Nominated Tract. The variance of nomination counts among nominated tracts provides additional information about the shape of the nomination distribution:

$$\widehat{\text{Var}}_{\text{nom}} = \frac{1}{|\mathcal{A}| - 1} \sum_{t \in \mathcal{A}} (M_t - \bar{m}_{\text{nom}})^2 .$$

Under the zero-truncated Poisson,

$$\text{Var}(M \mid M \geq 1) = \frac{\Lambda_{\text{nom}}}{1 - e^{-\Lambda_{\text{nom}}}} \left(1 - \frac{\Lambda_{\text{nom}} \cdot e^{-\Lambda_{\text{nom}}}}{1 - e^{-\Lambda_{\text{nom}}}}\right),$$

and the moment condition is

$$g_3(\boldsymbol{\psi}) = \widehat{\text{Var}}_{\text{nom}} - \text{Var}(M \mid M \geq 1; \boldsymbol{\psi}) = 0. \tag{59}$$

Because Moments 1–2 already pin Λ_{nom} under the Poisson structure, Moment 3 is not independently identifying, but it is retained as an active moment (unit weight) that disciplines the concentration of the nomination distribution.

Moment 4: Mean Number of Bidders per Triggered Auction. The average number of total bidders per triggered auction from data ($\bar{I}_{\text{bid}} = \frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} I_t$) identifies the overall level of competition via equation (57):

$$g_4(\boldsymbol{\psi}) = \bar{I}_{\text{bid}} - \left[\mathbb{E}[M \mid M \geq K] + (\Lambda^{\text{pio,bid}} - \Lambda_{\text{nom}}) + \Lambda^{l=2} + \Lambda^{d=1}\right] = 0 \tag{60}$$

where $\mathbb{E}[M \mid M \geq K] = \left(\Lambda_{\text{nom}} - \sum_{m=0}^{K-1} m \cdot \Pr(M = m)\right) / \Pr(M \geq K)$, and $\Lambda^{\text{pio,bid}} - \Lambda_{\text{nom}}$ is the expected count of early learners who bid without nominating ($r \leq v < v^*$). This moment brings in λ_{post} through $\Lambda^{d=1}$ and $\pi_0^{l=2}$ through $\Lambda^{l=2}$. In the absence of the coverage moment, however, it identifies chiefly the *product* of arrival rate and entry probability, $\Lambda_b = \lambda \, \pi_b(F_C)$; the level of λ_{post} and the cost scale are therefore only weakly separated (see the identification remark below).

Identification of the Arrival-Rate Level. Moments 1–4 identify the *products* $\Lambda_b = \lambda\,\pi_b(F_C)$ of arrival rate and entry probability. Consequently a high-λ/high-cost-barrier configuration (small π_b) and a low-λ/low-cost configuration (large π_b) generate nearly identical Λ_b and fit these moments comparably. Therefore, the *level* of $(\lambda_{\text{pre}}, \lambda_{\text{post}})$ relative to the cost scale is only weakly identified; the estimates below should be read with this caveat

B.6 Binning: Handling Variation in the Residual-Space Nomination Fee

After homogenization, the flat \$10 nomination fee becomes $\phi = 10/s_t$, which varies across auctions because s_t reflects observable tract heterogeneity. This variation means the equilibrium indifference conditions depend on the auction: a bidder with value v^* indifferent between nominating and not nominating at a high-s_t auction faces a lower residual-space fee than at a low-s_t auction. Treating all auctions as sharing a single equilibrium would therefore be incorrect.

I address this by grouping the 728 estimation auctions into 19 bins of s_b, with breakpoints derived from the full available-tract universe, i.e. all tracts available for nomination across the eight nomination seasons (2017Q1–2018Q4), rather than from the estimation sample alone. Starting from fine quantile breaks on the universe distribution of s_t, adjacent bins are iteratively merged until each bin contains at least the required minimum number of estimation auctions and the bin-average trigger rate is weakly declining in s_b where s_b is the within-bin median:

$$s_b = \text{median}_{t \in b}\big(s_t\big). \tag{61}$$

.

The binned MPEC estimates bin-specific equilibrium objects $(v_b^*, c_{Lb}^*, c_{Hb}^*)$ for each of the 19 bins while keeping the cost distribution $\text{SN}(\xi_c, \omega_c, \alpha_c)$ shared. All cross-bin heterogeneity in equilibrium cutoffs is therefore driven entirely by the bin-specific fees ϕ_b, not by different cost distributions.

The reason for per-bin moment matching is because nomination fee and learning costs do not scale in the same way as bids. The auction-level scaling factor s_t from bid homogenization should be divided from the two. As a result, the residual-space nomination fee, reserve price and cost, ϕ/s_t, r/s_t and c_i/s_t, is systematically smaller for high-s_t bins (large, high-quality tracts) and larger for low-s_t bins (small or marginal tracts). Bins therefore face genuinely different effective cost barriers, and the observable moment patterns (trigger rates, bidder counts, FR ratios) differ across bins for this economic reason. Matching each bin to its own empirical moments allows the model to fit those cross-bin differences directly, rather than

averaging them away into a single global target.

The GMM objective stacks the per-bin moment residuals into a single vector,

$$g(\boldsymbol{z}) = \big(g_{1,1},\, g_{2,1},\, g_{3,1},\, g_{4,1},\, \ldots,\, g_{1,19},\, g_{2,19},\, g_{3,19},\, g_{4,19}\big)^{\top},$$

where $g_{k,b}$ denotes the moment $k \in \{1,2,3,4\}$ residual for bin b. Each of the 19 bins contributes these four moment conditions, resulting in 76 active moments. The weighting matrix W is diagonal, with unit weight on the trigger, mean-nomination, and variance residuals and weight 1.6 on the mean-bidder residual.

B.7 Robustness: Nomination Counts via $\Pr(M \geq 2 \mid M \geq 1)$

The baseline specification summarizes the nomination-count distribution through its trigger rate (Moment 1), conditional mean (Moment 2), and conditional variance (Moment 3), each of which is a function of the single Poisson rate Λ_{nom}. As a robustness check I re-estimate the model with an alternative characterization that keeps the trigger (Moment 1) but replaces the conditional mean and variance (Moments 2–3) with the conditional multiple-nomination share

$$\Pr(M \geq 2 \mid M \geq 1) = \frac{1 - (1 + \Lambda_{\text{nom}})\, e^{-\Lambda_{\text{nom}}}}{1 - e^{-\Lambda_{\text{nom}}}}, \tag{62}$$

matched in log scale.[47] This moment targets the frequency of *multiple* nominations directly rather than the first two conditional moments of the count.

The re-estimated arrival rates are essentially unchanged from the baseline, where the pre-period arrival rate remains small ($\hat{\lambda}_{\text{pre}} \approx 0.0003$, though below the baseline ≈ 0.005) and the post-period rate moderate ($\hat{\lambda}_{\text{post}} \approx 4.1$) ($\hat{\lambda}_{\text{pre}} \approx 0.0003$, $\hat{\lambda}_{\text{post}} \approx 4.1$). However the fit deteriorates sharply: the GMM objective rises to $\approx 2{,}166$, against ≈ 35 at the baseline. The reason is structural rather than numerical. Under a single Poisson rate, the trigger and the multiple-nomination share are both governed by Λ_{nom}, so matching the low observed trigger ($\Pr(M \geq 1) \approx 1\%$) forces $\Pr(M \geq 2 \mid M \geq 1) \approx \Lambda_{\text{nom}}/2 \approx 0.4$–$0.8\%$. In the data, however, the conditional multiple-nomination share averages roughly 7.7% (and reaches 16% in the lowest-cost bins)—an order of magnitude larger than any Poisson consistent with the observed trigger can produce. A single-rate Poisson nomination process is therefore rejected by this moment: conditional on being nominated, tracts draw two or more nominations far more often than independent Poisson arrivals allow. This overdispersion (clustering) in nominations points toward a mixed-Poisson / negative-binomial arrival extension, in which

[47] The complementary single-nomination share $\Pr(M = 1 \mid M \geq 1) = 1 - \Pr(M \geq 2 \mid M \geq 1)$ carries no additional information and is not matched separately.

a dispersion parameter separates the overall nomination rate from the frequency of multiple nominations. I retain the Poisson baseline for the reported results and leave the overdispersed extension to future work.

B.8 MPEC Implementation

The goal is to estimate the structural primitives $\boldsymbol{\psi} = (\lambda_{pre}, \lambda_{post}, \xi_c, \omega_c, \alpha_c)$ using a GMM objective. Rather than the standard nested fixed point (NFXP) approach [Rust, 1987], I adopt the constrained optimization approach of Su and Judd [2012] (MPEC).[48] I augment the parameter vector with equilibrium objects and impose equilibrium conditions as explicit constraints, rather than solving for equilibrium at each candidate parameter value. Specifically, I solve

$$\begin{aligned} \hat{\boldsymbol{z}} &= \arg\min_{\boldsymbol{z}} \; g(\boldsymbol{z})' W g(\boldsymbol{z}) \\ \text{where } \boldsymbol{z} &= \Big(\underbrace{\lambda_{\text{pre}},\, \lambda_{\text{post}},\, \xi_c,\, \omega_c,\, \alpha_c}_{\text{5 structural}},\; \underbrace{v_1^*, \ldots, v_{19}^*}_{\text{19 bin thresholds}},\; \underbrace{c_{L,1}^*, \ldots, c_{L,19}^*}_{\text{19 early learner cutoffs}},\; \underbrace{c_{H,1}^*, \ldots, c_{H,19}^*}_{\text{19 } l_i = 2 \text{ cutoffs}} \Big) \end{aligned} \tag{63}$$

The decision vector $\boldsymbol{z}$ comprises five shared structural parameters $\psi = (\lambda_{\text{pre}}, \lambda_{\text{post}}, \xi_c, \omega_c, \alpha_c)$ plus per-bin equilibrium objects $(v_b^*,\, c_{Lb}^*,\, c_{Hb}^*)$ for bins $b = 1, \ldots, 19$, 62 elements in total.

$$\begin{aligned} \pi_{0,b}^{l=1} &= F_C(c_{Lb}^*;\, \xi_c, \omega_c, \alpha_c) \\ \pi_{0,b}^{l=2} &= F_C(c_{Hb}^*;\, \xi_c, \omega_c, \alpha_c) - F_C(c_{Lb}^*;\, \xi_c, \omega_c, \alpha_c) \\ \pi_{0,b}^{\text{bid}} &= F_C(c_{Hb}^*;\, \xi_c, \omega_c, \alpha_c) = \pi_{0,b}^{l=1} + \pi_{0,b}^{l=2} \\ \Lambda_b^{\text{nom}} &= \lambda_{\text{pre}} \cdot \pi_{0,b}^{l=1} \cdot [1 - F_V(v_b^*)] \\ \Lambda_b^{\text{pio,bid}} &= \lambda_{\text{pre}} \cdot \pi_{0,b}^{l=1} \cdot [1 - F_V(r_b)] \\ \Lambda_b^{FR+\text{post}} &= \lambda_{\text{pre}} \cdot \pi_{0,b}^{l=2} \cdot [1 - F_{\text{post}}(r_b)] \\ &\quad + \lambda_{\text{post}} \cdot \pi_{0,b}^{\text{bid}} \cdot [1 - F_{\text{post}}(r_b)] \\ F_{\text{post}}(r_b) &= \sum_{m \geq K} F(r_b \mid m)\, \Pr(M = m \mid M \geq K) \end{aligned}$$

Learning is required for auction entry; conditional on learning, every type bids whenever $v \geq r$. Early learners (including both nominators and non-nominators) value the tract from the prior, so their reserve-clearing probability is $[1 - F_V(r)]$. Non-nominating entrants learn

[48] See Dubé et al. [2012] for its computational advantages when estimating equilibrium models, particularly when the inner fixed point is expensive or exhibits convergence difficulties.

only after observing that the auction triggered; by positive correlation their values follow the posterior conditional on $M \geq K$, so their reserve-clearing probability is $[1 - F_{\text{post}}(r_b)]$. Accordingly Λ_b^{nom} $(v \geq v_b^*)$ governs only the *trigger* probability, while the competition/bidding objects use $[1 - F_V(r_b)]$ for early learners ($\Lambda_b^{\text{pio,bid}}$, which includes the early learning non-nominators $\Lambda_b^{\text{pio,bid}} - \Lambda_b^{\text{nom}}$) and $[1 - F_{\text{post}}(r_b)]$ for late learners ($\Lambda_b^{FR+\text{post}}$). The post-period term includes $\pi_{0,b}^{\text{bid}} = F_C(c_{Hb}^*)$ since $c_{\text{post}}^* = c_H^*$. The mean bidder count per triggered auction is $\mathbb{E}[M \mid M \geq K] + (\Lambda_b^{\text{pio,bid}} - \Lambda_b^{\text{nom}}) + \Lambda_b^{FR+\text{post}}$.

Nominators use $[1 - F_V(v_b^*)]$ since they only bid when they nominated instead of exitting; non-nominators and post-nomination arrivals use $[1 - F_V(r_b)]$ since they bid whenever their value exceeds the auction reserve. The post-period term includes $\pi_{0,b}^{\text{bid}} = F_C(c_{Hb}^*)$ reflecting the $c_{\text{post}}^* = c_H^*$ assumption.

The optimization is subject to 57 equality constraints (three per bin: $h_{1,b}$, $h_{2,b}$, and $h_{3,b}$) and 19 inequality constraints (one per bin, $g_{1,b}$, corresponding to the participation condition $A_b - B_b \geq 0$), encoding the equilibrium conditions from Definition 2. The learning cost F_C is parameterized as a skew-normal distribution $\text{SN}(\xi_c, \omega_c, \alpha_c)$ with support on $\mathbb{R}$, so realized costs may be negative; rather than truncating the distribution at zero, the inequality constraints $g_{1,b}$ keep the equilibrium cost cutoffs nonnegative, so that only the nonnegative part of the support is economically active.

The nomination threshold v_b^* is estimated jointly with $c_{L,b}^*$ and $c_{H,b}^*$ for each bin, with $h_{1,b}$ (the indifference condition $\tilde{\Pi}(v_b^*) = \phi_b$) imposed as an explicit equality constraint.

First, the **Nomination threshold** constraint for bin b, derived from equation 25, is characterized as

$$h_{1,b}(\boldsymbol{z}) = \tilde{\Pi}(v_b^*;\, m = K - 1, \Lambda_{\text{nn}b}, \hat{\rho}) - \phi_b = 0, \quad \phi_b = 10/\text{median}_{t \in b}(s_t) \tag{64}$$

where $\tilde{\Pi}(v_b^*; \cdot)$ is the expected auction surplus evaluated at the bin-b nomination threshold, and ϕ_b is the representative residual-space nomination fee for bin b. This ensures that a nominator with value exactly v_b^* earns auction surplus equal to bin b's residual-space nomination fee, and is therefore indifferent between nominating and not nominating.

The constraint $h_{1,b}$ is maintained as an active equality constraint for each bin in estimation. Because $\phi_b = 10/\text{median}_{t \in b}(s_t)$ differs across bins, the equilibrium threshold v_b^* is bin-specific: bins with a higher residual-space fee require a higher value threshold to maintain the nomination indifference condition, and vice versa. This cross-bin variation in v_b^* is identified from the bin-level indifference condition $\tilde{\Pi}(v_b^*; \cdot) = \phi_b$, without imposing any additional identifying assumption on the threshold.

Next, the $l = 1$ **cutoff** constraint for bin b, derived from $U_{l=1}(c^*_{Lb}) = U_{l=2}(c^*_{Lb})$, is

$$h_{2,b}(\boldsymbol{z}) = c^*_{Lb} - \frac{A_b - \phi_b\big[1 - F_V(v^*_b)\big] - B_b}{1 - \Pr(M_{-i} \geq K)} = 0 \tag{65}$$

where $A_b = \sum_{m=K-1}^{\infty} \Pr(M_{-i} = m) \cdot S_{l=1}(m)$ is the expected gross auction surplus for a early learner in bin b, $B_b = \Pr(M_{-i} \geq K) \cdot E[S_{l=2}(m) \mid M \geq K]$ is the expected gross surplus for a non-nominator, and $\phi_b[1 - F_V(v^*_b)]$ is the expected nomination fee (paid with probability $1 - F_V(v^*_b)$). This follows from setting $U_{l=1}(c^*_{Lb}) = U_{l=2}(c^*_{Lb})$ and solving for c^*_{Lb} using equation (11). $S_{l=1}(m)$ integrates $\tilde{\Pi}$ against the full $f_V(v)$; $S_{l=2}(m)$ integrates $\tilde{\Pi}_{FR}$ against the joint Gaussian copula posterior $f(v \mid v_1 \geq v^*_b, \ldots, v_m \geq v^*_b)$, precomputed on a (40×15) grid of (u^*, m) pairs and linearly interpolated at runtime.

The **Non-nominator ($l_i = 2$) cutoff** related constraint, derived from equation 18, is given as

$$h_3(\boldsymbol{z}) = c^*_H - \mathbb{E}[S_{l=2}(m) \mid M_{-i} \geq K] = 0 \tag{66}$$

where $S_{l=2}(m)$ uses the joint Gaussian copula posterior density for the non-nominator's own value, not the independently truncated marginal.

Lastly, for each bin b, two inequality constraints ensure the interior solution condition:

$$\begin{aligned} A_b - \phi_b\big[1 - F_V(v^*_b)\big] - B_b &\geq 0 \\ c^*_{Lb} &\leq c^*_{Hb} \end{aligned} \tag{67}$$

The first condition, $\Delta_{0,b} = U_{l=1}(0) - U_{l=2}(0) \geq 0$ within bin b, ensures that the zero-cost bidder prefers early learning over late-learning in that bin, guaranteeing an interior solution with positive nominations. The second ensures the bin-level cost cutoffs are correctly ordered.

I use inverse cross-bin variance of each moments as weights for GMM.

B.9 Per-Bin Equilibrium Objects and Moment Fit

Table 18: Per-Bin Equilibrium Objects

Bin	$\hat{v}_b^*$	$\hat{c}_b^{*L}$	$\hat{c}_b^{*H}$
1	0.0569	1.550	1.873
2	0.0798	1.545	1.914
3	0.0787	1.545	1.916
4	0.0894	1.543	1.896
5	0.0942	1.542	1.904
6	0.1224	1.536	1.965
7	0.1128	1.537	1.950
8	0.1375	1.532	1.986
9	0.1406	1.531	1.990
10	0.1293	1.532	1.975
11	0.2588	1.514	2.097
12	0.1185	1.532	1.960
13	0.1041	1.534	1.929
14	0.1692	1.524	2.028
15	0.2204	1.583	2.105
16	0.2191	1.651	2.131
17	0.2055	1.961	2.226
18	0.1424	2.207	2.207
19	2.7079	2.150	3.047

Notes: All quantities in residual units, from the estimated model. $\hat{v}_b^*$ is the nomination threshold solved from $h_{1,b}$; $\hat{c}_b^{*L}$ and $\hat{c}_b^{*H}$ are the early-learner and later-learner (non-nominator) cost cutoffs from $h_{2,b}$ and $h_{3,b}$. All 19 bins are reported.

Table 19: Moment Fit

Moment	**Description**	**Data**	**Model**
m_1	Mean trigger rate $\Pr(M \geq 1)$	0.00664	0.00325
m_2	ZTP mean nominations $\mathbb{E}[M \mid M \geq 1]$	1.056	1.002
m_3	ZTP variance of nominations $\mathrm{Var}[M \mid M \geq 1]$	0.0784	0.00163
m_4	Mean total bidders $\mathbb{E}[I \mid M \geq 1]$	3.318	3.240

Notes: The four active GMM moments (Appendix B.5), each reported as the equal-weight average across all 19 bins of s_t; "Data" are the empirical bin moments and "Model" the fitted values at the estimated parameters (Table 2). m_1 is matched in logs and m_4 carries weight 1.6; m_2 and m_3 carry unit weight. Every bin is matched, including bin 19 (two triggered auctions), whose conditional moments are defined.

B.10 Value Functions and Equilibrium Cutoffs

Throughout the counterfactual analysis, results are reported for Bin 5 as the main (median) specification, with Bins 1 and 9 providing a robustness range. The bins are ordered by their representative baseline fee ϕ_b in residual space: Bin 1 has the smallest ϕ_b (highest-scale auctions; lowest residual fee) and Bin 9 the largest. This ordering is in *parameter* space, not outcome space: Bin 5 is the median bin by fee level but is not guaranteed to produce the median outcome across all counterfactual scenarios. Accordingly, the shaded uncertainty band in all bin figures shows the true minimum and maximum across *all three* plotted bins (1, 5, and 9), so the Bin 5 line is guaranteed to fall within the shaded region by construction.

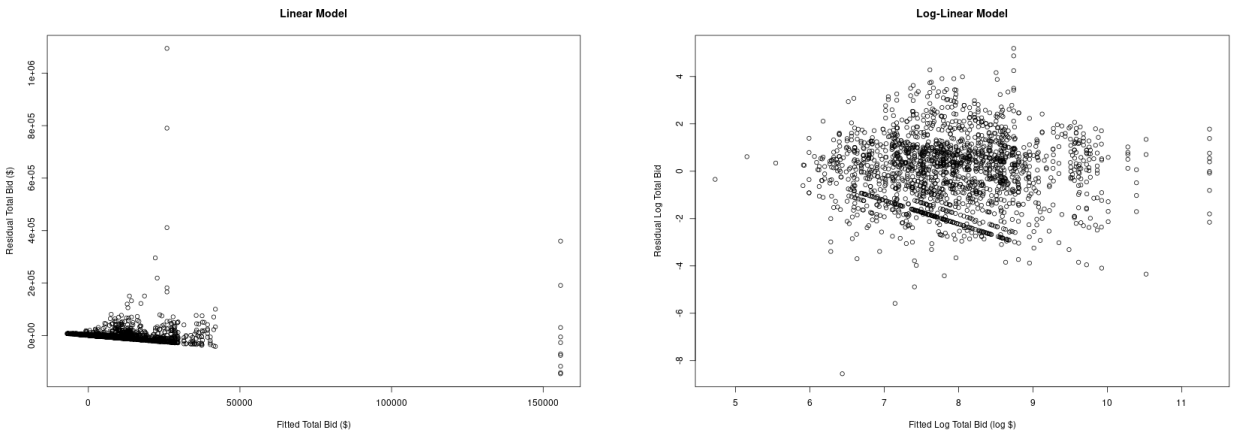
Figure 13 presents the structural estimation results for Bin 9 (the median decile of the residual-space). Panel A shows the estimated value functions $U_{l=1}(c)$ and $U_{l=2}(c)$. Both are linear and decreasing in learning cost c, but their slopes differ sharply. The early learner value function has slope -1: every unit increase in learning cost reduces the expected payoff one-for-one, because exploration is paid upfront and unconditionally. The non-nominator value function is nearly flat (slope ≈ -0.002): a non-nominator pays exploration cost only if the auction is triggered in Stage 2, so the expected cost is discounted by the small equilibrium trigger probability $\Pr(M \geq 1)$.

The two functions cross at the early learner cutoff c_L^* (in residual units); above c_L^*, late-learning dominates early learning, and the non-nominator value crosses zero at c_H^*, beyond which exit is optimal. Converted to dollars through the bin's scaling factor ($c^* \cdot s_b$), both thresholds lie in the right tail of the shared cost distribution, whose parameters $(\hat{\xi}_c, \hat{\omega}_c, \hat{\alpha}_c) = (1{,}712,\ 690,\ -126.9)$ imply a mean of $\approx$ \$1,161 and standard deviation $\approx$ \$416. Panel B shows the consequence *for this bin* (the median, high-value bin): the cost cutoffs are non-binding, so essentially all arriving bidders have exploration costs below the exploration threshold. This is representative of the high-value bins. In the high-s (high-value) bins, the dollar cutoffs fall into the body of the cost distribution and cost does screen participation (the early learning share falls to $\approx 3.5\%$ in the lowest s bin; see Section 6). Even where cost imposes no screen, however, the low nomination rate observed in the data is driven by the low pre-period arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$, which makes early (learning) arrivals rare even though exploration is individually worthwhile.

B.11 Residual Diagnostics for Bid Homogenization

Figure 9 compares residual-versus-fitted plots for two specifications of the bid-homogenization regression.

The formal Breusch-Pagan / White heteroskedasticity logic is as follows. Under a linear



(a) Linear Model: residual total bid ($) vs. fitted total bid ($).

(b) Log-Linear Model: residual log total bid vs. fitted log total bid.

Figure 9: Residual Diagnostic Plots for the Bid-Homogenization Regression.

Notes: Panel (a) plots residuals from a linear regression of total bid (per-acre bid $\times$ acreage, in dollars) on season fixed effects, cluster fixed effects, royalty rate, distance to nearest well, tract size, and spud-date age. The residuals fan out dramatically with fitted values, indicating severe multiplicative heteroskedasticity. Panel (b) plots residuals from the same specification with the log of total bid as the dependent variable (equation 31 in the text). Variance is approximately constant across the support, confirming the appropriateness of the multiplicative (log-linear) specification.

specification $b_{istc} = \exp(Z_i) \cdot v_{istc}$, the residual variance grows with $\exp(2Z_i)$, which depends on the fitted value; OLS is inefficient and standard errors are invalid. Under the log-linear specification, $\log b_{istc} = Z_i + \log v_{istc}$, the residual $\hat{\varepsilon}_{istc} = \log v_{istc} - \mathbb{E}[\log v_{istc}|X_t]$ has approximately constant variance. The diagnostic plots in Figure 9 confirm this pattern in the data.

B.12 Geographic Clustering

Using location data, I define a cluster to group auctions that are geographically close to one another, in order to control for geological similarity via cluster fixed effects in the bid-homogenization regression (Section B.1 below). Tracts that are closely located are more likely to share similar geological characteristics—such as subsurface composition and basin correlation—each of which affects learning costs and development revenue.

I implement a clustering algorithm using a 7.5-mile proximity threshold, following Hendricks et al. [2003] and Nguyen [2022]'s approaches to defining *neighboring tracts*.[49]

[49]Both Hendricks et al. [2003] and Nguyen [2022] define neighboring tracts using latitude and longitude thresholds (0.11 degrees of latitude and 0.12 degrees of longitude). I convert these to miles to provide more intuitive distance measures for the 7.5-mile threshold used here.

Definition 3 (Geographic Clusters)**.** *Two mineral rights auctions (tracts) X_1 and X_2 belong to the same cluster if there exists a sequence of auctions $\{y_1, y_2, \ldots, y_k\}$ such that the geographic distance between centroids of consecutive tracts y_i and y_{i+1} (for $i = 1, \ldots, k-1$), and the distances between centroids of y_1 and X_1 and between centroids of y_k and X_2, are all at most 7.5 miles.*

The algorithm computes Haversine distances between all tract centroids, initially assigns each tract to its own cluster, then iteratively merges clusters whenever any two tracts fall within the 7.5-mile threshold. This ensures transitivity: any two tracts in the same cluster are connected through a chain of neighbors each separated by at most 7.5 miles. Time is not considered—tracts from different auction seasons may belong to the same cluster if they are geographically proximate. The algorithm yields 36 clusters in total, ranging from single isolated tracts to dense concentrations of up to 179 tracts, with an average of 24.4 tracts per cluster.

B.13 Value of Nomination: Simulation Details

For each bin b I draw a single shared pool of $N_{\text{pool}} = 100{,}000$ candidate auctions and evaluate all five worlds on the *same* pool with common random numbers, so world-to-world differences are not contaminated by sampling noise. Each candidate auction t is built as follows.

- **Common component.** Draw one auction-specific factor W_t. Every bidder's value in auction t is generated through the estimated Gaussian equicorrelation copula ($\hat{\rho} = 0.8605$), $v = F_V^{-1}\big(\Phi(\sqrt{\rho}\, W_t + \sqrt{1-\rho}\, \varepsilon)\big)$, so values within an auction are correlated while W_t is independent across auctions. A fresh W_t is drawn for each of the 100,000 auctions.

- **Early (potential nominator).** One early value e_t off W_t; the tract is "nominated" if $e_t > v^*$ (worlds $\mathcal{W}_1, \mathcal{W}_2$ keep the nominated subset; $\mathcal{W}_3$–$\mathcal{W}_{4b}$ use the whole pool as arbitrary tracts).

- **Later arrivals.** A zero-truncated Poisson(λ_{post}) number of later bidders, each with a value off the same W_t and an independent learn draw.

The worlds differ only in three toggles—(i) whether later bidders price off the signal (F_{post}) or the prior (F_V); (ii) whether the tract is the nominated draw or an arbitrary draw; (iii) whether the early firm is a guaranteed bidder, an optional learner, or absent—and in the arrival rate (the no-nomination world $\mathcal{W}_{4b}$ uses $\lambda_{pre} + \lambda_{post}$). Participation uses re-solved cost cutoffs under the world's information: the learn-vs-exit cutoff $\bar{c}_{\text{sig}}$ under the posterior

and $\bar{c}_{\text{nos}} = \bar{c}_{\text{sig}} - \Delta$ under the prior, with Δ the posterior-minus-prior expected surplus from the analytic copula. A bidder is *active* (counts, and can win) only if it both learns and clears the reserve $v \geq r$.

Each auction is resolved as a second-price sale conditional on at least one active bidder: revenue is the second-highest active value (or the reserve with a single active bidder) and the winning bid is the highest active value. I report outcomes per sold auction and the sale rate separately.

Standard errors are from a two-phase bootstrap with $R = 2{,}000$ replications. Phase 1 computes each pool auction's outcome in each world once. Phase 2 resamples a fixed 800 auction *indices* per bin into those precomputed outcomes and re-averages; no re-simulation is needed, so large R is cheap. The reported s.e. is the standard deviation of the resampled effect across replications, and the pooled figures are tract-count-weighted across bins. The common resample size (800 per bin, rather than each bin's handful of observed auctions) puts every bin's sampling uncertainty on a comparable footing, so the per-bin decomposition is informative and not swamped by the smallest bins; it should be read as the precision one *would* attain with 800 auctions per bin, not the raw data count.

C Estimation and Counterfactual Figures and Tables

This appendix collects supplementary material for the Estimation Results (Section 6) and Counterfactual Analysis (Section 7): robustness of the attention function, the per-bin attention-implied arrival rates used in the attention counterfactual, per-bin welfare results, and the measurement of wasteful learning.

C.1 Attention Function: Robustness

Figure 10 reproduces the bidding-stage panel of Figure 5(b) after dividing each cell's bids by $\Pr(v > v_b^*)$, the model-implied probability that a draw clears the bin's nomination threshold; this nets out the mechanical tendency of low-v^* bins to attract more bidders. Re-estimating the pooled attention function with these adjusted bidding points leaves the elasticity essentially unchanged ($\hat{\kappa} \approx 1.15$, against 1.16 after the adjustment), confirming that the attention pattern is not an artifact of cross-bin differences in participation thresholds. Figure 11 reports the pooled regression of equation (30) on linear axes (the main-text Figure 6 uses log–log axes).

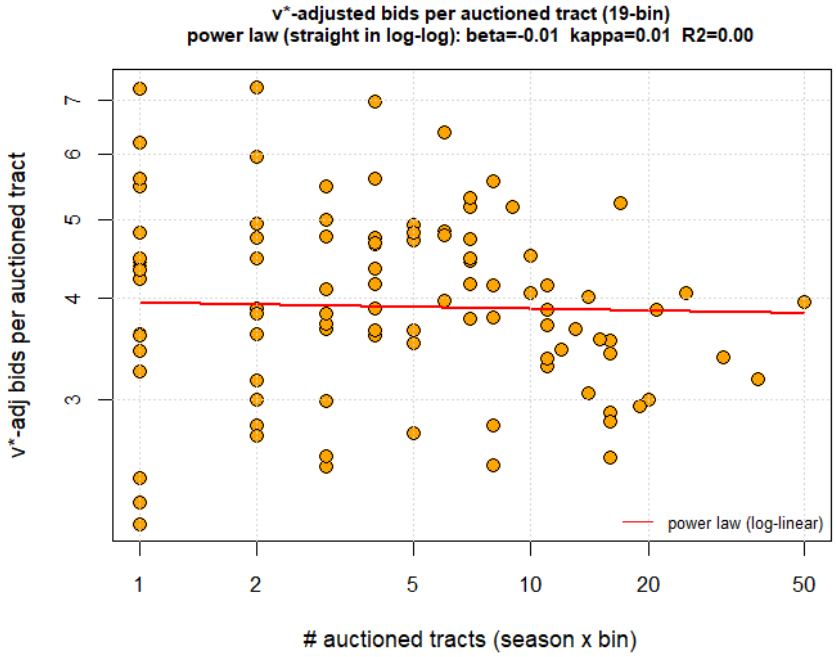


Figure 10: Bidding stage, v^*-adjusted. Bids per auctioned tract, divided by $\Pr(v > v_b^*)$, against the number of tracts auctioned (season × bin). The v^* adjustment leaves the pooled attention elasticity $\hat{\kappa} \approx 1.15$ essentially unchanged relative to the main-text fit.

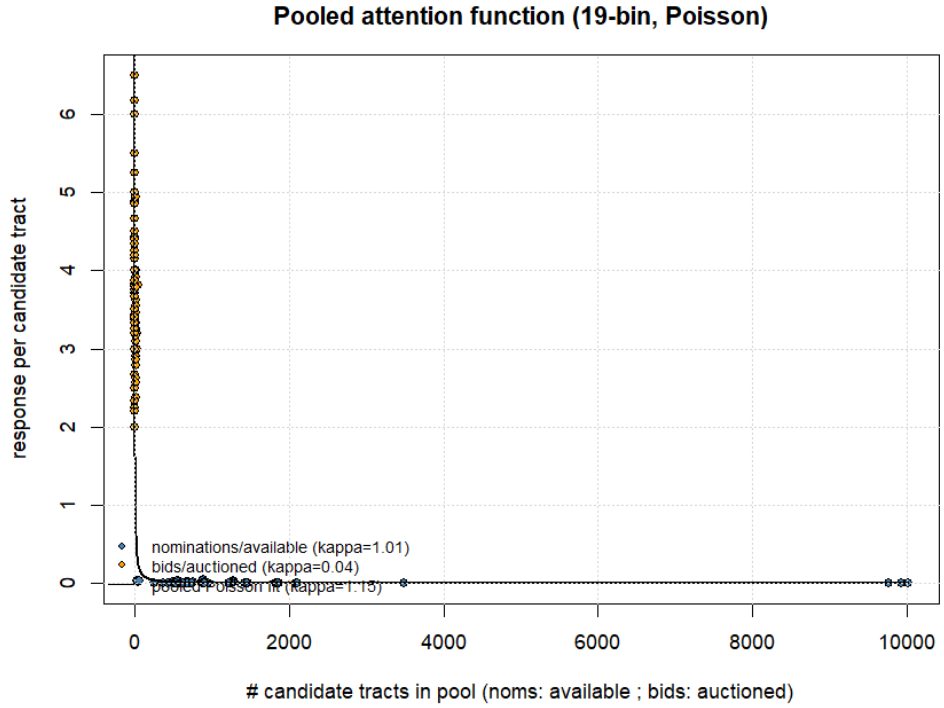


Figure 11: Pooled attention function, linear axes. Same data and pooled power-law fit as Figure 6, plotted on linear rather than log–log axes.

Attention also varies with tract value. Figure 12a (Appendix C.2) plots, for each value bin, the share of available tracts that receive at least one nomination. Because almost every nominated tract receives exactly one nomination, this share is essentially nominations per available tract. It is about 1–2% in the most valuable bins and falls to roughly 10^{-4} in the least valuable ones. The estimated model holds λ_{pre} fixed across bins, so it can generate cross-bin differences only through cost and value screening. Those forces barely bind in the high-value bins, so the model predicts a nearly flat trigger rate through bin 12: it under-predicts for the most valuable tracts and over-predicts for the least valuable ones. The remaining gradient is consistent with potential bidders directing their limited attention toward higher-stakes tracts [Caplin et al., 2019]. Letting the pre-nomination arrival rate vary with the tract's observable component s_b is a natural extension.

C.2 Per-Bin Moment Fit

Figure 12 compares each of the four estimation moments (Appendix B.5) with its fitted value, bin by bin; Table 19 reports the averages across bins. Panel (a), the trigger rate, is discussed in Appendix C.1.

Panels (b) and (c) show that nominated tracts almost always receive a single nomination.

With a small Poisson rate Λ_{nom}, the model predicts a mean of essentially one and a variance of essentially zero in every bin. The data agree wherever no tract drew a second nomination (bins 3, 10, and 13–19). In the other bins the mean is 1.04–1.27 and the variance 0.04–0.50, outside the model's intervals. This is the overdispersion in nomination counts discussed in Appendix B.7.

Panel (d) shows that the model fits the number of bidders well in the more valuable bins: in bins 1–14 the data lie inside the model's interval in every bin except bin 6. In the least valuable bins the model predicts fewer bidders—down to about one in bin 19—because the learning cost screens out most arrivals there (Figure 13). The data stay between 2.0 and 3.9 bidders and lie above the model's interval in bins 15, 17, 18, and 19, but each of these bins contains only 2 to 10 auctions.

C.3 Per-Bin Fitted Arrival Rate from the Attention Function

This appendix derives the per-bin fitted nomination arrival rates $\hat{\lambda}_b$, which read the estimated attention function off at each bin's available-tract pool.

Attention function. The attention function estimated in Section 6 (equation (30); pooled $\hat{A} \approx 23.9$, $\hat{\kappa} \approx 1.15$) gives the expected response per candidate tract, $r(n)$, as a function of the pool size n. Here I read it off at each bin's available-tract pool to obtain the per-bin fitted nomination arrival rate.

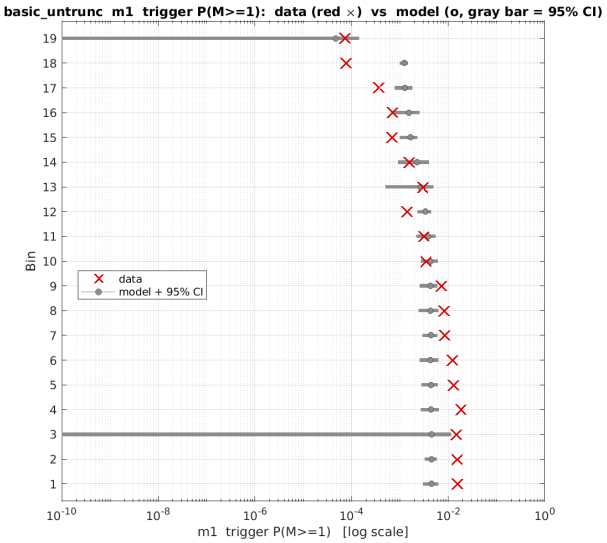
Nomination response. For an available tract the response is the *total number of nominations it receives*, so in bin b, season s,

$$r_{bs}^{\text{nom}} = \frac{\sum_{t \in b,s} M_t}{N_{bs}^{\text{avail}}}, \tag{68}$$

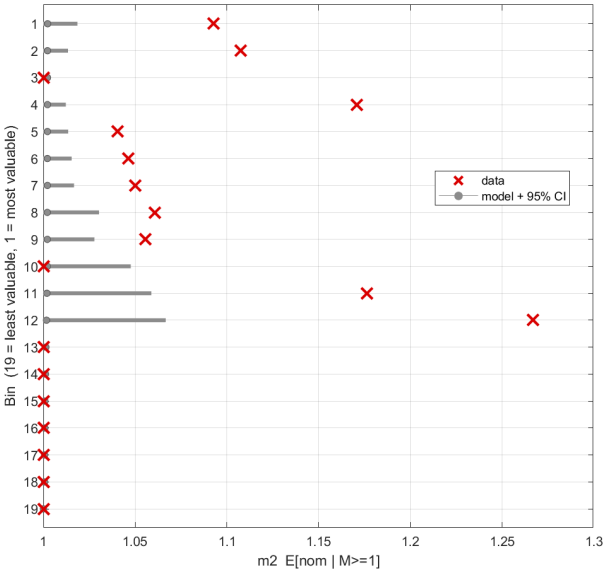
where M_t is tract t's nomination count ($M_t = 0$ for the vast majority of available tracts) and N_{bs}^{avail} is the number of available tracts. Equivalently $r_{bs}^{\text{nom}} = m_{bs}^{(1)}\, m_{bs}^{(2)}$, the product of the trigger rate $m_{bs}^{(1)} = \Pr(M \geq 1)$ and the mean nominations per nominated tract $m_{bs}^{(2)} = \mathbb{E}[M \mid M \geq 1]$.

Bin pool size. For each bin b define the representative available-tract pool as the median, across the eight seasons, of the number of available tracts in that bin,

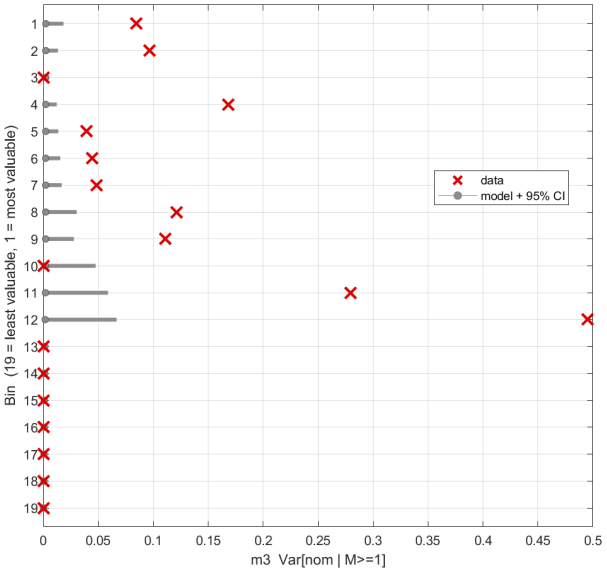
$$\bar{n}_b = \text{median}_s\ N_{bs}^{\text{avail}}. \tag{69}$$



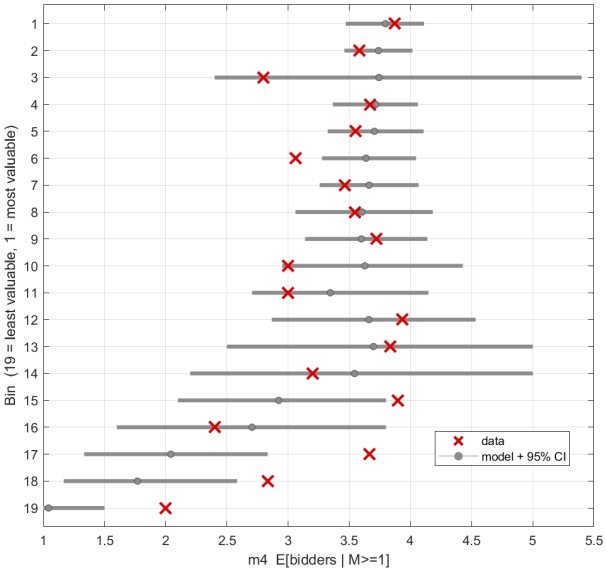
(a) Trigger rate $\Pr(M \geq 1)$, log scale



(b) Mean nominations $\mathbb{E}[M \mid M \geq 1]$



(c) Variance of nominations $\mathrm{Var}[M \mid M \geq 1]$



(d) Mean bidders $\mathbb{E}[I \mid M \geq 1]$

Figure 12: Per-bin moment fit, data vs. model. Red crosses: data moment in each bin. Gray dots: fitted moment at the estimated parameters (Table 2); gray bars: 95% intervals from a parametric bootstrap at the estimates, using each bin's sample size. Bins run from the least valuable tracts (bin 19, bottom) to the most valuable (bin 1, top). In panel (a) the intervals for bins 3 and 19 include zero and therefore extend to the left edge of the log axis.

Available counts are nearly constant across seasons, so the median is a robust representative pool size; summed over bins $\sum_b \bar{n}_b \approx 28{,}000$, partitioning the full available-tract universe.

Fitted per-bin arrival rate. Reading the attention function (30) off at $\bar{n}_b$,

$$\hat{\lambda}_b = \hat{A}\,\bar{n}_b^{-\hat{\kappa}} = 23.9\,\bar{n}_b^{-1.15}\,, \tag{70}$$

the fitted total-nominations-per-available-tract rate for a bin whose available pool is $\bar{n}_b$.

Low-attention arrival rate (aggregating the per-bin fits). The low-attention rate is the world where attention is diffused over the full $\approx 28{,}000$ number of tracts (n). Rather than evaluating the curve at $n = 28{,}000$ directly (a $\sim$3$\times$ out-of-range extrapolation that overstates dilution), I aggregate the *within-range* per-bin fits—predicted nominations $\sum_b \hat{\lambda}_b \bar{n}_b$ over total available tracts $\sum_b \bar{n}_b$:

$$\hat{\lambda}_{\text{low}} \;=\; \frac{\sum_b \hat{\lambda}_b\,\bar{n}_b}{\sum_b \bar{n}_b} \;=\; \frac{\sum_b \hat{A}\,\bar{n}_b^{\,1-\hat{\kappa}}}{\sum_b \bar{n}_b} \;\approx\; 0.0058. \tag{71}$$

Each $\bar{n}_b$ lies inside the 50–9,900 data range, so no extrapolation is used. The aggregate ≈ 0.0058 lands essentially on the structurally estimated pre-nomination arrival rate $\hat{\lambda}_{\text{pre}} \approx 0.005$ (Table 2).

High-attention arrival rate (concentrated pool). The high-attention rate is the counterfactual in which attention is concentrated on the smaller *offered* set, as in the auction stage. I again use the pooled attention function (70) and aggregate over the per-bin *offered* pools. Writing $\bar{n}_b^{\text{off}}$ for the median number of tracts offered in bin b,

$$\hat{\lambda}_{\text{high}} \;=\; \frac{\sum_b \hat{\lambda}_b^{\text{off}}\,\bar{n}_b^{\text{off}}}{\sum_b \bar{n}_b^{\text{off}}} \;=\; \frac{\sum_b \hat{A}\,(\bar{n}_b^{\text{off}})^{\,1-\hat{\kappa}}}{\sum_b \bar{n}_b^{\text{off}}} \;\approx\; 4.2, \qquad \hat{\lambda}_b^{\text{off}} = \hat{A}\,(\bar{n}_b^{\text{off}})^{-\hat{\kappa}}, \tag{72}$$

This is of the same order as the structural post-nomination arrival rate $\hat{\lambda}_{\text{post}} \approx 2.9$ (Table 2).

The two ends of the *same* pooled attention function therefore line up with the two structural arrival rates, $\hat{\lambda}_{\text{low}} \approx 0.006$ ($\approx \hat{\lambda}_{\text{pre}}$) $\longrightarrow$ $\hat{\lambda}_{\text{high}} \approx 4.2$ ($\sim \hat{\lambda}_{\text{post}}$). These are the pre-nomination arrival inputs used in the counterfactual analysis (Section 7).

Remark (zero cells). 31% of the nomination season$\times$bin cells receive *zero* nominations (a real response of zero per tract). The Poisson GLM keeps them; a log-log OLS of log(rate) on log(pool) must drop them and so understates dilution. The *pooled* $\hat{\kappa}$ is robust to this

choice of specification (1.13 OLS vs. 1.15 Poisson) because it is identified mainly by the large gap between the nomination and auction clusters

C.4 Seller-initiated Auctions With Information

C.5 Headline Totals Valuing Each Tract at Its Own Scaler

The abstract and the introduction report the value of nomination as totals over the 728 tracts in the estimation sample, valuing each tract at its own scaler s_t: each tract's outcome is its bin's simulated outcome in residual units multiplied by the tract's s_t, and outcomes are summed over tracts. Section 7 instead values each bin at its median tract (Table 3). Table 20 reports the totals on the own-scaler basis. The two bases give nearly the same percentage gain (44% versus 45%) and the same split across channels. The own-scaler totals are larger in dollars because the scaler is right-skewed within bins, so the average tract is worth more than the median tract, and because 23 tracts lie above the range of the top bin (see the table notes).

C.6 Per-Bin Welfare by Party and Concentrated Attention

Figures 15–16 report the per-bin welfare results underlying Section 7: the value-of-nomination decomposition by party and the attention-thickening channel ($\mathcal{W}_5 \to \mathcal{W}_6$), and the by-world seller revenue and sale rate. Bins run from the most valuable tracts (Bin 1, i.e. closest to existing wells and largest) to the least valuable (Bin 19).

C.7 Measuring Wasteful Learning off the $\mathcal{W}_1$ and $\mathcal{W}_{4b}$ Simulations

I measure *wasteful learning* by reading it directly off the $\mathcal{W}_1$ and $\mathcal{W}_{4b}$ simulations, rather than from a closed form. Each world is simulated as a large pool of tract draws; on every tract, post-nomination (*late*) arrivals each draw a private learning cost and, upon learning, a value. For each world $w \in \{\mathcal{W}_1, \mathcal{W}_{4b}\}$, I index the late arrivals i in the pool and record (i) the cost draw c_i; (ii) whether it learns, $0 < c_i < \bar{c}_w$, i.e. a positive cost below the world's learning cutoff ($\bar{c}_{\text{sig}}$ under $\mathcal{W}_1$, i.e.$= c_H^*$ from the original estimates), which conditions on the nomination signal, and $\bar{c}_{\text{nosig}}$ under $\mathcal{W}_{4b}$, which does not); and (iii) conditional on learning, whether it draws $v_i < r$ and so does not bid. The measure is

$$\text{WL}_w \;=\; \frac{\sum_{\text{late } i} \mathbb{1}\{0 < c_i < \bar{c}_w\}\,\mathbb{1}\{v_i < r\}\,c_i}{\sum_{\text{late } i} \mathbb{1}\{0 < c_i < \bar{c}_w\}},$$

Table 20: Value of nomination over the 728 tracts, valuing each tract at its own scaler s_t

Panel A: Outcomes

	No nomination ($\mathcal{W}_4$)	Nomination ($\mathcal{W}_1$)	Difference
Seller revenue ($ million)	6.86	9.91	3.05 (0.57)
Revenue per offered tract ($)	9,418	13,607	4,189
Revenue per sold tract ($)	10,472	13,607	3,135
Sale rate (%)	89.9	100.0	10.1
Bidders per tract	2.61	3.72	1.11

Panel B: Value of nomination by party and channel ($ million)

	Information spill-over ($\mathcal{W}_1-\mathcal{W}_2$)	More valuable tracts ($\mathcal{W}_2-\mathcal{W}_3$)	Guaranteed bidder ($\mathcal{W}_3-\mathcal{W}_4$)	Total (value of nom.) ($\mathcal{W}_1-\mathcal{W}_4$)
Seller	0.00 (0.00)	0.04 (0.60)	3.01 (0.54)	3.05 (0.57)
Nominator	0.00 (0.00)	0.00 (0.35)	2.15 (0.26)	2.15 (0.24)
Non-nominator	0.00 (0.00)	0.02 (0.50)	-1.93 (0.53)	-1.91 (0.53)
Total welfare	0.00 (0.00)	0.06 (1.00)	3.22 (0.93)	**3.28** (0.97)

Each tract's outcome is its bin's simulated outcome in residual units multiplied by the tract's own scaler s_t, summed over the 728 tracts in the estimation sample; Table 3 instead values each bin at its median tract. Worlds and channels are as in Table 3, and $\mathcal{W}_4$ counts tracts that draw no bid as zero revenue. Bootstrap standard errors in parentheses, from the same 1,000 resampling draws as Table 3, rescaled tract by tract. Revenue per sold tract rises by 30%. Twenty-three of the 728 tracts have a scaler (s_t between $7,466 and $9,332) above the range of the top bin; they are valued with Bin 1's outcomes, which is why the sale rate is 89.9% rather than the 89.8% in Section 7. Excluding them, seller revenue rises from $6.28 million to $9.10 million ($2.82 million, 45%). Rows may not sum exactly because of rounding.

Table 21: Seller-initiated cherry-picking by observable value versus nomination

Bins offered	Top % auctioned	Top % available	Bidders per tract	Winning bid ($)	Sale rate (%)	Revenue ($M)
1	15.3	3.3	2.78	17,184	93.0	11.63
1–2	36.5	7.9	2.74	14,453	92.2	9.71
1–3	37.2	8.0	2.74	14,390	92.2	9.66
1–4	48.8	10.0	2.72	13,401	92.0	8.98
1–5	59.3	12.5	2.72	12,578	91.9	8.41
1–6	68.5	15.0	2.70	11,941	91.6	7.96
1–7	77.0	18.2	2.69	11,321	91.5	7.54
1–8	81.7	20.0	2.69	11,007	91.3	7.32
1–9	86.8	22.5	2.68	10,660	91.2	7.08
1–10	89.8	25.0	2.68	10,459	91.2	6.94
1–11	92.2	27.5	2.67	10,309	91.0	6.83
1–12	94.3	32.8	2.67	10,153	91.0	6.72
1–13	95.2	33.5	2.67	10,089	91.0	6.68
1–14	95.9	35.0	2.66	10,038	90.9	6.65
1–15	97.3	41.7	2.65	9,936	90.8	6.57
1–16	98.0	45.0	2.64	9,886	90.7	6.53
1–17	98.9	52.5	2.63	9,834	90.4	6.47
1–18	99.7	87.5	2.61	9,788	90.1	6.42
1–19	100.0	100.0	2.61	9,788	89.8	6.40
Nomination ($\mathcal{W}_1$)			3.72	12,743	100.0	9.28

The seller runs no nomination and offers 728 tracts arbitrarily from the listed bins, in proportion to their auction shares ($\mathcal{W}_4$ restricted to those bins); row 1–19 is the no-nomination benchmark $\mathcal{W}_4$. "Top % auctioned" is the cumulative share of auctioned tracts in the listed bins; "top % available" is the cumulative share of tracts available in 2017Q1. Bidders are per offered tract; the winning bid is per sold tract. Each bin is valued at its median tract.

the average learning cost sunk by a late learner who ends up unable to bid. Only post-nomination arrivals enter the sum; the early nominator is excluded, since she has $v \geq v^* \geq r$ and never wastes. Costs are clamped at $\max(0, c)$ (the skew-normal left tail may draw $c < 0$ which I drop to focus on 'watesful' learning), and the denominator counts late learners with a strictly positive cost below the cutoff and therefore learned.

The two worlds differ only in the *distribution of tracts* their late learners face. In $\mathcal{W}_{4b}$ the seller offers arbitrary tracts—including those on which the early arrival drew $v < v^*$ (a low common value) but which are auctioned regardless—so late learners frequently draw $v < r$ and waste their cost. In $\mathcal{W}_1$ late learners are on nominated tracts, where an early arrival has already cleared $v \geq v^*$; they therefore draw from the *posterior* and usually find $v \geq r$.

I find $\mathrm{WL}_{\mathcal{W}_{4b}} \approx \$120$ per late learner, with 10.6% drawing $v < r$—about \$272 per auctioned tract (paid by roughly 2.3 late learners per tract). The reduction nomination delivers is clearest when I split the $\mathcal{W}_{4b}$ tracts by whether the early arrival cleared v^*. On the tracts it would not have been nominated (early learner draws $v < v^*$, those $\mathcal{W}_1$ never offers) 53.6% of late learners draw $v < r$, wasting about \$605 per learner. On the tracts it would have been nominated (early learner draws $v \geq v^*$, which $\mathcal{W}_1$ keeps) only 5.2% falls below the reserve, resulting in about \$59 wasteful learning per learner. Nomination's screen drops exactly the high-waste tracts, which is why wasteful learning roughly halves, from \$272 to \$140 per auctioned tract.[50]

I find $\mathrm{WL}_{\mathcal{W}_{4b}} \approx \$120$ per learner and 10.6% draw $v < r$—about \$272 per auctioned tract (roughly 2.3 late learners per tract). Under $\mathcal{W}_1$, only 5.2% draw $v < r$ and the waste falls to about \$59 per learner (\$140 per auctioned tract), whereas on the tracts nomination skips (early $v < v^*$) a striking 53.6% of late learners draw $v < r$, wasting about \$605 per learner. Nomination's screen removes exactly these high-waste tracts, which is why wasteful learning roughly halves, from \$272 to \$140 per auctioned tract. [51]

[50] Because learning cost and value are drawn independently, WL_w also equals the closed form $\Pr(v < r \mid w) \cdot \mathbb{E}[\, c \mid 0 < c < \bar{c}_w \,]$—the learning a late learner sinks, times the probability it is wasted. The simulation read-off and this closed form agree to Monte-Carlo precision (within about 0.4%). The split above holds the learning cutoff at the no-signal level, isolating the value-distribution (posterior) channel; allowing the with-signal cutoff leaves the \$272→\$140 conclusion unchanged.

[51] Because learning cost and value are drawn independently, WL_w also equals the closed form $\Pr(v < r \mid w) \cdot \mathbb{E}[\, c \mid 0 < c < \bar{c}_w \,]$. The simulation read-off and this expression coincide to Monte-Carlo precision.

D Theory and Proofs

D.1 Positive Correlation Details

Formal definition. Under affiliated private values (APV), the joint density f of H_V satisfies

$$f(\mathbf{v} \vee \mathbf{v}')\, f(\mathbf{v} \wedge \mathbf{v}') \;\geq\; f(\mathbf{v})\, f(\mathbf{v}') \qquad \text{for all valuation vectors } \mathbf{v}, \mathbf{v}',$$

where $\vee$ and $\wedge$ denote coordinate-wise maximum and minimum. The parametric form of H_V adopted in Section 5 generates joint distributions satisfying this property [Nelsen, 2006, Müller and Scarsini, 2005], ensuring the empirical findings are consistent with the theoretical framework underlying equilibrium existence and revenue ranking results in auction theory [Castro, 2007].

Posterior stochastic dominance. The stochastic dominance result in equation (2) is a consequence of this affiliation property: for affiliated random variables,

$$\Pr(V_i > v \mid V_j \geq v^*,\, j = 1, \ldots, m) \;\geq\; \Pr(V_i > v) \qquad \text{for all } v$$

[Milgrom and Weber, 1982]. This holds for *any* H_V satisfying **positive correlation**, not only for the Gaussian copula parameterization adopted in Section 5. Therefore, the equilibrium characterization in Section 4.4 does not rely on the specific copula choice.

D.2 Robustness to Noisy Nomination

The main text models nomination as a deterministic threshold rule: $n(v) = \mathbb{I}\{v \geq v^*\}$. Here I show the framework is robust to mixed or noisy nomination strategies.

Suppose a positive fraction $p \in (0, 1)$ of nominations are uninformative: a bidder with realized value below v^* randomizes and nominates with probability p, independent of her realized value (a bidder above v^* still nominates with probability 1, as in the main text). This dilutes the informational content of a nomination. The non-nominator's effective screening probability – the probability that a nominator's value actually clears v^*, given that she nominated – becomes

$$\pi \;\equiv\; \Pr(v \geq v^* \mid \text{nominate}) \;=\; \frac{1 - F_V(v^*)}{1 - F_V(v^*) + p\, F_V(v^*)},$$

rather than $\pi = 1$ under the pure-strategy rule of the main text.

D.3 Posterior Updating of the Non-Nominator's Value Distribution

This section derives the mechanism through which observing m nominators above threshold v^* shifts the non-nominator's own value distribution upward, as claimed in Section 4.3.1. The mechanism operates through the one-factor representation of the Gaussian equicorrelation copula. Each normal score $Z_i = \Phi^{-1}(U_i)$ decomposes as

$$Z_i = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \varepsilon_i, \qquad W, \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

where W is a shared common factor and ε_i is idiosyncratic. Conditional on $W = w$, the Z_i are independent with $Z_i \mid W = w \sim \mathcal{N}(\sqrt{\rho}\, w,\, 1-\rho)$, so each nominator's threshold-crossing probability is $P(Z_i > z^* \mid W = w) = 1 - \Phi\left(\frac{z^* - \sqrt{\rho}\, w}{\sqrt{1-\rho}}\right)$, where $z^* = \Phi^{-1}(u^*)$. Observing m threshold-crossings updates the posterior on W via Bayes' theorem:

$$p(w \mid m \text{ crossings}) \;\propto\; \phi(w) \cdot \left[1 - \Phi\left(\frac{z^* - \sqrt{\rho}\, w}{\sqrt{1-\rho}}\right)\right]^m.$$

This posterior shifts rightward as m increases — each additional nominator above z^* is independent evidence that W is high. Since the non-nominator's own normal score satisfies $Z_0 = \sqrt{\rho}\, W + \sqrt{1-\rho}\, \varepsilon_0$, the rightward shift in $p(w \mid m)$ stochastically increases Z_0, and therefore $U_0 = \Phi(Z_0)$ and the non-nominator's value $V_0 = F_V^{-1}(U_0)$.

D.4 Corner Equilibrium Cases

This appendix presents the two corner-solution equilibrium cases that arise in the nomination game when the interior-solution condition fails. Recall the definitions (equations 13–14):

$$\Delta_0 = A - \phi\big[1 - F_V(v^*)\big] - B \quad \text{(value difference at zero cost)}$$
$$\Delta_{l=2} = U_{l=1}(c_H^*) \quad \text{(value of early learning at the non-nominator cutoff)}$$

where $A = \sum_{m=K-1}^{\infty} \Pr(M_{-i}{=}m)\, S_{l=1}(m)$ is the expected gross auction surplus from early learning and $B = \Pr(M \geq K)\, \mathbb{E}[S_{l=2}(m) \mid M \geq K]$ is the expected gross surplus from late-learning.

Case 2: Corner Solution (All Wait and Learn Later)

Condition: $\Delta_0 < 0$ and $\Delta_{l=2} < 0$.

When $\Delta_0 < 0$, the value of early learning at zero cost falls below the value of late-learning: $U_{l=1}(0) < U_{l=2}(0)$. Combined with $\Delta_{l=2} < 0$ (late-learning is still preferable even at the non-nominator cutoff), the function $U_{l=1}(c_i) - U_{l=2}(c_i)$ is strictly negative for all $c_i \in [0, c_H^*]$. There is therefore no interior crossing point, and no bidder finds it optimal to nominate.

Formally, $\pi_0^{l=1} = 0$ and the equilibrium degenerates to a mass of non-nominators (those with $c_i \leq c_H^*$) and exits. This outcome is a *coordination failure*: each potential nominator would prefer to wait for someone else to incur the nomination cost and fee, and in equilibrium no one does.[52]

As shown in the center panel of Figure 4, the $U_{l=1}$ intercept lies below the $U_{l=2}$ intercept ($A - \phi[1 - F_V(v^*)] < B$), so $U_{l=2}(c_i) > U_{l=1}(c_i)$ for all c_i in the relevant range.

Case 3: Corner Solution (All Learn Early)

Condition: $\Delta_{l=2} \geq 0$.

When $\Delta_{l=2} = U_{l=1}(c_H^*) \geq 0$, the value of early learning remains non-negative at the non-nominator cutoff c_H^*. Since $U_{l=1}$ is weakly decreasing in c_i and $U_{l=2}(c_H^*) = 0$ by definition of c_H^*, this means $U_{l=1}(c_i) \geq U_{l=2}(c_i)$ for all $c_i \in [0, c_H^*]$. No bidder prefers late-learning over early learning within the participation range.

In this case the equilibrium threshold is defined by $U_{l=1}(c_L^*) = 0$ (where the early learning value just breaks even, not where it crosses the late-learning value). Any bidder with $c_i < c_L^*$ learns early; any bidder with $c_i > c_L^*$ exits. The late-learning strategy is dominated and $\pi_0^{l=2} = 0$.

The intuition is that either the nomination fee is negligible relative to the expected auction surplus (low ϕ) or the pivotal probability is very high, making the nomination benefit large enough that all participating bidders prefer to bear the cost and nominate outright. As shown in the right panel of Figure 4, the $U_{l=1}$ line lies above $U_{l=2}$ everywhere in the feasible cost range.

Empirical relevance. In the ND data, both nominators and non-nominators are observed in every auction season, consistent with the Case 1 interior solution. Cases 2 and 3 are not observed in the data but are relevant for counterfactual policy analysis: a sufficiently large increase in the nomination fee ϕ can push the equilibrium from Case 1 into Case 2, eliminating nominations entirely.

[52] The pivotal probability $\Pr(M_{-i} = K - 1)$ is the probability that a given bidder's nomination is decisive for triggering the auction. When this probability is low (many other nominators expected) or the fee ϕ is high relative to the nomination benefit, $\Delta_0 < 0$ and the coordination failure arises.

D.5 Proofs for Section 4.4

D.6 Proof of Lemma 1: Poisson Arrivals of Nominators

Proof. Let $N \sim \text{Poisson}(\Lambda)$ denote the total number of arriving bidders, where $\Lambda = \lambda_{pre}\cdot$. Each arriving bidder independently becomes a competing nominator with probability $p \equiv \pi_0^{l=1} \cdot [1 - F_V(v^*)]$, reflecting two independent thinning events: the bidder chooses the early learning ($l = 1$) strategy with probability $\pi_0^{l=1}$, and conditional on that, has a value exceeding the threshold v^* with probability $[1 - F_V(v^*)]$.

I derive the distribution of M_{-i}, the number of competing nominators excluding bidder i, directly. Conditioning on the total number of arrivals $N = j$ gives:

$$\Pr(M_{-i} = m) = \sum_{j=m}^{\infty} \Pr(M_{-i} = m \mid N = j)\ \Pr(N = j) \tag{73}$$

$$= \sum_{j=m}^{\infty} \binom{j}{m} p^m (1-p)^{j-m} \cdot \frac{e^{-\Lambda}\Lambda^j}{j!} \tag{74}$$

where the binomial factor $\binom{j}{m} p^m (1-p)^{j-m}$ is the probability that exactly m of the j arrivals become competing nominators. Rearranging terms,

$$\Pr(M_{-i} = m) = \sum_{j=m}^{\infty} \frac{j!}{m!\,(j-m)!} \cdot p^m (1-p)^{j-m} \cdot \frac{e^{-\Lambda}\Lambda^j}{j!} \tag{75}$$

$$= \frac{e^{-\Lambda} p^m}{m!} \sum_{j=m}^{\infty} \frac{\Lambda^j (1-p)^{j-m}}{(j-m)!} \tag{76}$$

Substituting $k = j - m$,

$$\Pr(M_{-i} = m) = \frac{e^{-\Lambda} p^m \Lambda^m}{m!} \sum_{k=0}^{\infty} \frac{[\Lambda(1-p)]^k}{k!} \tag{77}$$

$$= \frac{e^{-\Lambda}(\Lambda p)^m}{m!} \cdot e^{\Lambda(1-p)} \tag{78}$$

$$= \frac{e^{-\Lambda p}(\Lambda p)^m}{m!} \tag{79}$$

where the Taylor expansion $\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$ is used and combines $e^{-\Lambda} \cdot e^{\Lambda(1-p)} = e^{-\Lambda p}$. This is the PMF of Poisson(Λp), so

$$M_{-i} \sim \text{Poisson}(\Lambda p) = \text{Poisson}\big(\lambda_{pre} \cdot \pi_0^{l=1} \cdot [1 - F_V(v^*)]\big)\,. \quad \square$$

□

D.7 Proof of Proposition 1: Single Crossing

Proof. Both value functions are linear in c_i:

$$U_{l=1}(c_i) = A - \phi\big[1 - F_V(v^*)\big] - c_i \quad \text{with slope } -1 \tag{80}$$

$$U_{l=2}(c_i) = B - \Pr(M \geq K) \cdot c_i \quad \text{with slope } -\Pr(M \geq K) \tag{81}$$

where $A = \sum_{m=K-1}^{\infty} \Pr(M_{-i} = m) S_{l=1}(m)$ is the expected gross auction surplus for an early learner, $B = \Pr(M \geq K) \cdot \mathbb{E}[S_{l=2}(m) | M \geq K]$ is the expected gross surplus for a non-nominator, and $\phi[1 - F_V(v^*)]$ is the expected nomination fee paid with probability $1 - F_V(v^*)$.

Since $\Pr(M \geq K) < 1$,

$$|dU_{l=1}/dc_i| = 1 > \Pr(M \geq K) = |dU_{l=2}/dc_i|. \tag{82}$$

Two lines with different slopes can intersect at most once. If they intersect at c_L^*, then

$$\begin{cases} U_{l=1}(c_i) > U_{l=2}(c_i) & \text{(learn early) if } c_i < c_L^* \\ U_{l=1}(c_i) < U_{l=2}(c_i) & \text{(learn later) if } c_i > c_L^* \end{cases}$$

□

D.8 Numerical Implementation Details for Section B.4

I implement several numerical techniques to ensure stable computation across the parameter space. First, I compute Poisson probabilities in log space to avoid overflow for large n or λ

$$\log P(N = n) = -\lambda + n \log \lambda - \sum_{i=1}^{n} \log i \tag{83}$$

and then exponentiate only when summing across different values of n. This allows us to handle cases with expected numbers of competitors as large as 20 or more without numerical issues.

Second, when computing the copula density, I handle extreme values of the CDF. I clamp probabilities to the range $[\epsilon, 1 - \epsilon]$ with $\epsilon = 10^{-10}$ before taking logarithms, and I check for numerical overflow in the copula density formula, capping density values at 10^{10} to prevent propagation of numerical errors.

Third, for the numerical integration in equation (2), I use the trapezoidal rule with 500 evaluation points, which I verified provides sufficient accuracy by comparing with higher-resolution grids. I also normalize the conditional density to integrate to one over its support before computing the expected surplus, which ensures that any small errors in the density estimation do not compound during integration.

D.9 Joint GMM Estimation of F_V and ρ

Method. As a robustness check I re-estimate the copula parameter using a joint GMM estimator ('Joint GMM Strategy') that avoids the two-step structure of pairwise pseudo-MLE. The objective is

$$(\hat{\beta},\, \hat{\rho}) = \arg\min_{\beta,\,\rho} \sum_{i \in \mathcal{G}} \Big[G_k\big(B(x_i;\beta);\, k_i,\, I_i,\, \rho\big) - \hat{U}^{(r_i)}_{(k_i)} \Big]^2, \tag{84}$$

where $B(x;\beta)$ is a monotone B-spline (10 internal knots, degree 3) with coefficient vector β constrained to be non-decreasing, G_k is the copula order-statistic CDF (Equation 43), $\hat{U}$ are Weibull empirical ranks, and $\mathcal{G}$ is a 200-point evenly-spaced x-grid selected from the training bids. The B-spline basis is built on all training bids; only the SSR evaluation is restricted to the 200-point grid to reduce computation.

After minimising (84), the lognormal parameters $(\hat{\mu}, \hat{\sigma})$ are initialized by NLS: fitting $\Phi\big((\log x - \mu)/\sigma\big)$ to the spline values $B(x_i;\hat{\beta})$ for all training bids x_i. This provides a smooth, parametric extrapolation of F_V below v^*, but is then refined by the truncation-corrected lognormal MLE described in Appendix B.3 (Eq. 47); the MLE pair $(\hat{\mu}, \hat{\sigma}) = (0.032,\, 1.614)$

is the one used downstream.

Estimation uses 100 random 90/10 train/test splits of the 728 unviolated auctions. The test-set RMSE for each split is computed by evaluating the copula diagonal $D(\hat{F}_V(x_{\max});\, I_r{-}1,\, \hat{\rho})$ for each test-auction maximum bid $x_{\max}$ in an I_r-bidder auction, then comparing to the empirical Weibull rank of $x_{\max}$ among all observed winner bids.

Grid200 results. Table 22 reports the Joint GMM Strategy estimates for the full sample ($I_r \in \{2, \ldots, 11\}$, 728 auctions) that is used as the main estimation method in the main text.

Table 22: Joint GMM Strategy estimates: joint estimation of F_V (monotone B-spline) and ρ by SSR minimisation (200-point moment grid, 100 random 90/10 splits). RMSE is evaluated by applying the copula diagonal $D(\hat{F}_V(x_{\max}),\, I_r{-}1,\, \hat{\rho})$ to test-set maximum bids. Mean $\pm$ SD across 100 splits. Bold row: selected specification (lowest RMSE).

Copula	$\hat{\rho}$ (mean $\pm$ SD)	**RMSE** (mean $\pm$ SD)
Gaussian ($\hat{\rho}$)	**0.866 $\pm$ 0.096**	**0.183 $\pm$ 0.037**
Gumbel	2.404 ± 0.661	0.210 ± 0.036
Clayton	17.31 ± 4.71	0.212 ± 0.066
Independence	—	0.364 ± 0.043

Full-data fit (Gaussian, 728 auctions): $\hat{\rho} = 0.8605$. Lognormal tail: NLS stage $(\hat{\mu}, \hat{\sigma}) = (-0.715,\, 3.863)$, refined by truncation-corrected MLE (Eq. 47) to (0.032, 1.614) (final; used downstream).

The Gaussian copula achieves the lowest RMSE (0.183 ± 0.037). Independence is clearly dominated (RMSE 0.364, 2.0$\times$ larger), confirming that positive dependence is a significant feature of the bid distribution. Clayton is unstable ($\hat{\rho} = 17.31 \pm 4.71$, near the upper bound of 20), which reflects boundary behavior rather than well-identified upper-tail dependence.

Monte Carlo evidence for median aggregation. This section provide formal evidence to why I use median rather than the mean across the 100 train/test splits as main specification of the correlation parameter, I run a Monte Carlo study in which I simulate fresh datasets from the correctly-specified model across all three copula families and apply the same Joint GMM estimator, then compare the mean and median as aggregation strategies across replications. I document that under Gumbel and Clayton dependence structures, the distribution of the aggregated $\hat{\rho}$ is right-skewed shape, providing justification to why median is a better predictor of the true correlation parameter.

Design. For each of six DGP cells (Gumbel $\rho_0 \in \{1.5, 4.0\}$, Clayton $\rho_0 \in \{0.5, 3.0\}$, Gaussian $\rho_0 \in \{0.3, 0.8\}$) crossed with six lognormal marginal variants ($\mu \in \{-0.5, 0.0, 0.5\}$,

$\sigma \in \{0.3, 0.7\}$), I draw $R = 100$ independent datasets for each auction-count level $S \in \{50, 100, 700\}$. Each dataset consists of S auctions with $I_s \sim \text{Uniform}\{2, \ldots, 11\}$ bidders and one observed bid vector per auction (mimicking the single-observation-per-auction structure of the empirical data). The Gaussian copula objective uses a 20-node Gauss-Hermite quadrature approximation to evaluate the copula diagonal. I apply the same Joint GMM procedure to each dataset and record $\hat{\rho}^{(r)}$ for $r = 1, \ldots, 100$.

The key measure for comparison is the *normalised deviation* $\delta^{(r)} = (\hat{\rho}^{(r)} - \rho_0)/\rho_0$, and I compareOk bu these two different aggregation method across $r = 1, \ldots, 100$ datasets: mean($= \bar{\rho} = \frac{1}{100}\sum_r \hat{\rho}^{(r)}$) and median($= \tilde{\rho} = \text{median}(\hat{\rho}^{(1)}, \ldots, \hat{\rho}^{(100)})$).

Note that the root mean squared error RMSE $= \sqrt{\frac{1}{100}\sum_i (\delta^{(i)})^2}$ characterizes the *full distribution* of individual estimates and is the same regardless of which aggregator is reported.

Results. Figure 18 shows the RMSE of the normalised deviation $(\hat{\rho} - \rho_0)/\rho_0$ as a function of S for each copula family and dependence level. All six DGP cells converge toward zero as S grows, confirming consistency of the Joint GMM estimator. The Gaussian high cell ($\rho_0 = 0.8$) converges fastest, while Clayton low ($\rho_0 = 0.5$) and Gaussian low ($\rho_0 = 0.3$) show the highest RMSE at $S = 700$ (25% and 38% respectively), reflecting the flat GMM objective surface at weak dependence.[53]

Table 23 summarises the bias of each aggregator, pooled over the six lognormal marginal variants (600 replications per row).

Interpretation. Four findings emerge from Table 23 and Figures 18–19.

First, for Gumbel and Clayton families the median aggregator is consistently less biased than the mean at every sample size. At $S = 50$, the mean–median gap reaches 35 percentage points for Gumbel high dependence (+109% vs. +74%) and 18 points for Clayton high (+109% vs. +91%); by $S = 700$ the gap shrinks to $\lesssim 5$ points as the objective surface concentrates and optimizer outliers vanish. This pattern is the hallmark of a right-skewed estimator distribution driven by occasional L-BFGS-B convergence to high local minima, not by systematic model misspecification.

Second, the Gaussian copula behaves differently across the two dependence regimes. For strong dependence ($\rho_0 = 0.8$), the estimator is remarkably well-concentrated even at $S = 50$: mean and median bias are both around +16% with RMSE of 16%. By $S = 700$, bias falls to +3% and RMSE to 4% — the tightest convergence of any copula family. The bounded

[53] For all three copula families, the copula diagonal evaluates to the independence case $C_r(u, \ldots, u) = u^r$ as the dependence parameter approaches its independence limit. Since the moment conditions $G_k(u; I, \rho)$ are built from sums of copula diagonals, it follows that $\partial G_k/\partial\rho \to 0$ as dependence weakens: the moment conditions stop responding to ρ near independence, making the GMM objective surface flat in the ρ direction.

Table 23: Monte Carlo bias of mean and median $\hat{\rho}$ aggregators under the Joint GMM estimator, pooled over six lognormal marginal variants (600 replications per row). Bias is $(\hat{\rho} - \rho_0)/\rho_0 \times 100\%$. RMSE is the root mean squared normalised deviation of the *individual* $\hat{\rho}$ draws — it is the same regardless of which aggregator is reported. Gaussian results use the vectorised 20-node Gauss-Hermite evaluator (`strategy3_sim_validation_fast.py`).

DGP	ρ_0	S	**Mean bias**	**Median bias**	**RMSE**
Gumbel low	1.5	50	+57%	+50%	66%
		100	+28%	+25%	32%
		700	+5%	+5%	8%
Gumbel high	4.0	50	+109%	+74%	148%
		100	+48%	+36%	75%
		700	+14%	+10%	24%
Clayton low	0.5	50	+211%	+208%	229%
		100	+112%	+105%	126%
		700	+17%	+16%	25%
Clayton high	3.0	50	+109%	+91%	134%
		100	+60%	+54%	76%
		700	+11%	+9%	20%
Gaussian low	0.3	50	+110%	+110%	114%
		100	+65%	+67%	71%
		700	+10%	+15%	38%
Gaussian high	0.8	50	+16%	+16%	16%
		100	+9%	+10%	11%
		700	+3%	+2%	4%

parameter space $\rho \in (-1, 1)$ prevents the runaway outliers seen in Gumbel and Clayton, so mean and median agree. For weak dependence ($\rho_0 = 0.3$), the Gaussian objective is flatter: RMSE remains elevated at $S = 700$ (38%), and the mean–median gap is *reversed* (mean +10%, median +15%), consistent with occasional near-boundary estimates at $\hat{\rho} \approx 0$ pulling the mean below the median rather than above it.

Third, the estimator converges to the truth as S grows across all families. By $S = 700$ (the actual sample size in this paper), Gumbel and Clayton cells show median bias of at most 16% and RMSE of at most 25%, and Gaussian high shows median bias of 2% with RMSE of 4%.

Fourth, the empirical application uses the Gaussian copula family with an estimated $\hat{\rho} = 0.861$, placing it in the "Gaussian high" regime ($\rho_0 = 0.8$). In that regime, mean and median are nearly identical at $S = 700$ and both provide low-bias estimates. The residual mean–median gap observed in the 100 train/test splits (mean 0.866, median 0.861, SD 0.096) therefore likely reflects a combination of (i) data heterogeneity across splits and (ii) a modest right-skew from splits with fewer unviolated auctions, where the $\hat{\rho}$ distribution behaves more

like the small-S case. The **median** $\hat{\rho} = 0.861$ is the preferred point estimate as it is more robust to this residual skew.

D.10 Implementation: Handling Mixed Competition

I compute $\tilde{\Pi}(v; m, \Lambda, \rho)$, where m is the deterministic component (known competitors) and Λ is the stochastic component that captures the random variable $n \sim \text{Poisson}(\Lambda)$. First, for m known competitors and n potential competition, I determine 90th percentile support ($n_{\max} = \text{poissinv}(0.90, \Lambda) + 3\sqrt{\Lambda}$) and compute Poisson probabilities as $p_n = \frac{e^{-\Lambda}\Lambda^n}{n!}$ for $n \in \{0, ..., n_{\max}\}$. This is then normalized as $p_n \leftarrow p_n / \sum p_n$. Next, for each n, I compute surplus with $m+n$ total competitors and take expectation as $\tilde{\Pi}(v; m, \Lambda, \rho) = \sum_n p_n \cdot \text{surplus}(v, m+n, \rho)$.

Simulation of Gaussian Equicorrelation Copula

I estimate the conditional density $f_{V_{(n:n)}|V_0}(y \mid v, \hat{\rho})$ via Monte Carlo simulation from the $(n+1)$-dimensional Gaussian equicorrelation copula. The simulation exploits the one-factor ("compound symmetric") structure.

Simulation algorithm.

1. **Generate common factor:** Draw $W \sim N(0,1)$ independently for each simulation replicate. This shared factor induces equicorrelation across all dimensions.

2. **Generate idiosyncratic shocks:** Draw $\varepsilon_j \overset{\text{i.i.d.}}{\sim} N(0,1)$, $j = 1, \ldots, n+1$, independent of W.

3. **Construct correlated normals:** $Z_j = \sqrt{\hat{\rho}}\, W + \sqrt{1-\hat{\rho}}\, \varepsilon_j$. By construction, $\text{Cov}(Z_j, Z_k) = \hat{\rho}$ for $j \neq k$.

4. **Transform to uniform marginals:** $U_j = \Phi(Z_j)$, where Φ is the standard normal CDF. The vector $(U_1, \ldots, U_{n+1})$ follows the Gaussian equicorrelation copula with parameter $\hat{\rho}$.

5. **Transform to bid scale:** $V_j = F_V^{-1}(U_j)$ for $j = 1$ (own value, full distribution) and $V_j = F_{V|\text{trunc}}^{-1}(U_j)$ for $j = 2, \ldots, n+1$ (competitors, truncated distribution $F_V(\cdot \mid v \geq v^*)$).

6. **Record maximum competitor bid:** $V_{(n:n)} = \max\{V_2, \ldots, V_{n+1}\}$.

7. **Conditional sampling:** Retain draws where the own bid $|U_1 - F_V(v)| < \varepsilon$ (adaptive tolerance $\varepsilon = 0.02$, doubled if fewer than 100 draws survive).

8. **Kernel density estimation:** Estimate $\hat{f}_{V_{(n:n)}|V_0}(y \mid v)$ from retained $\{V_{(n:n)}^{(j)}\}$ via Gaussian KDE with Silverman bandwidth. Normalize to integrate to one over $[v^*, v]$.

E Toy Model and Supplementary Analysis

E.1 Simplified Nomination Model: Coordination Without Late-Learners

This appendix presents a simplified version of the nomination game that abstracts from value uncertainty and strategic late-learning. The purpose is to isolate the pure coordination problem inherent in the nomination threshold and to provide geometric intuition for equilibrium multiplicity. The full structural model (Section 7) adds learning costs, endogenous late-learning, and Poisson arrival rates, which substantially amplify coordination failure relative to the analysis here.

E.2 Simplifications Relative to Full Model

The following simplifications transform the two-stage game with early learning ($l = 1$) to nominate and late-learning into a one-shot coordination game where the only decision is whether to pay cost c_i to nominate a parcel of known value v_i.

Assumption 1: No Value Uncertainty. All bidders know their private values v_i ex ante, eliminating learning costs and the late-learning motive. This allows us to focus purely on the coordination problem inherent in the nomination threshold.

Assumption 2: Fixed Competition. The number of potential bidders I is known and fixed (not Poisson-distributed). This removes uncertainty about the number of competitors and isolates the strategic interaction in nomination decisions.

Assumption 3: Nomination Costs Only. Bidders face only nomination costs $c_i \sim \text{Gamma}(\theta_{c1}, \theta_{c2})$, which rationalize the observed low nomination rates $\frac{M_t}{I_t}$ despite the absence of value uncertainty.

These simplifications transform the model from a complex game with sequential exploration and entry decisions into a cleaner coordination game where the trade-off between auction occurrence (extensive margin) and revenue conditional on occurrence (intensive margin) is transparent.

E.3 Nomination Decision and Equilibrium

A bidder with valuation v_i and nomination cost c_i nominates if and only if

$$h(\pi_0; K, I) \cdot \Delta(v_i; I) \geq c_i \tag{85}$$

where $h(\pi_0; K, I) = \binom{I-1}{K-1} \pi_0^{K-1} (1-\pi_0)^{I-K}$ is the probability of being pivotal and $\Delta(v_i; I) = \int_0^{v_i} F_V(y)^{I-1} dy$ is the expected surplus from a second-price auction with $I-1$ competitors. The symmetric equilibrium nomination probability π_0^* satisfies

$$\pi_0^* = \int_{\underline{v}}^{\overline{v}} F_C\left(h(\pi_0^*; K, I) \cdot \Delta(v; I)\right) f_V(v)\, dv. \tag{86}$$

When $K = 1$, nominations are strategic substitutes: $h(\pi_0; 1, I) = (1-\pi_0)^{I-1}$ is strictly decreasing, ensuring a unique equilibrium. When $K \geq 2$, nominations become strategic complements and multiple equilibria, including a coordination-failure equilibrium at $\pi_0^* = 0$, can arise (Figure 20).

E.4 Estimation of Cost Distribution

I estimate the nomination cost distribution $F_C(c; \theta_c)$ using a Gamma specification

$$F_C(c; \alpha, \beta) = \frac{\gamma(\alpha, c/\beta)}{\Gamma(\alpha)} \tag{87}$$

where $\theta_c = (\alpha, \beta)$ are shape and rate parameters. Using the observed nomination rate $\bar{\pi}_{\text{obs}}$ and the variance of nomination rates, I form two moment conditions and estimate via GMM:

$$\hat{\theta}_c = \arg\min_{\theta_c \in \Theta_c} \mathbf{g}(\theta_c)' \mathbf{W} \mathbf{g}(\theta_c). \tag{88}$$

Table 24 presents the estimates.

E.5 Counterfactual: Nomination Threshold K

Table 25 presents counterfactual results varying $I \in \{1, 2, 3\}$ at mean $I = 4$ bidders. In this simplified model (no late-learning), $K = 2$ produces a reduced but positive trigger rate and doubles conditional revenue through selection. $K = 3$ produces complete coordination failure. The full structural model (Section 7) finds complete failure already at $K = 2$, because late-learning compounds the coordination problem.

Table 24: Toy Model: Estimated Parameters

Parameter	**Estimate**	**Std. Error**
Shape (α)	0.82	(0.09)
Scale (β)	34.44	(3.12)
Implied Moments		
Mean cost (\$/acre)	28.09	(3.05)
Std. dev. cost (\$/acre)	31.11	(4.21)

Standard errors in parentheses, computed via bootstrap with 1,000 replications.

Table 25: Toy Model Counterfactual: Effect of Nomination Threshold

Metric	$K = 1$ **(Baseline)**	$K = 2$	$K = 3$
Nomination rate (π_0^*)	19.2%	15.4%	0%
Auction occurrence $\Pr(M \geq K)$	58%	24%	0%
# Bidders (if occurs)	3.56	3.56	—
Revenue | occurs (\$/acre)	120	250	—
Revenue unconditional (\$/acre)	70	60	0
Change vs. baseline:			
Δ Revenue | occurs	—	+108%	—
Δ Revenue unconditional	—	−14%	−100%

All results at mean $N = 4$ bidders. Revenue is per-acre winning bid. "—" indicates auction never occurs. Contrast with the full structural model (Section 7), where $K = 2$ already produces zero trigger rate.

E.6 Equilibrium Multiplicity Under $K \geq 2$

The hump-shaped $G(\pi)$ function when $K \geq 2$ reflects strategic complementarities: the pivotal probability initially increases with π (making nomination more attractive), peaks at $\pi \approx \frac{K-1}{I-1}$, then declines. This non-monotonicity generates multiple equilibria, with the coordination-failure equilibrium ($\pi = 0$) always present when $K \geq 2$.

E.7 Proofs for Toy Example

E.8 Proof of Uniqueness when $K = 1$

Proposition 4 (Uniqueness for $K = 1$)**.** *When $K = 1$, there exists at most one equilibrium nomination probability $\pi^* \in [0, 1]$.*

Proof. When $K = 1$, the pivotal probability becomes

$$h(\pi) = (1-\pi)^{I-1} \tag{89}$$

Taking the derivative

$$h'(\pi) = -(I-1)(1-\pi)^{I-2} < 0 \quad \forall \pi \in [0,1) \tag{90}$$

The operator $G(\pi, I)$ is defined as

$$G(\pi, I) = \int_{\underline{v}}^{\bar{v}} F_C\left((1-\pi)^{I-1} \cdot \Delta(v; I)\right) f_V(v)\, dv \tag{91}$$

Since F_C is monotonically non-decreasing and $h(\pi)$ is strictly decreasing, $G(\pi, I)$ is strictly decreasing in π. Therefore, the fixed point equation $G(\pi^*, I) = \pi^*$ has at most one solution.

If $G(0, I) \leq 0$, then $\pi^* = 0$ is the unique equilibrium. If $G(0, I) > 0$, then by the Intermediate Value Theorem and monotonicity, there exists a unique $\pi^* \in (0, 1)$ satisfying $G(\pi^*, I) = \pi^*$. □

E.9 Proof of Multiple Equilibria when $K > 1$

Proposition 5 (Multiple Equilibria for $K > 1$)**.** *When $K > 1$, multiple equilibria may exist, including*

1. $\pi^* = 0$ *(coordination failure equilibrium)*
2. *One or more interior equilibria* $\pi^* \in (0, 1)$

Proof. For $K > 1$, the pivotal probability is

$$h(\pi) = \binom{I-1}{K-1} \pi^{K-1} (1-\pi)^{I-K} \tag{92}$$

Step 1: Properties of $h(\pi)$.

Taking the derivative

$$h'(\pi) = \binom{I-1}{K-1} \left[(K-1)\pi^{K-2}(1-\pi)^{I-K} - (I-K)\pi^{K-1}(1-\pi)^{I-K-1}\right] \tag{93}$$

$$= \binom{I-1}{K-1} \pi^{K-2}(1-\pi)^{I-K-1} \left[(K-1)(1-\pi) - (I-K)\pi\right] \tag{94}$$

Setting $h'(\pi) = 0$

$$(K-1)(1-\pi) = (I-K)\pi \implies \pi_{l=1} = \frac{K-1}{I-1} \tag{95}$$

This is the unique interior critical point. Since $h(0) = h(1) = 0$ and $h(\pi_{l=1}) > 0$, $h(\pi)$ is single-peaked.

Step 2: Properties of $G(\pi, I)$.

Because F_C is non-decreasing and $\Delta(v; I) \geq 0$, for each v the map $\pi \mapsto F_C(h(\pi)\,\Delta(v; I))$ is single-peaked at the same point as $h(\pi)$; integrating over v preserves this, so

$$G(\pi, I) = \int_{\underline{v}}^{\bar{v}} F_C(h(\pi) \cdot \Delta(v; I)) f_V(v)\, dv \tag{96}$$

is also single-peaked at $\pi_{l=1}$, with $G(0, I) = G(1, I) = F_C(0)$.

Step 3: Equilibrium multiplicity.

The fixed point equation $G(\pi, I) = \pi$ can have multiple solutions depending on the height of the peak $G(\pi_{l=1}, I)$. If $G(\pi_{l=1}, I) < \pi_{l=1}$, only $\pi^* = 0$ exists (coordination failure). If $G(\pi_{l=1}, I) > \pi_{l=1}$, the 45-degree line intersects $G(\cdot)$ at $\pi^* = 0$ and at least one interior point. The number of interior equilibria depends on the curvature of $G(\cdot)$ near $\pi_{l=1}$

□

F Posterior Density Ratio: $f_{\text{post}}(v \mid m)/f_V(v)$

Figure 21 provides direct evidence for the nature of the Bayesian update at the estimated parameters ($\hat{\rho} = 0.861$, $v^* \approx r$, $u^* = F_V(v^*) \approx 0$). The left panel overlays the copula posterior density $f_{\text{post}}(v \mid m)$ at $m \in \{1, 2, 4, 8\}$ against the prior density truncated at v^* (black dashed, the $m = 0$ baseline). The right panel plots the pointwise ratio $f_{\text{post}}(v \mid m)/f_{\text{prior}}(v)$.

Across the mass-bearing region the ratio is indistinguishable from 1: the posterior and truncated-prior densities in the left panel are visually almost identical, so the Bayesian update leaves the conditional distribution unchanged.

This confirms that the near-zero nomination threshold $u^* \approx 0$ renders the conditioning event $\{V_j \geq v^*\}$ nearly certain: it conveys almost no information about the common factor W through the *shape* of the conditional distribution within $[v^*, \infty)$. The operative information channel at the estimated parameters is instead the *entry probability*: the mass $\int_{v^*}^{\infty} f_{\text{post}}(v \mid m)\, dv = \Pr(V \geq v^* \mid m)$ rises above $\Pr(V \geq v^*)$ via the strong copula correlation ($\hat{\rho} = 0.861$), even though the shape of the conditional density is unchanged.

G The Nomination Fee Counterfactual

In this counterfactual, nomination fee is raised from its current level, holding the single-nomination rule ($K = 1$) fixed. A dollar fee ϕ enters each tract's equilibrium as the residual fee $\tilde{\phi} = \phi/s_b$, so a common dollar fee is a larger burden on smaller tracts. I re-solve the equilibrium objects (v^{**}, c_L^*, c_H^*) at each fee for the median value bin, with the most expensive and cheapest bins as bounds, and report revenue both *conditional* on a triggered auction and *unconditional* (per offered tract, i.e. trigger rate $\times$ conditional revenue), each net of the fee transfer collected from nominators.

A higher fee raises the nomination-value threshold v^{**}, so the firms that still nominate are more strongly selected on value. Conditional on an auction occurring, this *raises* the winning bid: at the median value tracts, revenue per triggered auction rises from about \$2,900 at the current fee toward \$9,800 as the fee climbs (and analogously \$10,400 $\rightarrow$ \$36,400 among the most expensive tracts). But the same increase drives the trigger rate $\Pr(M \geq 1) = 1 - e^{-\Lambda_{nom}}$ down monotonically, because $\Lambda_{nom} = \lambda_{pre}\, \pi_0^L\, [1 - F_V(v^{**})]$ falls as v^{**} rises. The two effects work against each other.

Netting the two, expected revenue *per offered tract* is maximized at $\phi = 0$ in every value class and declines as the fee rises. For most expensive tracts the seller's net revenue (auction revenue plus the fee transfer) stays roughly flat over a wide range because the fee it collects offsets the lost auctions. It never exceeds the zero-fee level and the underlying auction revenue (excluding the transfer) falls throughout. There is thus no revenue-improving interior fee: the current nominal fee is already essentially optimal, and the flat range means the seller is close to indifferent over modest fees.

The reason the fee cannot help is that the constraint on market thickness is the scarcity of early, informed arrivals ($\hat{\lambda}_{pre} \approx 0.005$) together with the learning-cost distribution, not the level of the \$10 fee. Raising the fee can only screen the extensive margin further and cannot manufacture the lacking early interest.

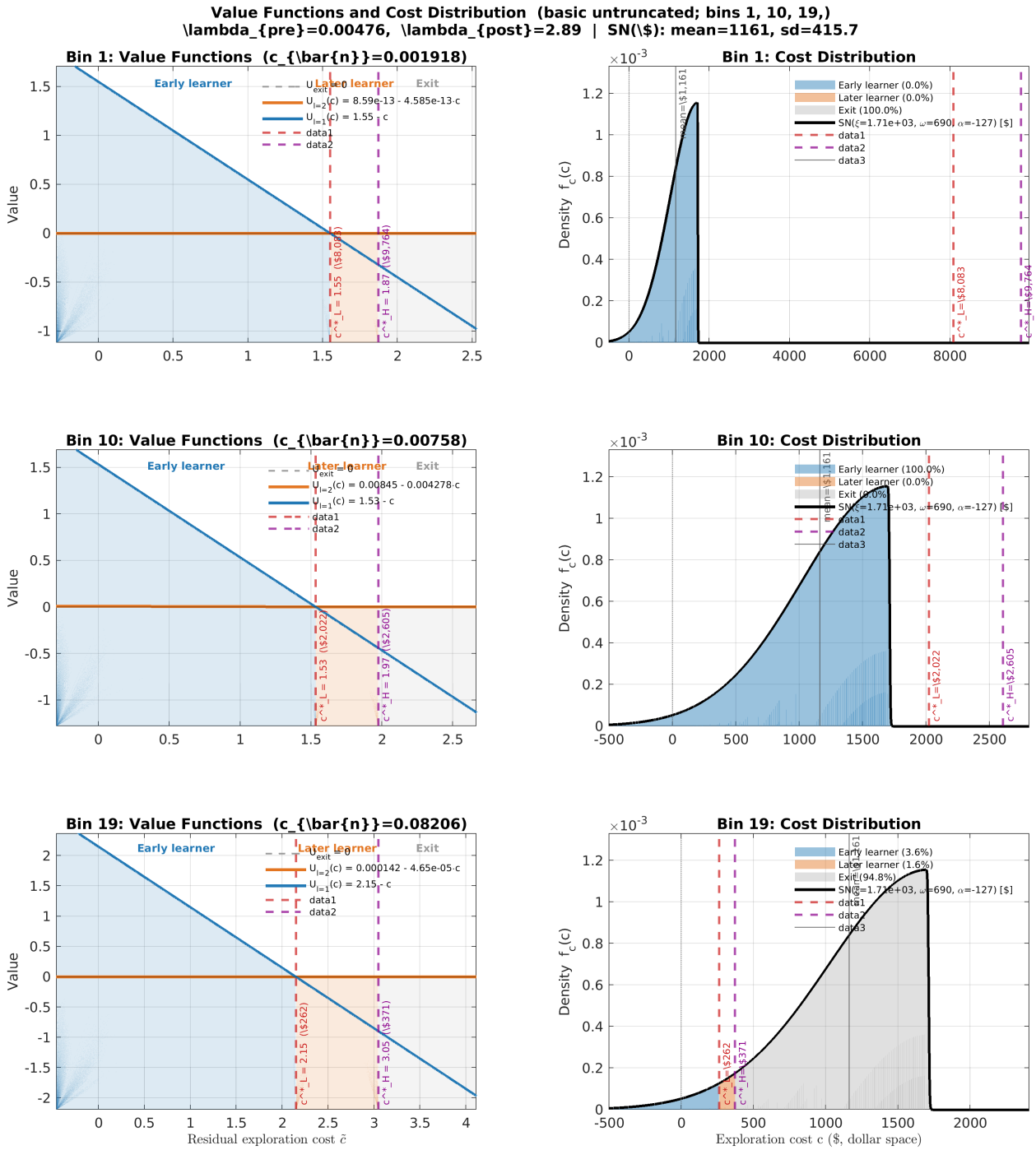


Figure 13: Value Functions and Cost Distribution (Bins 1, 10, and 19)

Notes: Panel A: Early learner value $U_{l=1}(c)$ (blue) and non-nominator value $U_{l=2}(c)$ (orange) as functions of learning cost c, evaluated at Bin 5 estimated parameters. Vertical dashed lines mark c_L^* and c_H^*. Panel B: Estimated Skew-Normal cost density $F_C(c)$ with early learner and $l_i = 2$ non-nominator cutoffs marked. Early learner region: $\approx 0.1\%$; non-nominator region: $\approx 0.2\%$; exit: $\approx 99.8\%$. Costs are in residual units; dollar costs equal $c \times s_t$ where $s_t = \exp(\hat{y}_t)$ is the auction-specific hedonic scaling factor.

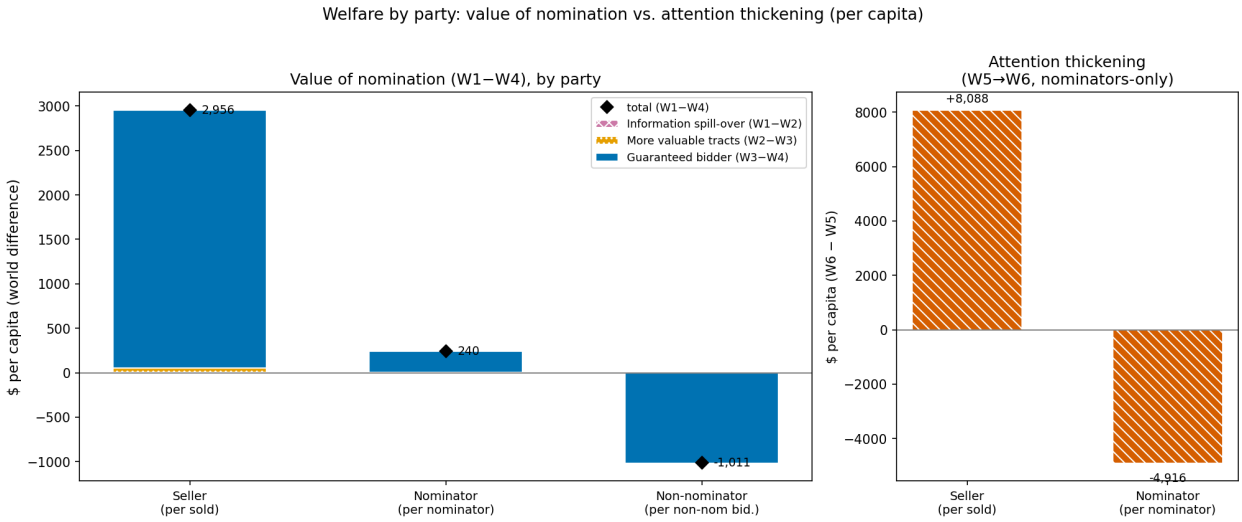


Figure 14: **Welfare by party: value of nomination vs. concentrated attention (per offered tract).** *Left:* the value of nomination ($\mathcal{W}_1 - \mathcal{W}_4$), per offered tract, for the seller, nominator, and non-nominator, decomposed into information spill-over ($\mathcal{W}_1 - \mathcal{W}_2$), the selection of more valuable tracts ($\mathcal{W}_2 - \mathcal{W}_3$), and the guaranteed bidder ($\mathcal{W}_3 - \mathcal{W}_4$); the white/red diamond marks each party's total (the value of nomination). *Right:* concentrated attention ($\mathcal{W}_5 \to \mathcal{W}_6$), per tract. Per offered tract, $\mathcal{W}_4$ counts tracts that receive no bid as zero. Bars value each bin at its median tract; valuing each tract at its own scaler s_t instead gives a seller gain of \$4,189 per offered tract (\$9,418 → \$13,607; Appendix Table 20), the basis of the abstract and the introduction. Bootstrap standard errors from the resampling bootstrap.

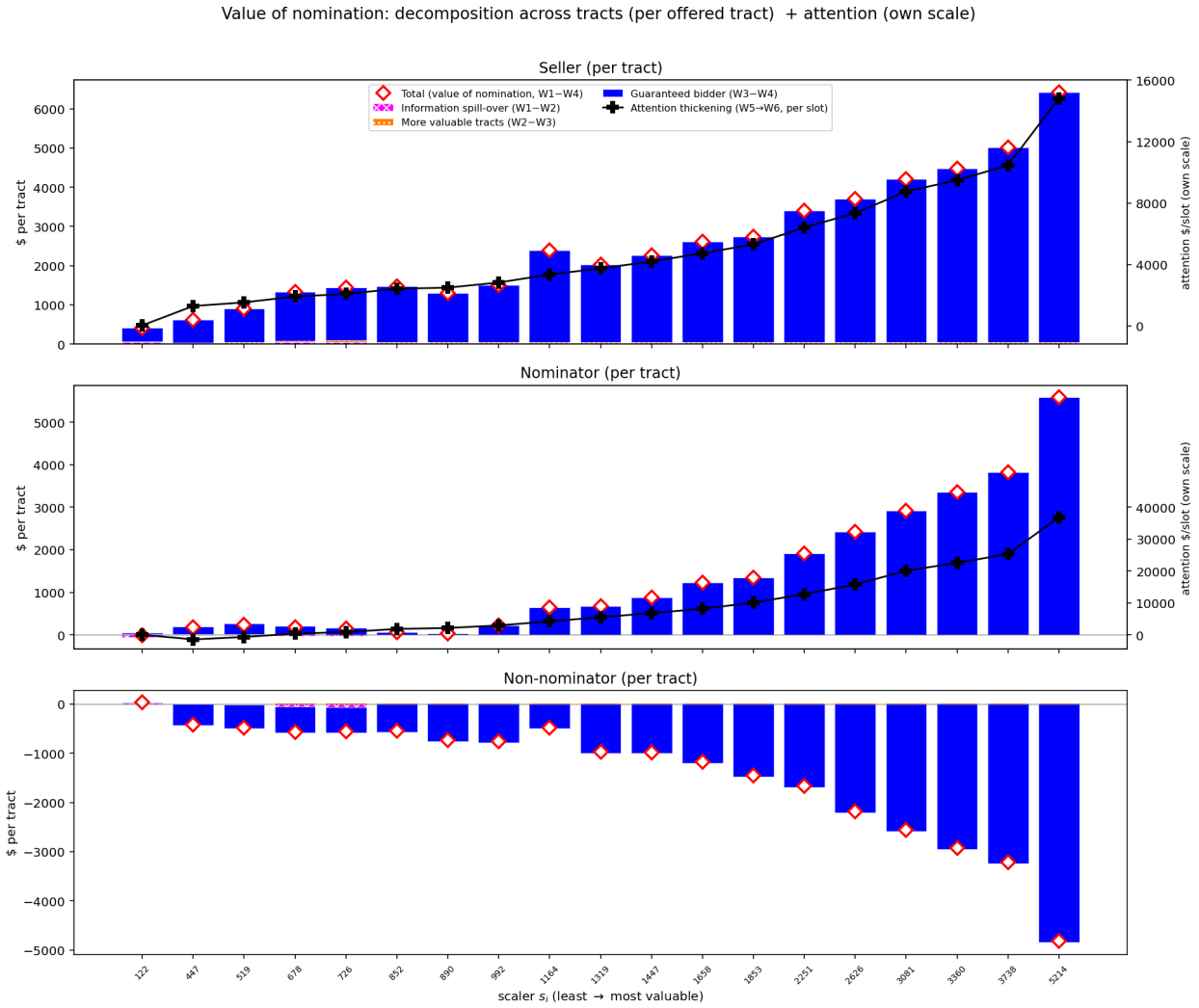


Figure 15: **Welfare across tracts (bins), by party.** Four-channel decomposition of the value of nomination per bin for the seller, nominator, and non-nominator, each **per offered tract** ($\mathcal{W}_{4b}$ counts non-entered tracts as zero); the white/red diamond is the total before attention thickening and the black "+" line is attention thickening ($\mathcal{W}_5 \to \mathcal{W}_6$, per tract) on a zero-aligned secondary axis. Bins run from the most valuable tracts (Bin 1) to the least valuable (Bin 19).

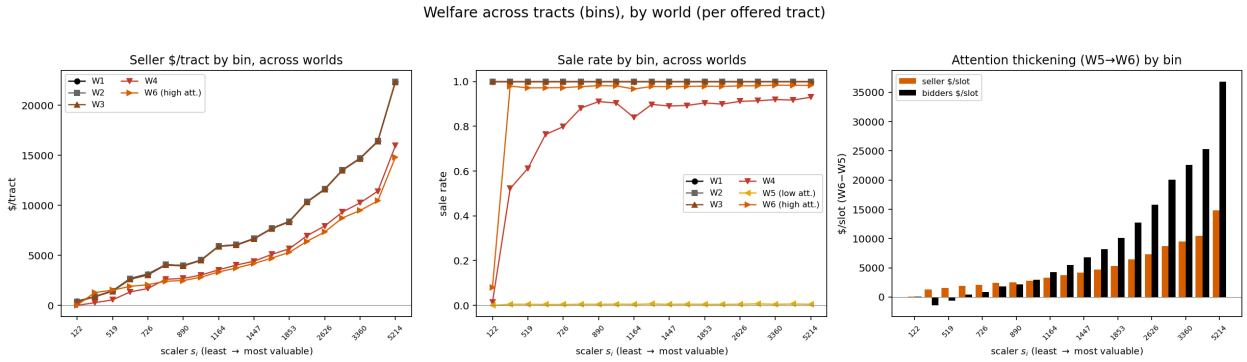


Figure 16: **Welfare across tracts, by world.** Seller revenue per offered auction and the sale rate in each counterfactual world, and attention thickening ($\mathcal{W}_5 \to \mathcal{W}_6$), by bin. Under the thin fitted rate $\hat{\lambda}_b$ ($\mathcal{W}_5$) the sale rate is near zero except in the highest-attention bin; thickening ($\mathcal{W}_6$) restores near-certain sale throughout.

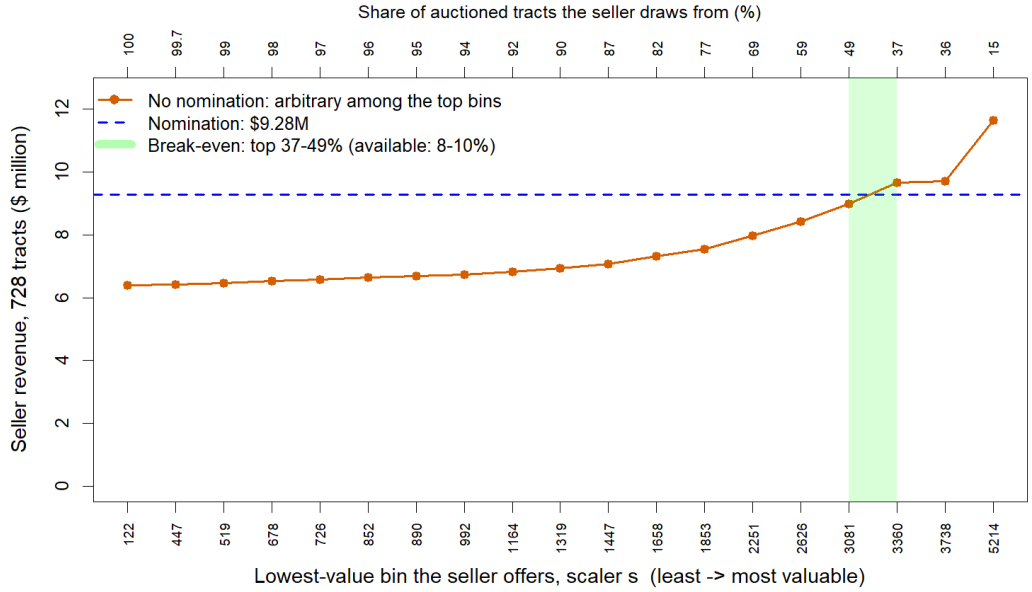


Figure 17: **How much would the seller need to know?** Seller revenue over 728 offered tracts when the seller runs no nomination and offers tracts arbitrarily from the top bins by observable value (scaler s_b), with the included bins in proportion to their auction shares, against revenue under nomination (dashed). The horizontal axis is the lowest-value bin the seller still offers, labelled by its scaler s_b and running from the least to the most valuable; the top axis gives the share of auctioned tracts in the included bins. Offering all bins (the leftmost point) reproduces the no-nomination benchmark $\mathcal{W}_4$. The shaded band marks the break-even: offering only Bins 1–3 (8% of available tracts; 37% of auctioned tracts) beats nomination, while offering Bins 1–4 (10% of available tracts; 49% of auctioned) does not. Each bin is valued at its median tract.

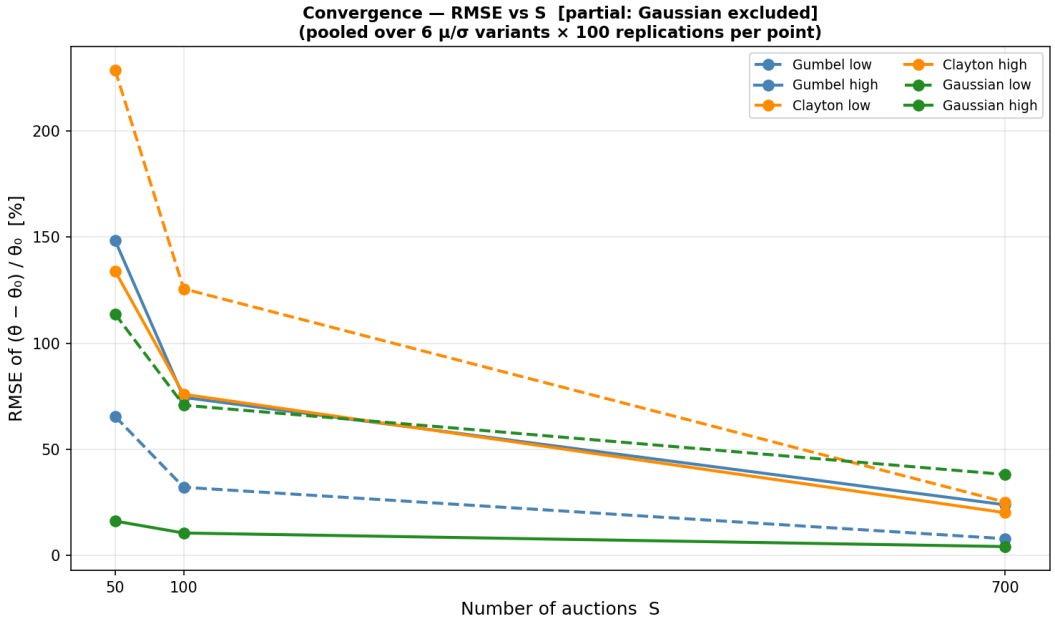


Figure 18: RMSE of the normalised deviation $(\hat{\rho} - \rho_0)/\rho_0$ versus number of auctions S, pooled over six lognormal marginal variants (600 replications per point). Solid lines = high dependence; dashed lines = low dependence. All copula families converge toward zero as S grows.

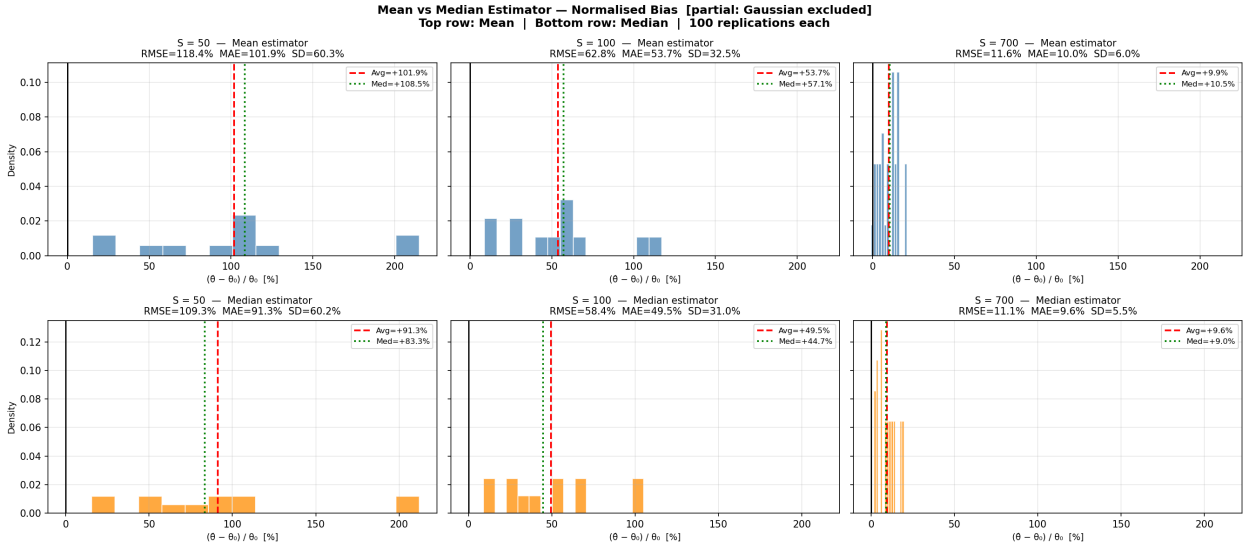
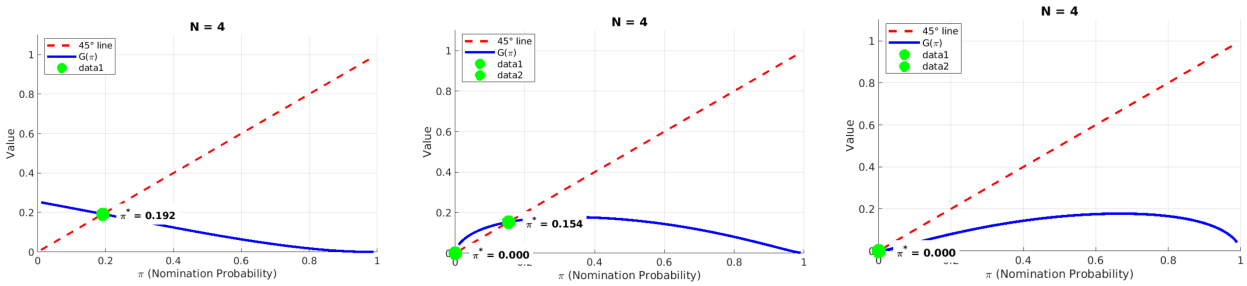


Figure 19: Distribution of the normalised deviation $(\hat{\rho} - \rho_0)/\rho_0$ across all 36 DGP cells, separately for the mean aggregator (top row) and the median aggregator (bottom row), at $S \in \{50, 100, 700\}$ (columns left to right). Red dashed line = average of the displayed distribution; green dotted line = median of the displayed distribution. The mean distribution has a heavier right tail at small S (driven by Gumbel and Clayton cells), which the median aggregator substantially dampens.



(a) $K = 1$: Unique equilibrium (b) $K = 2$: Multiple equilibria (c) $K = 3$: Coordination failure

Figure 20: Fixed Point Problem $G(\pi; K, I) = \pi$ for $I = 4$ Bidders (Simplified Model)

Each panel plots the operator $G(\pi; K, I)$ (blue curve) against the 45-degree line (red dashed). Equilibria correspond to intersections. Panel (a): strategic substitutes under $K = 1$, unique equilibrium. Panel (b): strategic complements under $K = 2$, multiple equilibria including coordination failure. Panel (c): $K = 3$, only coordination failure. This simplified model abstracts from late-learning; the full model (Section 7) finds coordination failure already at $K = 2$.

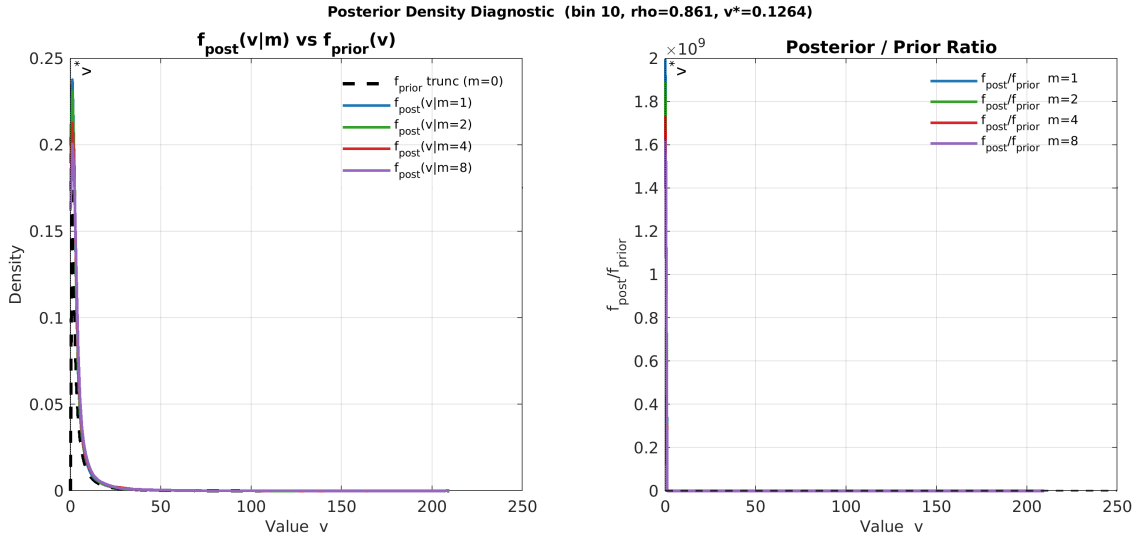


Figure 21: Posterior Density vs. Prior, and Their Ratio (Estimated Parameters)

Notes: Left panel: prior density truncated at v^* (black dashed, the $m = 0$ baseline) and copula posterior $f_{\text{post}}(v \mid m)$ at $m \in \{1, 2, 4, 8\}$ (colored solid). Right panel: point-wise ratio $f_{\text{post}}(v \mid m)/f_{\text{prior}}(v)$. Vertical dotted line marks v^*. Away from the boundary the ratio is indistinguishable from 1, so the conditional density shape is essentially unchanged by the Bayesian update at $u^* \approx 0$; the large spike near the lower boundary $v \approx v^*$ (of order 10^9) is a numerical artifact of dividing by the sharply truncated prior density there and carries negligible probability mass. Reported for Bin 10 (median bin), $\hat{\rho} = 0.861$, $v^* = 0.1264$ ($u^* \approx 0$).